SQL MASTERY

1000 Multiple Choice Questions

Complete Study Guide with Answers

Covers: Basics • DDL/DML/DCL/TCL • Joins • Subqueries • Window Functions

Indexes • Stored Procedures • Triggers • Transactions • Normalization

MySQL • PostgreSQL • SQL Server • Oracle • Performance Tuning

✦ All 1000 questions include 4 options with correct answer ✦

Total Questions: 1000	**Format:** MCQ (4 options)	**Answer Key:** Included	**Difficulty:** Beginner to Expert

Section	Topics Covered
Q 1–80	SQL Basics, DDL, DML, Constraints, Data Types, Functions
Q 81–160	Joins (INNER, OUTER, CROSS, SELF, JOIN Algorithms)
Q 161–240	Subqueries, CTEs, Set Operations (UNION/INTERSECT/EXCEPT)
Q 241–310	Indexes, Query Optimization & Performance
Q 311–380	Stored Procedures, Functions, Triggers, Cursors
Q 381–440	Transactions, ACID, Concurrency & Locking
Q 441–500	Window Functions (ROW_NUMBER, RANK, LAG/LEAD, Frames)
Q 501–570	Normalization, Database Design, Data Warehousing
Q 571–700	Advanced SQL, MERGE, JSON/XML, Security, ORM, Temporal
Q 701–760	MySQL Specific
Q 761–820	PostgreSQL Specific
Q 821–880	SQL Server Specific
Q 881–1000	Oracle Specific & Expert-Level Optimization

Section: SQL Basics, DDL, DML, Constraints & Built-in Functions (Q 1–80)

Q1. What does SQL stand for?

(A) Structured Query Language
(B) Simple Query Language
(C) Standard Query Logic
(D) Sequential Query Language

✓ **Answer: (A) Structured Query Language**

Q2. Which SQL statement is used to create a new table?

(A) CREATE TABLE
(B) MAKE TABLE
(C) NEW TABLE
(D) BUILD TABLE

✓ **Answer: (A) CREATE TABLE**

Q3. Which clause is used to filter records?

(A) WHERE
(B) HAVING
(C) FILTER
(D) SELECT

✓ **Answer: (A) WHERE**

Q4. Which command is used to remove a table permanently?

(A) DROP TABLE
(B) DELETE TABLE
(C) REMOVE TABLE
(D) ERASE TABLE

✓ **Answer: (A) DROP TABLE**

Q5. Which SQL statement modifies an existing table structure?

(A) ALTER TABLE
(B) MODIFY TABLE
(C) CHANGE TABLE
(D) UPDATE TABLE

✓ **Answer: (A) ALTER TABLE**

Q6. Which data type stores variable-length character strings?

(A) VARCHAR
(B) CHAR
(C) TEXT
(D) STRING

✓ **Answer: (A) VARCHAR**

Q7. What is the default sort order of ORDER BY?

(A) Ascending
(B) Descending
(C) Random
(D) Alphabetical

✓ **Answer: (A) Ascending**

Q8. Which keyword removes duplicate rows from a result set?

(A) DISTINCT
(B) UNIQUE
(C) DIFFERENT
(D) SINGLE

✓ Answer: (A) DISTINCT

Q9. Which constraint ensures a column cannot have NULL values?

(A) NOT NULL
(B) PRIMARY KEY
(C) UNIQUE
(D) CHECK

✓ Answer: (A) NOT NULL

Q10. Which SQL command is used to insert data into a table?

(A) INSERT INTO
(B) ADD INTO
(C) PUT INTO
(D) STORE INTO

✓ Answer: (A) INSERT INTO

Q11. Which SQL command is used to update existing data?

(A) UPDATE
(B) MODIFY
(C) CHANGE
(D) ALTER

✓ Answer: (A) UPDATE

Q12. Which SQL command deletes rows from a table?

(A) DELETE
(B) REMOVE
(C) ERASE
(D) DROP

✓ Answer: (A) DELETE

Q13. What does DML stand for?

(A) Data Manipulation Language
(B) Data Management Language
(C) Data Merging Language
(D) Data Modification Logic

✓ Answer: (A) Data Manipulation Language

Q14. What does DDL stand for?

(A) Data Definition Language
(B) Data Deletion Language
(C) Data Design Language
(D) Data Distribution Language

✓ Answer: (A) Data Definition Language

Q15. Which DDL command removes all rows but keeps the table structure?

(A) TRUNCATE
(B) DELETE
(C) DROP
(D) CLEAR

✓ Answer: (A) TRUNCATE

Q16. Which constraint uniquely identifies each record?

(A) PRIMARY KEY
(B) UNIQUE
(C) NOT NULL
(D) INDEX

✓ **Answer: (A) PRIMARY KEY**

Q17. Which constraint links two tables together?

(A) FOREIGN KEY
(B) PRIMARY KEY
(C) UNIQUE
(D) CHECK

✓ **Answer: (A) FOREIGN KEY**

Q18. What does the UNIQUE constraint do?

(A) Ensures all values in a column are different
(B) Makes a column primary key
(C) Prevents NULL values
(D) Creates an index

✓ **Answer: (A) Ensures all values in a column are different**

Q19. Which data type stores whole numbers?

(A) INT
(B) FLOAT
(C) VARCHAR
(D) DATE

✓ **Answer: (A) INT**

Q20. Which data type stores decimal numbers?

(A) DECIMAL
(B) INT
(C) VARCHAR
(D) BOOLEAN

✓ **Answer: (A) DECIMAL**

Q21. How do you add a column to an existing table?

(A) ALTER TABLE ADD COLUMN
(B) INSERT COLUMN
(C) UPDATE TABLE ADD
(D) CREATE COLUMN

✓ **Answer: (A) ALTER TABLE ADD COLUMN**

Q22. Which statement is used to rename a table?

(A) ALTER TABLE RENAME TO
(B) RENAME TABLE
(C) CHANGE TABLE NAME
(D) UPDATE TABLE NAME

✓ **Answer: (A) ALTER TABLE RENAME TO**

Q23. Which SQL keyword is used to create a view?

(A) CREATE VIEW
(B) MAKE VIEW
(C) BUILD VIEW
(D) NEW VIEW

✓ **Answer: (A) CREATE VIEW**

Q24. What is a schema in SQL?

(A) A logical container for database objects
(B) A type of index
(C) A query plan
(D) A stored procedure

✓ **Answer: (A) A logical container for database objects**

Q25. Which command grants privileges to a user?

(A) GRANT
(B) ALLOW
(C) PERMIT
(D) GIVE

✓ **Answer: (A) GRANT**

Q26. Which command revokes privileges?

(A) REVOKE
(B) DENY
(C) REMOVE
(D) DELETE

✓ **Answer: (A) REVOKE**

Q27. What does DCL stand for?

(A) Data Control Language
(B) Data Configuration Language
(C) Data Computation Language
(D) Data Conversion Language

✓ **Answer: (A) Data Control Language**

Q28. What does TCL stand for?

(A) Transaction Control Language
(B) Table Control Language
(C) Tuple Control Language
(D) Type Control Language

✓ **Answer: (A) Transaction Control Language**

Q29. Which TCL command saves a transaction?

(A) COMMIT
(B) SAVE
(C) STORE
(D) END

✓ **Answer: (A) COMMIT**

Q30. Which TCL command undoes a transaction?

(A) ROLLBACK
(B) UNDO
(C) CANCEL
(D) REVERT

✓ **Answer: (A) ROLLBACK**

Q31. Which TCL command sets a point within a transaction?

(A) SAVEPOINT
(B) CHECKPOINT
(C) MARKER
(D) ANCHOR

✓ **Answer: (A) SAVEPOINT**

Q32. Which keyword begins a transaction explicitly?

(A) BEGIN TRANSACTION
(B) START TRANSACTION
(C) OPEN TRANSACTION
(D) INIT TRANSACTION

✓ **Answer: (A) BEGIN TRANSACTION**

Q33. Which constraint checks a condition before inserting data?

(A) CHECK
(B) VERIFY
(C) VALIDATE
(D) ASSERT

✓ **Answer: (A) CHECK**

Q34. What is the maximum length of a CHAR data type in most databases?

(A) 255
(B) 64
(C) 512
(D) 1024

✓ **Answer: (A) 255**

Q35. Which data type stores TRUE/FALSE values?

(A) BOOLEAN
(B) BIT
(C) INT
(D) CHAR

✓ **Answer: (A) BOOLEAN**

Q36. Which clause is used with GROUP BY to filter groups?

(A) HAVING
(B) WHERE
(C) FILTER
(D) SELECT

✓ **Answer: (A) HAVING**

Q37. What is a NULL value?

(A) Absence of a value
(B) Zero
(C) Empty string
(D) False

✓ **Answer: (A) Absence of a value**

Q38. Which operator checks for NULL?

(A) IS NULL
(B) = NULL
(C) == NULL
(D) NULL =

✓ **Answer: (A) IS NULL**

Q39. Which operator checks for non-NULL?

(A) IS NOT NULL
(B) != NULL

(C) <> NULL

(D) NOT NULL =

✓ Answer: (A) IS NOT NULL

Q40. Which function returns the number of rows?

(A) COUNT

(B) SUM

(C) TOTAL

(D) NUM

✓ Answer: (A) COUNT

Q41. Which function returns the highest value?

(A) MAX

(B) TOP

(C) HIGHEST

(D) UPPER

✓ Answer: (A) MAX

Q42. Which function returns the lowest value?

(A) MIN

(B) BOTTOM

(C) LOWEST

(D) LOWER

✓ Answer: (A) MIN

Q43. Which function calculates the average?

(A) AVG

(B) MEAN

(C) AVERAGE

(D) MID

✓ Answer: (A) AVG

Q44. Which function sums values?

(A) SUM

(B) ADD

(C) TOTAL

(D) PLUS

✓ Answer: (A) SUM

Q45. What does SELECT * FROM table return?

(A) All columns

(B) Only first column

(C) Only last column

(D) Primary key column

✓ Answer: (A) All columns

Q46. Which clause limits rows returned?

(A) LIMIT / TOP

(B) RESTRICT

(C) CAP

(D) STOP

✓ Answer: (A) LIMIT / TOP

Q47. Which clause skips rows in a result?

(A) OFFSET
(B) SKIP
(C) JUMP
(D) PASS

✓ Answer: (A) OFFSET

Q48. Which operator is used for pattern matching?

(A) LIKE
(B) MATCH
(C) SIMILAR
(D) REGEX

✓ Answer: (A) LIKE

Q49. Which wildcard represents any single character in LIKE?

(A) _
(B) *
(C) %
(D) ?

✓ Answer: (A) _

Q50. Which wildcard represents zero or more characters in LIKE?

(A) %
(B) _
(C) *
(D) ?

✓ Answer: (A) %

Q51. Which operator checks membership in a list?

(A) IN
(B) AMONG
(C) CONTAINS
(D) WITHIN

✓ Answer: (A) IN

Q52. Which operator checks a value within a range?

(A) BETWEEN
(B) RANGE
(C) FROM-TO
(D) WITHIN

✓ Answer: (A) BETWEEN

Q53. Which operator combines two conditions both must be true?

(A) AND
(B) OR
(C) NOT
(D) XOR

✓ Answer: (A) AND

Q54. Which operator at least one condition must be true?

(A) OR
(B) AND
(C) NOT
(D) XOR

✓ Answer: (A) OR

Q55. Which operator negates a condition?

(A) NOT
(B) NO
(C) NEGATE
(D) REVERSE

✓ **Answer: (A) NOT**

Q56. What is an alias in SQL?

(A) A temporary name for a column or table
(B) A stored procedure
(C) An index
(D) A constraint

✓ **Answer: (A) A temporary name for a column or table**

Q57. Which keyword creates an alias?

(A) AS
(B) ALIAS
(C) NAME
(D) CALL

✓ **Answer: (A) AS**

Q58. Which statement retrieves data from a database?

(A) SELECT
(B) GET
(C) FETCH
(D) READ

✓ **Answer: (A) SELECT**

Q59. Which clause specifies the table to query?

(A) FROM
(B) IN
(C) AT
(D) USING

✓ **Answer: (A) FROM**

Q60. Which clause sorts the result?

(A) ORDER BY
(B) SORT BY
(C) ARRANGE BY
(D) GROUP BY

✓ **Answer: (A) ORDER BY**

Q61. Which clause groups rows sharing a property?

(A) GROUP BY
(B) ORDER BY
(C) CLUSTER BY
(D) COLLECT BY

✓ **Answer: (A) GROUP BY**

Q62. Which SQL comment syntax is single-line?

(A) --
(B) //
(C) #
(D) /* */

✓ **Answer: (A) --**

Q63. Which SQL comment syntax is multi-line?

(A) /* */
(B) --
(C) //
(D) ##

✓ Answer: (A) /* */

Q64. Which function returns the current date and time?

(A) NOW()
(B) TODAY()
(C) GETDATE()
(D) CURRENT()

✓ Answer: (A) NOW()

Q65. Which function returns the current date only?

(A) CURDATE()
(B) NOW()
(C) DATE()
(D) TODAY()

✓ Answer: (A) CURDATE()

Q66. Which function returns the length of a string?

(A) LEN() / LENGTH()
(B) SIZE()
(C) COUNT()
(D) CHARS()

✓ Answer: (A) LEN() / LENGTH()

Q67. Which function converts a string to uppercase?

(A) UPPER()
(B) TOUPPER()
(C) CAPS()
(D) UPPERCASE()

✓ Answer: (A) UPPER()

Q68. Which function converts a string to lowercase?

(A) LOWER()
(B) TOLOWER()
(C) SMALL()
(D) LOWERCASE()

✓ Answer: (A) LOWER()

Q69. Which function removes leading/trailing spaces?

(A) TRIM()
(B) STRIP()
(C) CLEAN()
(D) REMOVE()

✓ Answer: (A) TRIM()

Q70. Which function concatenates strings?

(A) CONCAT()
(B) JOIN()
(C) MERGE()
(D) COMBINE()

✓ **Answer: (A) CONCAT()**

Q71. Which function returns part of a string?

(A) SUBSTRING()
(B) PART()
(C) SLICE()
(D) EXTRACT()

✓ **Answer: (A) SUBSTRING()**

Q72. Which function replaces occurrences in a string?

(A) REPLACE()
(B) SWAP()
(C) CHANGE()
(D) MODIFY()

✓ **Answer: (A) REPLACE()**

Q73. Which function rounds a number?

(A) ROUND()
(B) CEIL()
(C) FLOOR()
(D) TRUNCATE()

✓ **Answer: (A) ROUND()**

Q74. Which function rounds up to the nearest integer?

(A) CEIL() / CEILING()
(B) FLOOR()
(C) ROUND()
(D) TRUNC()

✓ **Answer: (A) CEIL() / CEILING()**

Q75. Which function rounds down to the nearest integer?

(A) FLOOR()
(B) CEIL()
(C) ROUND()
(D) TRUNC()

✓ **Answer: (A) FLOOR()**

Q76. Which function returns the absolute value?

(A) ABS()
(B) MOD()
(C) SIGN()
(D) NEG()

✓ **Answer: (A) ABS()**

Q77. Which function returns the power of a number?

(A) POWER()
(B) EXP()
(C) POW()
(D) SQRT()

✓ **Answer: (A) POWER()**

Q78. Which function returns the square root?

(A) SQRT()
(B) ROOT()

(C) POWER()

(D) RADICAL()

✓ **Answer: (A) SQRT()**

Q79. What is the result of MOD(10,3)?

(A) 1

(B) 2

(C) 3

(D) 0

✓ **Answer: (A) 1**

Q80. Which function converts data types?

(A) CAST() / CONVERT()

(B) CHANGE()

(C) TRANSFORM()

(D) MORPH()

✓ **Answer: (A) CAST() / CONVERT()**

Section: Joins (Q 81–160)

Q81. Which JOIN returns rows that have matching values in both tables?

(A) INNER JOIN
(B) LEFT JOIN
(C) RIGHT JOIN
(D) FULL JOIN

✓ **Answer: (A) INNER JOIN**

Q82. Which JOIN returns all rows from the left table?

(A) LEFT JOIN
(B) RIGHT JOIN
(C) INNER JOIN
(D) CROSS JOIN

✓ **Answer: (A) LEFT JOIN**

Q83. Which JOIN returns all rows from the right table?

(A) RIGHT JOIN
(B) LEFT JOIN
(C) INNER JOIN
(D) CROSS JOIN

✓ **Answer: (A) RIGHT JOIN**

Q84. Which JOIN returns all rows from both tables?

(A) FULL OUTER JOIN
(B) INNER JOIN
(C) CROSS JOIN
(D) SELF JOIN

✓ **Answer: (A) FULL OUTER JOIN**

Q85. Which JOIN produces a Cartesian product?

(A) CROSS JOIN
(B) INNER JOIN
(C) OUTER JOIN
(D) NATURAL JOIN

✓ **Answer: (A) CROSS JOIN**

Q86. Which JOIN joins a table with itself?

(A) SELF JOIN
(B) INNER JOIN
(C) CROSS JOIN
(D) LOOP JOIN

✓ **Answer: (A) SELF JOIN**

Q87. What is a NATURAL JOIN?

(A) A join that automatically uses matching column names
(B) A join using WHERE clause
(C) A join using CROSS product
(D) A join using aliases

✓ **Answer: (A) A join that automatically uses matching column names**

Q88. Which keyword is used to specify join conditions?

(A) ON
(B) WHERE
(C) USING
(D) WITH

✓ Answer: (A) ON

Q89. What happens to non-matching rows in LEFT JOIN?

(A) They are included with NULL for right-table columns
(B) They are excluded
(C) They produce an error
(D) They are duplicated

✓ Answer: (A) They are included with NULL for right-table columns

Q90. Which join is used when no ON clause is specified?

(A) CROSS JOIN
(B) INNER JOIN
(C) OUTER JOIN
(D) SELF JOIN

✓ Answer: (A) CROSS JOIN

Q91. Which join type is the default when JOIN is written?

(A) INNER JOIN
(B) LEFT JOIN
(C) RIGHT JOIN
(D) FULL JOIN

✓ Answer: (A) INNER JOIN

Q92. What is the result of INNER JOIN when no rows match?

(A) Empty result set
(B) NULL rows
(C) Error
(D) All rows from both tables

✓ Answer: (A) Empty result set

Q93. Which SQL clause can replace INNER JOIN?

(A) WHERE with comma-separated tables
(B) OUTER JOIN
(C) CROSS JOIN
(D) NATURAL JOIN

✓ Answer: (A) WHERE with comma-separated tables

Q94. Can we join more than two tables?

(A) Yes
(B) No
(C) Only in stored procedures
(D) Only with UNION

✓ Answer: (A) Yes

Q95. Which type of join is a LEFT JOIN without matching rows from the right?

(A) LEFT ANTI JOIN / LEFT EXCLUDING JOIN
(B) RIGHT JOIN
(C) INNER JOIN
(D) CROSS JOIN

✓ Answer: (A) LEFT ANTI JOIN / LEFT EXCLUDING JOIN

Q96. What does USING keyword do in a JOIN?

(A) Specifies common column name for the join
(B) Replaces ON clause with a subquery
(C) Creates an alias
(D) Adds a filter

✓ **Answer: (A) Specifies common column name for the join**

Q97. Which JOIN is equivalent to FULL OUTER JOIN minus INNER JOIN?

(A) FULL OUTER JOIN with IS NULL filter
(B) CROSS JOIN
(C) SELF JOIN
(D) NATURAL JOIN

✓ **Answer: (A) FULL OUTER JOIN with IS NULL filter**

Q98. In a CROSS JOIN of 4 rows and 5 rows, how many result rows?

(A) 20
(B) 9
(C) 1
(D) 45

✓ **Answer: (A) 20**

Q99. Which clause filters rows AFTER a JOIN?

(A) WHERE
(B) ON
(C) HAVING
(D) FILTER

✓ **Answer: (A) WHERE**

Q100. Which condition in ON clause filters rows similar to WHERE?

(A) Both perform filtering, but ON applies during join
(B) ON applies after join
(C) WHERE applies during join
(D) They are identical

✓ **Answer: (A) Both perform filtering, but ON applies during join**

Q101. What is an equijoin?

(A) A join using equality operator
(B) A join with no condition
(C) A join using LIKE
(D) A join using BETWEEN

✓ **Answer: (A) A join using equality operator**

Q102. What is a non-equijoin?

(A) A join using operators other than =
(B) A join with no ON clause
(C) A CROSS JOIN
(D) A FULL OUTER JOIN

✓ **Answer: (A) A join using operators other than =**

Q103. Which type of JOIN preserves all records from both tables?

(A) FULL OUTER JOIN
(B) INNER JOIN
(C) LEFT JOIN
(D) RIGHT JOIN

✓ **Answer: (A) FULL OUTER JOIN**

Q104. Can INNER JOIN produce NULL values?

(A) No
(B) Yes
(C) Only for VARCHAR columns
(D) Only for numeric columns

✓ **Answer: (A) No**

Q105. What is the purpose of table aliases in JOINs?

(A) Shorten table names for readability
(B) Rename tables permanently
(C) Create new tables
(D) Add constraints

✓ **Answer: (A) Shorten table names for readability**

Q106. Which join can be used to find records that exist in one table but not another?

(A) LEFT JOIN with IS NULL
(B) INNER JOIN
(C) CROSS JOIN
(D) FULL JOIN

✓ **Answer: (A) LEFT JOIN with IS NULL**

Q107. In a LEFT JOIN, which table's all rows are preserved?

(A) Left table
(B) Right table
(C) Both tables
(D) Neither table

✓ **Answer: (A) Left table**

Q108. Which join is most memory-intensive?

(A) CROSS JOIN
(B) INNER JOIN
(C) LEFT JOIN
(D) RIGHT JOIN

✓ **Answer: (A) CROSS JOIN**

Q109. What does ON t1.id = t2.id define?

(A) The join condition
(B) The filter condition
(C) The group condition
(D) The sort condition

✓ **Answer: (A) The join condition**

Q110. Which join can replace a RIGHT JOIN by swapping table order?

(A) LEFT JOIN
(B) INNER JOIN
(C) CROSS JOIN
(D) FULL JOIN

✓ **Answer: (A) LEFT JOIN**

Q111. What is the output of LEFT JOIN when right table has no match?

(A) NULLs for right table columns
(B) Error
(C) Empty row
(D) Skipped row

✓ **Answer: (A) NULLs for right table columns**

Q112. Which keyword can be used instead of INNER JOIN for theta joins?

(A) WHERE with table comma list
(B) OUTER JOIN
(C) USING
(D) NATURAL

✓ **Answer: (A) WHERE with table comma list**

Q113. A FULL OUTER JOIN is combination of which two joins?

(A) LEFT JOIN + RIGHT JOIN
(B) INNER JOIN + CROSS JOIN
(C) LEFT JOIN + CROSS JOIN
(D) RIGHT JOIN + CROSS JOIN

✓ **Answer: (A) LEFT JOIN + RIGHT JOIN**

Q114. Which databases support FULL OUTER JOIN natively?

(A) PostgreSQL, SQL Server, Oracle
(B) MySQL only
(C) SQLite only
(D) None

✓ **Answer: (A) PostgreSQL, SQL Server, Oracle**

Q115. How to simulate FULL OUTER JOIN in MySQL?

(A) UNION of LEFT JOIN and RIGHT JOIN
(B) CROSS JOIN
(C) INNER JOIN with OR
(D) Using subquery

✓ **Answer: (A) UNION of LEFT JOIN and RIGHT JOIN**

Q116. Which join condition uses multiple columns?

(A) Compound join condition using AND
(B) Single column join
(C) CROSS JOIN
(D) NATURAL JOIN

✓ **Answer: (A) Compound join condition using AND**

Q117. What is an implicit join?

(A) Join using WHERE clause without JOIN keyword
(B) NATURAL JOIN
(C) CROSS JOIN
(D) SELF JOIN

✓ **Answer: (A) Join using WHERE clause without JOIN keyword**

Q118. Which join is preferred for performance: explicit or implicit?

(A) Explicit JOIN (same performance but clearer)
(B) Implicit is faster
(C) Explicit is always faster
(D) Depends on table size

✓ **Answer: (A) Explicit JOIN (same performance but clearer)**

Q119. Which join type is used in hierarchical queries?

(A) SELF JOIN
(B) INNER JOIN

(C) OUTER JOIN
(D) CROSS JOIN

✓ **Answer: (A) SELF JOIN**

Q120. What does a SELF JOIN require?

(A) Table aliases to distinguish instances
(B) No aliases
(C) Two separate tables
(D) A FOREIGN KEY

✓ **Answer: (A) Table aliases to distinguish instances**

Q121. Which JOIN result contains the most rows (assuming no filters)?

(A) CROSS JOIN
(B) INNER JOIN
(C) LEFT JOIN
(D) FULL JOIN

✓ **Answer: (A) CROSS JOIN**

Q122. INNER JOIN returns rows where condition evaluates to?

(A) TRUE
(B) FALSE
(C) NULL
(D) UNKNOWN

✓ **Answer: (A) TRUE**

Q123. Which clause filters groups after GROUP BY in a joined query?

(A) HAVING
(B) WHERE
(C) ON
(D) FILTER

✓ **Answer: (A) HAVING**

Q124. Can you use aggregate functions in a JOIN condition?

(A) No
(B) Yes
(C) Only with GROUP BY
(D) Only with HAVING

✓ **Answer: (A) No**

Q125. Which JOIN is typically fastest?

(A) INNER JOIN
(B) FULL OUTER JOIN
(C) CROSS JOIN
(D) SELF JOIN

✓ **Answer: (A) INNER JOIN**

Q126. Which statement about LEFT JOIN is true?

(A) Returns all rows from left, matching rows from right
(B) Returns all rows from right
(C) Returns only matching rows
(D) Returns Cartesian product

✓ **Answer: (A) Returns all rows from left, matching rows from right**

Q127. Which keyword can be used to make a RIGHT JOIN into LEFT JOIN?

(A) Swap table order
(B) REVERSE
(C) FLIP
(D) INVERT

✓ **Answer: (A) Swap table order**

Q128. NATURAL JOIN automatically joins on?

(A) Columns with same name and type
(B) Primary keys only
(C) Foreign keys only
(D) Any columns

✓ **Answer: (A) Columns with same name and type**

Q129. What is the danger of NATURAL JOIN?

(A) Unintended column matching
(B) Missing rows
(C) Duplicate columns
(D) Slow performance

✓ **Answer: (A) Unintended column matching**

Q130. Which join type creates all combinations of rows?

(A) CROSS JOIN
(B) INNER JOIN
(C) LEFT JOIN
(D) SELF JOIN

✓ **Answer: (A) CROSS JOIN**

Q131. Can you use subqueries as join targets?

(A) Yes
(B) No
(C) Only in FROM clause
(D) Only in WHERE clause

✓ **Answer: (A) Yes**

Q132. Which keyword is used to join a subquery?

(A) AS (alias)
(B) ON
(C) USING
(D) JOIN only

✓ **Answer: (A) AS (alias)**

Q133. What is a lateral join?

(A) A join that references columns of preceding tables
(B) A CROSS JOIN
(C) A SELF JOIN
(D) A NATURAL JOIN

✓ **Answer: (A) A join that references columns of preceding tables**

Q134. Which databases support LATERAL JOIN?

(A) PostgreSQL, MySQL 8.0+
(B) Only Oracle
(C) Only SQL Server
(D) All databases

✓ **Answer: (A) PostgreSQL, MySQL 8.0+**

Q135. Which SQL clause can perform the same as INNER JOIN?

(A) WHERE with = condition
(B) HAVING
(C) GROUP BY
(D) ORDER BY

✓ **Answer: (A) WHERE with = condition**

Q136. What is the result of joining a table on a NULL column?

(A) NULLs do not match; rows excluded from INNER JOIN
(B) NULLs always match
(C) Error
(D) Duplicate rows

✓ **Answer: (A) NULLs do not match; rows excluded from INNER JOIN**

Q137. Which join is best for finding orphaned records?

(A) LEFT JOIN with IS NULL check
(B) INNER JOIN
(C) CROSS JOIN
(D) FULL JOIN

✓ **Answer: (A) LEFT JOIN with IS NULL check**

Q138. In SQL which join is most commonly used in real-world queries?

(A) INNER JOIN
(B) CROSS JOIN
(C) FULL OUTER JOIN
(D) SELF JOIN

✓ **Answer: (A) INNER JOIN**

Q139. Which keyword makes a join condition case-insensitive (MySQL)?

(A) COLLATE
(B) BINARY
(C) CHARSET
(D) ILIKE

✓ **Answer: (A) COLLATE**

Q140. Which type of join is used in star schema queries?

(A) INNER JOIN between fact and dimension tables
(B) CROSS JOIN
(C) SELF JOIN
(D) NATURAL JOIN

✓ **Answer: (A) INNER JOIN between fact and dimension tables**

Q141. What is the ON clause equivalent of USING (col)?

(A) ON t1.col = t2.col
(B) ON col = col
(C) USING = ON
(D) They are not equivalent

✓ **Answer: (A) ON t1.col = t2.col**

Q142. Can OUTER JOINs be used with aggregate functions?

(A) Yes
(B) No
(C) Only with HAVING
(D) Only with WHERE

✓ **Answer: (A) Yes**

Q143. Which join eliminates duplicate column display?

(A) NATURAL JOIN / USING
(B) INNER JOIN
(C) OUTER JOIN
(D) CROSS JOIN

✓ **Answer: (A) NATURAL JOIN / USING**

Q144. Which is true about FULL OUTER JOIN?

(A) Includes unmatched rows from both tables
(B) Only matched rows
(C) Only left unmatched
(D) Only right unmatched

✓ **Answer: (A) Includes unmatched rows from both tables**

Q145. Which join type returns the intersection of two sets?

(A) INNER JOIN
(B) FULL OUTER JOIN
(C) CROSS JOIN
(D) LEFT ANTI JOIN

✓ **Answer: (A) INNER JOIN**

Q146. What is the syntax for joining three tables A, B, C?

(A) FROM A JOIN B ON ... JOIN C ON ...
(B) FROM A,B,C WHERE
(C) FROM A INNER B INNER C
(D) FROM A+B+C

✓ **Answer: (A) FROM A JOIN B ON ... JOIN C ON ...**

Q147. Which keyword is optional in INNER JOIN syntax?

(A) INNER (can write just JOIN)
(B) ON
(C) FROM
(D) SELECT

✓ **Answer: (A) INNER (can write just JOIN)**

Q148. Does ORDER BY work after a JOIN?

(A) Yes
(B) No
(C) Only with GROUP BY
(D) Only with HAVING

✓ **Answer: (A) Yes**

Q149. Can you use WHERE after a JOIN?

(A) Yes
(B) No
(C) Only for LEFT JOINs
(D) Only for INNER JOINs

✓ **Answer: (A) Yes**

Q150. Which is NOT a valid SQL join type?

(A) EXCLUSIVE JOIN
(B) INNER JOIN
(C) LEFT JOIN
(D) CROSS JOIN

✓ **Answer: (A) EXCLUSIVE JOIN**

Q151. Which join returns rows only when there is no match?

(A) ANTI JOIN (via LEFT JOIN + IS NULL)
(B) INNER JOIN
(C) FULL JOIN
(D) CROSS JOIN

✓ **Answer: (A) ANTI JOIN (via LEFT JOIN + IS NULL)**

Q152. What is a hash join?

(A) A join execution algorithm using hash tables
(B) A join using # symbol
(C) A join on hash columns
(D) A type of OUTER JOIN

✓ **Answer: (A) A join execution algorithm using hash tables**

Q153. What is a nested loop join?

(A) For each row in outer table, scan inner table
(B) A CROSS JOIN
(C) A SELF JOIN
(D) A NATURAL JOIN

✓ **Answer: (A) For each row in outer table, scan inner table**

Q154. What is a merge join?

(A) Join on sorted inputs by merging
(B) A FULL OUTER JOIN
(C) A CROSS JOIN
(D) A NATURAL JOIN

✓ **Answer: (A) Join on sorted inputs by merging**

Q155. Which join algorithm is best for large unsorted tables?

(A) Hash join
(B) Nested loop join
(C) Merge join
(D) Block nested loop

✓ **Answer: (A) Hash join**

Q156. Which algorithm is best when one table is small?

(A) Nested loop join
(B) Hash join
(C) Merge join
(D) Sort merge join

✓ **Answer: (A) Nested loop join**

Q157. What does query optimizer do for joins?

(A) Chooses best join algorithm and order
(B) Executes joins manually
(C) Sorts tables
(D) Removes duplicates

✓ **Answer: (A) Chooses best join algorithm and order**

Q158. Which hint forces a specific join type in MySQL?

(A) STRAIGHT_JOIN
(B) FORCE JOIN

(C) USE JOIN

(D) JOIN HINT

✓ **Answer: (A) STRAIGHT_JOIN**

Q159. What is an index join?

(A) Uses index to speed up join condition lookup

(B) A join on indexed columns only

(C) A NATURAL JOIN

(D) A SELF JOIN

✓ **Answer: (A) Uses index to speed up join condition lookup**

Q160. Which database uses MERGE JOIN as its default?

(A) SQL Server (for sorted data)

(B) MySQL

(C) SQLite

(D) MongoDB

✓ **Answer: (A) SQL Server (for sorted data)**

Section: Subqueries, CTEs & Set Operations (Q 161–240)

Q161. A subquery is also called?

(A) Inner query / nested query
(B) Outer query
(C) Correlated query
(D) Sub-select

✓ **Answer: (A) Inner query / nested query**

Q162. Which clause can contain a subquery?

(A) WHERE, FROM, SELECT, HAVING (all)
(B) Only WHERE
(C) Only FROM
(D) Only SELECT

✓ **Answer: (A) WHERE, FROM, SELECT, HAVING (all)**

Q163. A correlated subquery references?

(A) The outer query
(B) Another database
(C) A stored procedure
(D) A view

✓ **Answer: (A) The outer query**

Q164. Which operator is used to compare a value to any value in a subquery?

(A) ANY / SOME
(B) ALL
(C) IN
(D) EXISTS

✓ **Answer: (A) ANY / SOME**

Q165. Which operator compares a value to all values in a subquery?

(A) ALL
(B) ANY
(C) IN
(D) EXISTS

✓ **Answer: (A) ALL**

Q166. Which operator checks if a subquery returns any rows?

(A) EXISTS
(B) IN
(C) ANY
(D) ALL

✓ **Answer: (A) EXISTS**

Q167. Which is generally faster: EXISTS or IN for large subqueries?

(A) EXISTS
(B) IN
(C) They are identical
(D) Depends only on indexes

✓ **Answer: (A) EXISTS**

Q168. What is a scalar subquery?

(A) Returns exactly one row and one column
(B) Returns many rows
(C) Returns many columns
(D) Returns no rows

✓ **Answer: (A) Returns exactly one row and one column**

Q169. What is an inline view?

(A) A subquery in the FROM clause
(B) A subquery in WHERE
(C) A stored view
(D) A materialized view

✓ **Answer: (A) A subquery in the FROM clause**

Q170. What does UNION do?

(A) Combines result sets removing duplicates
(B) Combines result sets keeping duplicates
(C) Joins two tables
(D) Filters duplicates in one table

✓ **Answer: (A) Combines result sets removing duplicates**

Q171. What does UNION ALL do?

(A) Combines result sets keeping all duplicates
(B) Removes duplicates
(C) Joins tables
(D) Filters rows

✓ **Answer: (A) Combines result sets keeping all duplicates**

Q172. What does INTERSECT do?

(A) Returns rows common to both queries
(B) Returns all rows from both
(C) Returns unique rows from first query
(D) Subtracts second from first

✓ **Answer: (A) Returns rows common to both queries**

Q173. What does EXCEPT / MINUS do?

(A) Returns rows in first query not in second
(B) Returns common rows
(C) Returns all rows
(D) Unions two queries

✓ **Answer: (A) Returns rows in first query not in second**

Q174. How many columns must each query in UNION have?

(A) Same number
(B) Different is OK
(C) At least one
(D) Exactly one

✓ **Answer: (A) Same number**

Q175. Must data types match in UNION queries?

(A) Yes (compatible types)
(B) No
(C) Only for numeric columns
(D) Only for string columns

✓ **Answer: (A) Yes (compatible types)**

Q176. Which set operator does Oracle use instead of EXCEPT?

(A) MINUS
(B) EXCEPT
(C) SUBTRACT
(D) REMOVE

✓ **Answer: (A) MINUS**

Q177. Can a subquery return multiple columns in WHERE clause?

(A) Yes with row constructors
(B) No
(C) Only in FROM
(D) Only in SELECT

✓ **Answer: (A) Yes with row constructors**

Q178. What is a CTE?

(A) Common Table Expression
(B) Correlated Table Expression
(C) Combined Table Entry
(D) Calculated Table Expression

✓ **Answer: (A) Common Table Expression**

Q179. Which keyword introduces a CTE?

(A) WITH
(B) CTE
(C) DEFINE
(D) DECLARE

✓ **Answer: (A) WITH**

Q180. Can a CTE be recursive?

(A) Yes
(B) No
(C) Only in Oracle
(D) Only in SQL Server

✓ **Answer: (A) Yes**

Q181. What is the purpose of a recursive CTE?

(A) Query hierarchical data
(B) Perform aggregation
(C) Replace joins
(D) Create indexes

✓ **Answer: (A) Query hierarchical data**

Q182. Which keyword is required in a recursive CTE?

(A) UNION ALL
(B) RECURSIVE
(C) REPEAT
(D) LOOP

✓ **Answer: (A) UNION ALL**

Q183. What is the anchor member in a recursive CTE?

(A) The base non-recursive query
(B) The recursive query
(C) The final select
(D) The termination condition

✓ **Answer: (A) The base non-recursive query**

Q184. What prevents infinite recursion in a CTE?

(A) A termination condition / MAXRECURSION
(B) LIMIT
(C) ROWNUM
(D) FETCH

✓ **Answer: (A) A termination condition / MAXRECURSION**

Q185. Which subquery clause returns TRUE if subquery has rows?

(A) EXISTS
(B) IN
(C) ANY
(D) SOME

✓ **Answer: (A) EXISTS**

Q186. Which is more efficient for checking existence: EXISTS or COUNT(*)?

(A) EXISTS
(B) COUNT(*)
(C) IN
(D) ANY

✓ **Answer: (A) EXISTS**

Q187. Can a subquery appear in a SELECT list?

(A) Yes (scalar subquery)
(B) No
(C) Only in WHERE
(D) Only in FROM

✓ **Answer: (A) Yes (scalar subquery)**

Q188. What is a derived table?

(A) Subquery in FROM clause with alias
(B) A view
(C) A CTE
(D) A temporary table

✓ **Answer: (A) Subquery in FROM clause with alias**

Q189. Can you use ORDER BY inside a subquery (non-standard)?

(A) Generally no (without LIMIT/TOP)
(B) Yes always
(C) Yes in all databases
(D) No never

✓ **Answer: (A) Generally no (without LIMIT/TOP)**

Q190. Which clause can use a subquery for row filtering?

(A) WHERE
(B) SELECT
(C) FROM
(D) All of the above

✓ **Answer: (D) All of the above**

Q191. What does NOT IN do?

(A) Returns rows where value not in subquery list
(B) Returns NULL rows
(C) Returns all rows
(D) Performs a join

✓ **Answer: (A) Returns rows where value not in subquery list**

Q192. What is the risk of NOT IN with NULL values in subquery?

(A) Returns no rows (NULL comparison)
(B) Returns all rows
(C) Error
(D) Duplicate rows

✓ **Answer: (A) Returns no rows (NULL comparison)**

Q193. Which is safer than NOT IN when NULLs may exist?

(A) NOT EXISTS
(B) IN
(C) ANY
(D) ALL

✓ **Answer: (A) NOT EXISTS**

Q194. Can multiple CTEs be chained?

(A) Yes, separated by commas
(B) No
(C) Only in Oracle
(D) Only in PostgreSQL

✓ **Answer: (A) Yes, separated by commas**

Q195. Which set operation is not supported in all databases?

(A) INTERSECT (not in MySQL < 8.0)
(B) UNION
(C) UNION ALL
(D) SELECT

✓ **Answer: (A) INTERSECT (not in MySQL < 8.0)**

Q196. What does UNION do with ORDER BY?

(A) Applies ORDER BY to final result
(B) Orders each sub-query
(C) Not allowed
(D) Requires subquery

✓ **Answer: (A) Applies ORDER BY to final result**

Q197. Which clause makes a CTE recursive in SQL Server?

(A) WITH RECURSIVE / MAXRECURSION option
(B) LOOP
(C) ITERATE
(D) REPEAT

✓ **Answer: (A) WITH RECURSIVE / MAXRECURSION option**

Q198. What is a materialized CTE?

(A) CTE result stored temporarily in memory
(B) A view saved to disk
(C) An indexed CTE
(D) A permanent table

✓ **Answer: (A) CTE result stored temporarily in memory**

Q199. Can CTEs replace views?

(A) For single-query use, yes
(B) No, views are permanent

(C) Only in PostgreSQL
(D) Only in Oracle

✓ **Answer: (A) For single-query use, yes**

Q200. Which subquery type executes once for the whole outer query?

(A) Non-correlated subquery
(B) Correlated subquery
(C) Recursive CTE
(D) Scalar subquery

✓ **Answer: (A) Non-correlated subquery**

Q201. Which subquery type executes once per row of outer query?

(A) Correlated subquery
(B) Non-correlated subquery
(C) CTE
(D) Derived table

✓ **Answer: (A) Correlated subquery**

Q202. What is the performance concern with correlated subqueries?

(A) Executes per outer row, potentially slow
(B) Always fast
(C) Requires index
(D) Uses extra memory

✓ **Answer: (A) Executes per outer row, potentially slow**

Q203. Can a subquery be used in an UPDATE statement?

(A) Yes
(B) No
(C) Only in WHERE
(D) Only in SET

✓ **Answer: (A) Yes**

Q204. Can a subquery be used in a DELETE statement?

(A) Yes
(B) No
(C) Only in FROM
(D) Only in HAVING

✓ **Answer: (A) Yes**

Q205. Can a subquery be used in an INSERT statement?

(A) Yes (INSERT INTO ... SELECT)
(B) No
(C) Only with VALUES
(D) Only with CTE

✓ **Answer: (A) Yes (INSERT INTO ... SELECT)**

Q206. What is the result of INTERSECT if one set is empty?

(A) Empty result set
(B) All rows from other set
(C) Error
(D) NULL

✓ **Answer: (A) Empty result set**

Q207. What is the result of UNION of two identical result sets?

(A) One copy (duplicates removed)
(B) Two copies
(C) Error
(D) NULL

✓ Answer: (A) One copy (duplicates removed)

Q208. What is the result of UNION ALL of two identical result sets?

(A) Two copies (all rows kept)
(B) One copy
(C) Error
(D) NULL

✓ Answer: (A) Two copies (all rows kept)

Q209. Can you nest subqueries?

(A) Yes, multiple levels
(B) No
(C) Only two levels
(D) Only in FROM

✓ Answer: (A) Yes, multiple levels

Q210. Which keyword limits CTE recursion depth in SQL Server?

(A) MAXRECURSION
(B) LIMIT
(C) DEPTH
(D) MAXDEPTH

✓ Answer: (A) MAXRECURSION

Q211. What is the default MAXRECURSION in SQL Server?

(A) 100
(B) 1000
(C) 10
(D) Unlimited

✓ Answer: (A) 100

Q212. Which SQL clause is equivalent to EXCEPT?

(A) MINUS (Oracle)
(B) UNION
(C) INTERSECT
(D) NOT IN

✓ Answer: (A) MINUS (Oracle)

Q213. What does a subquery in FROM clause require?

(A) An alias
(B) A HAVING clause
(C) ORDER BY
(D) GROUP BY

✓ Answer: (A) An alias

Q214. Can UNION include tables from different databases?

(A) Yes (with full path)
(B) No
(C) Only same schema
(D) Only same server

✓ Answer: (A) Yes (with full path)

Q215. Which is NOT a set operation in SQL?

(A) MERGE
(B) UNION
(C) INTERSECT
(D) EXCEPT

✓ **Answer: (A) MERGE**

Q216. What is a lateral subquery?

(A) Can reference columns from preceding FROM items
(B) A CROSS JOIN
(C) A SELF JOIN
(D) A correlated WHERE subquery

✓ **Answer: (A) Can reference columns from preceding FROM items**

Q217. Which syntax is used for lateral subqueries in PostgreSQL?

(A) LATERAL
(B) CROSS APPLY
(C) OUTER APPLY
(D) LOOP

✓ **Answer: (A) LATERAL**

Q218. Which SQL Server syntax is equivalent to LATERAL?

(A) CROSS APPLY / OUTER APPLY
(B) LATERAL
(C) SIDE JOIN
(D) APPLY

✓ **Answer: (A) CROSS APPLY / OUTER APPLY**

Q219. What does OUTER APPLY return for non-matching rows?

(A) NULLs (like LEFT JOIN)
(B) No rows (like INNER JOIN)
(C) Error
(D) Duplicate rows

✓ **Answer: (A) NULLs (like LEFT JOIN)**

Q220. What does CROSS APPLY return for non-matching rows?

(A) No rows (like INNER JOIN)
(B) NULLs
(C) Error
(D) All rows

✓ **Answer: (A) No rows (like INNER JOIN)**

Q221. Can a CTE reference itself?

(A) Yes (recursive CTE)
(B) No
(C) Only once
(D) Only in PostgreSQL

✓ **Answer: (A) Yes (recursive CTE)**

Q222. Which keyword separates multiple CTEs?

(A) Comma (,)
(B) Semicolon (;)
(C) AND
(D) THEN

✓ **Answer: (A) Comma (,)**

Q223. What is the scope of a CTE?

(A) Single SQL statement
(B) Entire session
(C) Entire database
(D) Entire transaction

✓ **Answer: (A) Single SQL statement**

Q224. Which is faster: CTE or subquery?

(A) Generally same; CTE improves readability
(B) CTE always faster
(C) Subquery always faster
(D) Depends on rows only

✓ **Answer: (A) Generally same; CTE improves readability**

Q225. Can a CTE be indexed?

(A) No (use temp table for indexing)
(B) Yes
(C) Only in SQL Server
(D) Only in Oracle

✓ **Answer: (A) No (use temp table for indexing)**

Q226. Which statement can follow a CTE?

(A) SELECT, INSERT, UPDATE, DELETE
(B) Only SELECT
(C) Only INSERT
(D) Only DELETE

✓ **Answer: (A) SELECT, INSERT, UPDATE, DELETE**

Q227. What happens if subquery returns more than one row in a scalar context?

(A) Error
(B) Returns first row
(C) Returns NULL
(D) Returns last row

✓ **Answer: (A) Error**

Q228. Can EXISTS subquery use SELECT *?

(A) Yes (common practice)
(B) No
(C) Only SELECT 1
(D) Only SELECT column

✓ **Answer: (A) Yes (common practice)**

Q229. What is the advantage of EXISTS over IN?

(A) Stops at first match, more efficient
(B) Always returns more rows
(C) Simpler syntax
(D) Uses less memory always

✓ **Answer: (A) Stops at first match, more efficient**

Q230. Which SQL construct is best for paginated queries?

(A) OFFSET FETCH / LIMIT OFFSET
(B) Subquery
(C) CTE
(D) UNION

✓ **Answer: (A) OFFSET FETCH / LIMIT OFFSET**

Q231. Can UNION be used in a subquery?

(A) Yes
(B) No
(C) Only in FROM
(D) Only in WHERE

✓ **Answer: (A) Yes**

Q232. What does INTERSECT ALL do (PostgreSQL)?

(A) Returns common rows including duplicates
(B) Removes duplicates
(C) Returns all rows
(D) Same as INTERSECT

✓ **Answer: (A) Returns common rows including duplicates**

Q233. Which database introduced WITH clause for CTEs?

(A) Oracle 9i (first widely adopted)
(B) MySQL
(C) SQLite
(D) Access

✓ **Answer: (A) Oracle 9i (first widely adopted)**

Q234. Can a subquery reference outer query columns?

(A) Yes (correlated subquery)
(B) No
(C) Only in SELECT
(D) Only in FROM

✓ **Answer: (A) Yes (correlated subquery)**

Q235. Which clause in a subquery filters the outer query rows?

(A) WHERE with outer column reference
(B) HAVING
(C) GROUP BY
(D) ORDER BY

✓ **Answer: (A) WHERE with outer column reference**

Q236. What is a semi-join?

(A) Returns rows from one table that have a match in another without including other table columns
(B) A FULL JOIN
(C) A CROSS JOIN
(D) A SELF JOIN

✓ **Answer: (A) Returns rows from one table that have a match in another without including other table columns**

Q237. Which SQL construct implements semi-join?

(A) EXISTS / IN
(B) LEFT JOIN
(C) CROSS JOIN
(D) NATURAL JOIN

✓ **Answer: (A) EXISTS / IN**

Q238. What is an anti semi-join?

(A) Returns rows with NO match in the other table
(B) Returns rows with match

(C) A FULL JOIN

(D) A CROSS JOIN

✓ **Answer: (A) Returns rows with NO match in the other table**

Q239. Which construct implements anti semi-join?

(A) NOT EXISTS / NOT IN

(B) INNER JOIN

(C) LEFT JOIN

(D) UNION

✓ **Answer: (A) NOT EXISTS / NOT IN**

Q240. What is the difference between UNION and JOIN?

(A) UNION combines rows vertically; JOIN combines columns horizontally

(B) They are the same

(C) JOIN removes duplicates

(D) UNION joins on conditions

✓ **Answer: (A) UNION combines rows vertically; JOIN combines columns horizontally**

Section: Indexes & Query Performance (Q 241–310)

Q241. What is an index in SQL?

(A) A data structure to speed up queries
(B) A primary key
(C) A constraint
(D) A stored procedure

✓ **Answer: (A) A data structure to speed up queries**

Q242. Which index type is the default in most databases?

(A) B-Tree index
(B) Hash index
(C) Bitmap index
(D) Full-text index

✓ **Answer: (A) B-Tree index**

Q243. Which index is best for equality searches?

(A) Hash index
(B) B-Tree index
(C) Bitmap index
(D) Full-text index

✓ **Answer: (A) Hash index**

Q244. Which index is best for range queries?

(A) B-Tree index
(B) Hash index
(C) Bitmap index
(D) Spatial index

✓ **Answer: (A) B-Tree index**

Q245. Which index is used for low-cardinality columns?

(A) Bitmap index
(B) B-Tree index
(C) Hash index
(D) Full-text index

✓ **Answer: (A) Bitmap index**

Q246. Which index type supports text search?

(A) Full-text index
(B) B-Tree index
(C) Hash index
(D) Bitmap index

✓ **Answer: (A) Full-text index**

Q247. What is a clustered index?

(A) Determines physical order of data in table
(B) Separate structure from table data
(C) Always on primary key
(D) A composite index

✓ **Answer: (A) Determines physical order of data in table**

Q248. How many clustered indexes can a table have?

(A) One
(B) Many
(C) Two
(D) Unlimited

✓ **Answer: (A) One**

Q249. What is a non-clustered index?

(A) Separate structure that points to data rows
(B) Determines physical order
(C) Always unique
(D) A bitmap index

✓ **Answer: (A) Separate structure that points to data rows**

Q250. How many non-clustered indexes can a table have?

(A) Many (hundreds)
(B) One
(C) Two
(D) Only one per table

✓ **Answer: (A) Many (hundreds)**

Q251. Which SQL command creates an index?

(A) CREATE INDEX
(B) MAKE INDEX
(C) BUILD INDEX
(D) ADD INDEX

✓ **Answer: (A) CREATE INDEX**

Q252. Which command removes an index?

(A) DROP INDEX
(B) DELETE INDEX
(C) REMOVE INDEX
(D) ERASE INDEX

✓ **Answer: (A) DROP INDEX**

Q253. What is a covering index?

(A) An index that includes all columns needed by a query
(B) An index on all columns
(C) A clustered index
(D) A unique index

✓ **Answer: (A) An index that includes all columns needed by a query**

Q254. What is index selectivity?

(A) Ratio of distinct values to total rows
(B) Number of indexes
(C) Index size
(D) Index type

✓ **Answer: (A) Ratio of distinct values to total rows**

Q255. High selectivity means?

(A) More distinct values; index more useful
(B) Fewer distinct values
(C) More duplicate values
(D) Index is less useful

✓ **Answer: (A) More distinct values; index more useful**

Q256. Which operation on a column prevents index usage?

(A) Applying a function to indexed column
(B) Using = operator
(C) Using BETWEEN
(D) Using IN

✓ **Answer: (A) Applying a function to indexed column**

Q257. What is a composite index?

(A) An index on multiple columns
(B) An index with multiple types
(C) A clustered index
(D) A unique index

✓ **Answer: (A) An index on multiple columns**

Q258. What is the leftmost prefix rule for composite indexes?

(A) Query must use leading columns of composite index
(B) Query can use any column
(C) Only last column matters
(D) Order doesn't matter

✓ **Answer: (A) Query must use leading columns of composite index**

Q259. What is an index scan?

(A) Scanning all index entries
(B) Using index to find specific rows
(C) A full table scan
(D) A hash lookup

✓ **Answer: (A) Scanning all index entries**

Q260. What is an index seek?

(A) Navigating directly to specific rows using index
(B) Scanning all index entries
(C) A full table scan
(D) A sort operation

✓ **Answer: (A) Navigating directly to specific rows using index**

Q261. What is a table scan?

(A) Reading all rows without using index
(B) Using index
(C) A partial read
(D) A clustered scan

✓ **Answer: (A) Reading all rows without using index**

Q262. When does the optimizer prefer a full table scan?

(A) When a large percentage of rows are returned
(B) When a few rows are returned
(C) When there is no WHERE clause
(D) When table has no indexes

✓ **Answer: (A) When a large percentage of rows are returned**

Q263. What is query optimization?

(A) Finding the most efficient execution plan
(B) Writing queries faster
(C) Adding indexes automatically
(D) Compressing data

✓ **Answer: (A) Finding the most efficient execution plan**

Q264. What does EXPLAIN / EXPLAIN PLAN do?

(A) Shows query execution plan
(B) Executes the query
(C) Creates an index
(D) Analyzes the query syntax

✓ **Answer: (A) Shows query execution plan**

Q265. What does cardinality mean in indexes?

(A) Number of unique values in a column
(B) Number of rows
(C) Number of indexes
(D) Table size

✓ **Answer: (A) Number of unique values in a column**

Q266. What is index fragmentation?

(A) Index pages out of logical order due to DML operations
(B) An error
(C) A missing index
(D) A dropped index

✓ **Answer: (A) Index pages out of logical order due to DML operations**

Q267. How to fix index fragmentation?

(A) REBUILD or REORGANIZE index
(B) DROP index
(C) CREATE new index
(D) UPDATE statistics

✓ **Answer: (A) REBUILD or REORGANIZE index**

Q268. What are statistics in SQL?

(A) Metadata about data distribution used by optimizer
(B) Index metadata
(C) Table metadata
(D) Query logs

✓ **Answer: (A) Metadata about data distribution used by optimizer**

Q269. Which command updates statistics?

(A) UPDATE STATISTICS / ANALYZE
(B) REBUILD INDEX
(C) DROP INDEX
(D) VACUUM

✓ **Answer: (A) UPDATE STATISTICS / ANALYZE**

Q270. What is a partial index?

(A) An index on a subset of rows
(B) An incomplete index
(C) A half-built index
(D) A fragmented index

✓ **Answer: (A) An index on a subset of rows**

Q271. What is an expression index?

(A) Index on a computed expression
(B) A partial index
(C) A composite index
(D) A clustered index

✓ **Answer: (A) Index on a computed expression**

Q272. Which index helps with LIKE 'abc%' searches?

(A) B-Tree index (prefix match)
(B) Hash index
(C) Bitmap index
(D) Full-text index

✓ **Answer: (A) B-Tree index (prefix match)**

Q273. Which index CANNOT help with LIKE '%abc%'?

(A) B-Tree index
(B) Full-text index
(C) Expression index
(D) None

✓ **Answer: (A) B-Tree index**

Q274. What is a filtered index (SQL Server)?

(A) Non-clustered index with WHERE filter
(B) A clustered index
(C) A composite index
(D) A hash index

✓ **Answer: (A) Non-clustered index with WHERE filter**

Q275. What is an invisible index (MySQL)?

(A) Index ignored by optimizer unless hinted
(B) A dropped index
(C) A partial index
(D) A composite index

✓ **Answer: (A) Index ignored by optimizer unless hinted**

Q276. What is a functional index?

(A) Index on a function applied to a column
(B) An expression index
(C) A partial index
(D) A clustered index

✓ **Answer: (A) Index on a function applied to a column**

Q277. Which type of index is used for geospatial data?

(A) Spatial index (R-Tree)
(B) B-Tree
(C) Hash
(D) Bitmap

✓ **Answer: (A) Spatial index (R-Tree)**

Q278. What is index-only scan?

(A) Query answered entirely from index without accessing table
(B) A table scan
(C) A hash join
(D) A full index scan

✓ **Answer: (A) Query answered entirely from index without accessing table**

Q279. What is a hot spot in indexing?

(A) Frequently accessed index page causing contention
(B) A highly selective index

(C) A clustered index
(D) An expression index

✓ **Answer: (A) Frequently accessed index page causing contention**

Q280. What is the fill factor of an index?

(A) Percentage of index page filled during creation
(B) The number of columns
(C) The index size
(D) The selectivity

✓ **Answer: (A) Percentage of index page filled during creation**

Q281. What fill factor reduces page splits?

(A) Lower fill factor (leaves space for inserts)
(B) Higher fill factor
(C) 100% fill factor
(D) 0% fill factor

✓ **Answer: (A) Lower fill factor (leaves space for inserts)**

Q282. What is a primary index?

(A) Index on the primary key
(B) Any clustered index
(C) Any unique index
(D) First created index

✓ **Answer: (A) Index on the primary key**

Q283. What is a secondary index?

(A) Any index other than primary/clustered
(B) The second created index
(C) A non-unique index
(D) A hash index

✓ **Answer: (A) Any index other than primary/clustered**

Q284. What is a dense index?

(A) An entry for every row
(B) An entry for every block
(C) A sparse index
(D) A partial index

✓ **Answer: (A) An entry for every row**

Q285. What is a sparse index?

(A) An entry for every block/page
(B) An entry for every row
(C) A dense index
(D) A partial index

✓ **Answer: (A) An entry for every block/page**

Q286. Which index type stores ROWID pointers?

(A) Non-clustered index
(B) Clustered index
(C) Hash index
(D) Bitmap index

✓ **Answer: (A) Non-clustered index**

Q287. What is an online index rebuild?

(A) Rebuilding index without locking table
(B) Rebuilding with table lock
(C) A partial rebuild
(D) An index update

✓ **Answer: (A) Rebuilding index without locking table**

Q288. Which statement is true about indexes and DML?

(A) Indexes slow down INSERT/UPDATE/DELETE
(B) Indexes always speed up DML
(C) Indexes have no effect on DML
(D) Indexes are updated manually

✓ **Answer: (A) Indexes slow down INSERT/UPDATE/DELETE**

Q289. What is the purpose of INCLUDE columns in an index?

(A) Add non-key columns to leaf level for covering index
(B) Add key columns
(C) Create composite key
(D) Reduce index size

✓ **Answer: (A) Add non-key columns to leaf level for covering index**

Q290. What is an adaptive index (some databases)?

(A) Index that adjusts based on query patterns
(B) A partial index
(C) A filtered index
(D) A functional index

✓ **Answer: (A) Index that adjusts based on query patterns**

Q291. Which command forces use of an index in MySQL?

(A) USE INDEX / FORCE INDEX
(B) FORCE INDEX HINT
(C) INDEX HINT
(D) USE INDEX ALL

✓ **Answer: (A) USE INDEX / FORCE INDEX**

Q292. What is the difference between REBUILD and REORGANIZE index?

(A) REBUILD recreates fully; REORGANIZE defragments in-place
(B) They are identical
(C) REORGANIZE is offline only
(D) REBUILD is online only

✓ **Answer: (A) REBUILD recreates fully; REORGANIZE defragments in-place**

Q293. Which is more expensive: REBUILD or REORGANIZE?

(A) REBUILD
(B) REORGANIZE
(C) Both equal
(D) Neither costs resources

✓ **Answer: (A) REBUILD**

Q294. What is a bitmap join index (Oracle)?

(A) Index on join columns using bitmaps
(B) A join without index
(C) A composite bitmap index
(D) A B-Tree join index

✓ **Answer: (A) Index on join columns using bitmaps**

Q295. Which index is best for columns with few distinct values?

(A) Bitmap index
(B) B-Tree index
(C) Hash index
(D) Full-text index

✓ **Answer: (A) Bitmap index**

Q296. What is index skip scan?

(A) Using composite index by skipping leading column
(B) A table scan
(C) A hash scan
(D) A full index scan

✓ **Answer: (A) Using composite index by skipping leading column**

Q297. What causes an index not to be used?

(A) Implicit type conversion, functions on indexed column, low selectivity
(B) Using = operator
(C) Using BETWEEN
(D) Having many rows

✓ **Answer: (A) Implicit type conversion, functions on indexed column, low selectivity**

Q298. Which tool shows missing index suggestions in SQL Server?

(A) sys.dm_db_missing_index_details
(B) EXPLAIN
(C) ANALYZE
(D) QUERY PLAN

✓ **Answer: (A) sys.dm_db_missing_index_details**

Q299. What is a reverse key index (Oracle)?

(A) Reverses bytes of key to distribute inserts
(B) A descending index
(C) A backward scan
(D) A composite index

✓ **Answer: (A) Reverses bytes of key to distribute inserts**

Q300. What is an IOT (Index-Organized Table) in Oracle?

(A) Table stored in index structure
(B) A clustered table
(C) A partitioned table
(D) A materialized view

✓ **Answer: (A) Table stored in index structure**

Q301. What is a heap table?

(A) Unordered table without clustered index
(B) A clustered table
(C) A partitioned table
(D) A temp table

✓ **Answer: (A) Unordered table without clustered index**

Q302. What is the purpose of query hints?

(A) Force optimizer to use specific plan
(B) Automatically optimize queries
(C) Create indexes
(D) Update statistics

✓ **Answer: (A) Force optimizer to use specific plan**

Q303. What does NOLOCK hint do (SQL Server)?

(A) Reads without shared locks (dirty read)
(B) Acquires exclusive lock
(C) Prevents reads
(D) Locks the table

✓ **Answer: (A) Reads without shared locks (dirty read)**

Q304. What is a dirty read?

(A) Reading uncommitted data
(B) Reading committed data
(C) A full table scan
(D) A deadlock

✓ **Answer: (A) Reading uncommitted data**

Q305. What is a deadlock?

(A) Two transactions blocking each other indefinitely
(B) A long-running query
(C) A missing index
(D) A table lock

✓ **Answer: (A) Two transactions blocking each other indefinitely**

Q306. Which isolation level prevents dirty reads?

(A) READ COMMITTED
(B) READ UNCOMMITTED
(C) SERIALIZABLE
(D) REPEATABLE READ

✓ **Answer: (A) READ COMMITTED**

Q307. What is a phantom read?

(A) New rows appear in same transaction re-read
(B) Reading dirty data
(C) A lost update
(D) A deadlock

✓ **Answer: (A) New rows appear in same transaction re-read**

Q308. Which isolation level prevents phantom reads?

(A) SERIALIZABLE
(B) READ COMMITTED
(C) REPEATABLE READ
(D) READ UNCOMMITTED

✓ **Answer: (A) SERIALIZABLE**

Q309. What is a covering query?

(A) All columns needed are in the index
(B) A query without WHERE clause
(C) A query with subquery
(D) A recursive query

✓ **Answer: (A) All columns needed are in the index**

Q310. What is query parameterization?

(A) Using parameters instead of literals for plan reuse
(B) Hard-coding values
(C) Using variables
(D) Using CTEs

✓ **Answer: (A) Using parameters instead of literals for plan reuse**

Section: Stored Procedures, Functions, Triggers & Cursors (Q 311–380)

Q311. What is a stored procedure?

(A) A precompiled set of SQL statements stored in the database
(B) A temporary table
(C) A view
(D) A trigger

✓ **Answer: (A) A precompiled set of SQL statements stored in the database**

Q312. Which keyword creates a stored procedure?

(A) CREATE PROCEDURE
(B) MAKE PROCEDURE
(C) BUILD PROCEDURE
(D) NEW PROCEDURE

✓ **Answer: (A) CREATE PROCEDURE**

Q313. Which keyword executes a stored procedure?

(A) EXEC / EXECUTE / CALL
(B) RUN
(C) START
(D) BEGIN

✓ **Answer: (A) EXEC / EXECUTE / CALL**

Q314. What is the advantage of stored procedures?

(A) Reduced network traffic, code reuse, security
(B) Only code reuse
(C) Only security
(D) Only speed

✓ **Answer: (A) Reduced network traffic, code reuse, security**

Q315. What is an IN parameter?

(A) Input parameter passed to procedure
(B) Output parameter
(C) Both input and output
(D) Return value

✓ **Answer: (A) Input parameter passed to procedure**

Q316. What is an OUT parameter?

(A) Returns value from procedure to caller
(B) Input only
(C) Both input and output
(D) Return value

✓ **Answer: (A) Returns value from procedure to caller**

Q317. What is an INOUT parameter?

(A) Both input and output
(B) Input only
(C) Output only
(D) Return value

✓ **Answer: (A) Both input and output**

Q318. What is a user-defined function (UDF)?

(A) A reusable function that returns a value
(B) A stored procedure
(C) A trigger
(D) A view

✓ **Answer: (A) A reusable function that returns a value**

Q319. What is the main difference between a function and a stored procedure?

(A) Function must return a value; procedure may not
(B) Procedure must return a value
(C) They are identical
(D) Functions cannot be called in SQL

✓ **Answer: (A) Function must return a value; procedure may not**

Q320. Can a function be used in a SELECT statement?

(A) Yes
(B) No
(C) Only scalar functions
(D) Only table functions

✓ **Answer: (A) Yes**

Q321. Can a stored procedure be used in a SELECT statement?

(A) No (generally)
(B) Yes
(C) Only in FROM
(D) Only in WHERE

✓ **Answer: (A) No (generally)**

Q322. What is a scalar function?

(A) Returns a single value
(B) Returns a table
(C) Returns multiple values
(D) Returns nothing

✓ **Answer: (A) Returns a single value**

Q323. What is a table-valued function?

(A) Returns a result set (table)
(B) Returns a scalar
(C) Returns nothing
(D) Returns a cursor

✓ **Answer: (A) Returns a result set (table)**

Q324. What is a trigger?

(A) Automatic action executed when DML event occurs
(B) A stored procedure
(C) A scheduled job
(D) A view

✓ **Answer: (A) Automatic action executed when DML event occurs**

Q325. Which events can fire a trigger?

(A) INSERT, UPDATE, DELETE
(B) Only INSERT
(C) Only DELETE
(D) SELECT

✓ **Answer: (A) INSERT, UPDATE, DELETE**

Q326. What is a BEFORE trigger?

(A) Fires before the DML event
(B) Fires after the DML event
(C) Fires instead of DML
(D) Fires on SELECT

✓ **Answer: (A) Fires before the DML event**

Q327. What is an AFTER trigger?

(A) Fires after the DML event
(B) Fires before DML
(C) Fires instead of DML
(D) Fires on DDL

✓ **Answer: (A) Fires after the DML event**

Q328. What is an INSTEAD OF trigger?

(A) Replaces the DML event
(B) Fires before DML
(C) Fires after DML
(D) Fires on SELECT

✓ **Answer: (A) Replaces the DML event**

Q329. Where are INSTEAD OF triggers commonly used?

(A) Views to handle complex DML
(B) Tables
(C) Stored procedures
(D) Functions

✓ **Answer: (A) Views to handle complex DML**

Q330. What are the virtual tables available in triggers (SQL Server)?

(A) INSERTED and DELETED
(B) BEFORE and AFTER
(C) OLD and NEW
(D) INPUT and OUTPUT

✓ **Answer: (A) INSERTED and DELETED**

Q331. What does the INSERTED table contain in triggers?

(A) New rows being inserted or updated
(B) Old rows
(C) All table rows
(D) NULL rows

✓ **Answer: (A) New rows being inserted or updated**

Q332. What does the DELETED table contain in triggers?

(A) Old rows being deleted or updated
(B) New rows
(C) All table rows
(D) NULL rows

✓ **Answer: (A) Old rows being deleted or updated**

Q333. Which pseudo-records are used in Oracle/PostgreSQL triggers?

(A) NEW and OLD
(B) INSERTED and DELETED
(C) BEFORE and AFTER
(D) IN and OUT

✓ **Answer: (A) NEW and OLD**

Q334. What is a DDL trigger?

(A) Fires on DDL events like CREATE, DROP, ALTER
(B) Fires on DML events
(C) Fires on SELECT
(D) Fires on COMMIT

✓ **Answer: (A) Fires on DDL events like CREATE, DROP, ALTER**

Q335. Can a trigger call a stored procedure?

(A) Yes
(B) No
(C) Only in SQL Server
(D) Only in Oracle

✓ **Answer: (A) Yes**

Q336. What is a row-level trigger?

(A) Fires once per affected row
(B) Fires once per statement
(C) Fires once per transaction
(D) Fires once per table

✓ **Answer: (A) Fires once per affected row**

Q337. What is a statement-level trigger?

(A) Fires once per DML statement
(B) Fires once per row
(C) Fires per column
(D) Fires per transaction

✓ **Answer: (A) Fires once per DML statement**

Q338. What is a compound trigger (Oracle)?

(A) Combines multiple timing points in one trigger
(B) A trigger with multiple events
(C) A recursive trigger
(D) A DDL trigger

✓ **Answer: (A) Combines multiple timing points in one trigger**

Q339. What is a logon trigger (SQL Server)?

(A) Fires when a user connects
(B) Fires on DML
(C) Fires on DDL
(D) Fires on SELECT

✓ **Answer: (A) Fires when a user connects**

Q340. Can triggers be recursive?

(A) Yes, but should be controlled
(B) No
(C) Only in Oracle
(D) Only in SQL Server

✓ **Answer: (A) Yes, but should be controlled**

Q341. What is a cursor?

(A) A database object to retrieve rows one at a time
(B) A stored procedure
(C) A trigger
(D) An index

✓ **Answer: (A) A database object to retrieve rows one at a time**

Q342. Which statement opens a cursor?

(A) OPEN cursor_name
(B) START cursor
(C) BEGIN cursor
(D) EXECUTE cursor

✓ **Answer: (A) OPEN cursor_name**

Q343. Which statement fetches a row from a cursor?

(A) FETCH cursor_name INTO variables
(B) GET cursor
(C) READ cursor
(D) NEXT cursor

✓ **Answer: (A) FETCH cursor_name INTO variables**

Q344. Which statement closes a cursor?

(A) CLOSE cursor_name
(B) STOP cursor
(C) END cursor
(D) RELEASE cursor

✓ **Answer: (A) CLOSE cursor_name**

Q345. Which statement deallocates a cursor?

(A) DEALLOCATE cursor_name
(B) DROP cursor
(C) DELETE cursor
(D) REMOVE cursor

✓ **Answer: (A) DEALLOCATE cursor_name**

Q346. What is an implicit cursor?

(A) Automatically created by database for DML
(B) Explicitly declared
(C) A named cursor
(D) A global cursor

✓ **Answer: (A) Automatically created by database for DML**

Q347. What is an explicit cursor?

(A) Declared by developer using DECLARE CURSOR
(B) Automatically created
(C) A named cursor only
(D) A global cursor

✓ **Answer: (A) Declared by developer using DECLARE CURSOR**

Q348. What is a cursor attribute %FOUND (Oracle/PL-SQL)?

(A) TRUE if last FETCH returned a row
(B) TRUE if cursor is open
(C) TRUE if cursor is closed
(D) Number of rows fetched

✓ **Answer: (A) TRUE if last FETCH returned a row**

Q349. What does %NOTFOUND return?

(A) TRUE if last FETCH returned no row
(B) TRUE if rows found

(C) Number of rows

(D) TRUE if cursor open

✓ **Answer: (A) TRUE if last FETCH returned no row**

Q350. What does %ROWCOUNT return?

(A) Number of rows fetched so far

(B) TRUE/FALSE

(C) Cursor status

(D) Row size

✓ **Answer: (A) Number of rows fetched so far**

Q351. What is a REF CURSOR (Oracle)?

(A) A cursor variable

(B) A static cursor

(C) A system cursor

(D) A table cursor

✓ **Answer: (A) A cursor variable**

Q352. What is dynamic SQL?

(A) SQL constructed and executed at runtime

(B) Static SQL

(C) Precompiled SQL

(D) Stored procedure SQL

✓ **Answer: (A) SQL constructed and executed at runtime**

Q353. Which statement executes dynamic SQL?

(A) EXECUTE / EXEC sp_executesql

(B) DYNAMIC SQL

(C) RUN SQL

(D) CALL SQL

✓ **Answer: (A) EXECUTE / EXEC sp_executesql**

Q354. What is the risk of dynamic SQL?

(A) SQL injection

(B) Syntax errors only

(C) Performance issues only

(D) Memory leaks

✓ **Answer: (A) SQL injection**

Q355. What is a parameterized query?

(A) Query using parameters to prevent injection

(B) A dynamic query

(C) A stored procedure

(D) A cursor-based query

✓ **Answer: (A) Query using parameters to prevent injection**

Q356. What is sp_executesql advantage over EXEC?

(A) Allows parameterized execution and plan reuse

(B) Faster always

(C) Simpler syntax

(D) No advantage

✓ **Answer: (A) Allows parameterized execution and plan reuse**

Q357. What is an anonymous block (Oracle/PostgreSQL)?

(A) PL/SQL block without a name
(B) A stored procedure
(C) A trigger
(D) A function

✓ **Answer: (A) PL/SQL block without a name**

Q358. Which keyword starts a PL/SQL block?

(A) BEGIN
(B) START
(C) OPEN
(D) DECLARE

✓ **Answer: (A) BEGIN**

Q359. Which keyword ends a PL/SQL block?

(A) END
(B) CLOSE
(C) FINISH
(D) STOP

✓ **Answer: (A) END**

Q360. What is exception handling in stored procedures?

(A) Catching and handling runtime errors
(B) Preventing SQL injection
(C) Optimizing queries
(D) Managing transactions

✓ **Answer: (A) Catching and handling runtime errors**

Q361. Which block handles exceptions in PL/SQL?

(A) EXCEPTION
(B) CATCH
(C) ERROR
(D) HANDLE

✓ **Answer: (A) EXCEPTION**

Q362. What is RAISE in PL/SQL?

(A) Throws an exception
(B) Logs an error
(C) Ends a procedure
(D) Starts a transaction

✓ **Answer: (A) Throws an exception**

Q363. What is RAISE_APPLICATION_ERROR in Oracle?

(A) Creates custom error message
(B) Logs to error table
(C) Ends transaction
(D) Raises system error

✓ **Answer: (A) Creates custom error message**

Q364. What is a package in Oracle?

(A) A group of related procedures and functions
(B) A stored procedure
(C) A schema
(D) A trigger

✓ **Answer: (A) A group of related procedures and functions**

Q365. What are the two parts of an Oracle package?

(A) Specification and Body
(B) Header and Code
(C) Interface and Implementation
(D) Declaration and Execution

✓ **Answer: (A) Specification and Body**

Q366. What is a package specification?

(A) Declares public elements (interface)
(B) Contains implementation
(C) Stores data
(D) Contains indexes

✓ **Answer: (A) Declares public elements (interface)**

Q367. What is package body?

(A) Contains implementation of declared elements
(B) Contains only declarations
(C) Stores package data
(D) Contains indexes

✓ **Answer: (A) Contains implementation of declared elements**

Q368. What is an overloaded procedure?

(A) Multiple procedures with same name but different parameters
(B) A recursive procedure
(C) A trigger
(D) A function

✓ **Answer: (A) Multiple procedures with same name but different parameters**

Q369. What is a pipelined table function (Oracle)?

(A) Returns rows one at a time as produced
(B) Returns all rows at once
(C) A scalar function
(D) A stored procedure

✓ **Answer: (A) Returns rows one at a time as produced**

Q370. What is a deterministic function?

(A) Always returns same result for same input
(B) Returns random results
(C) Returns multiple values
(D) A non-pure function

✓ **Answer: (A) Always returns same result for same input**

Q371. What is the RETURN statement in a function?

(A) Returns a value to the caller
(B) Ends a procedure
(C) Starts a transaction
(D) Logs a message

✓ **Answer: (A) Returns a value to the caller**

Q372. Can a stored procedure return multiple result sets?

(A) Yes (SQL Server, MySQL)
(B) No
(C) Only one result set
(D) Only with cursors

✓ **Answer: (A) Yes (SQL Server, MySQL)**

Q373. What is a global temporary table?

(A) Temp table visible to all sessions, data session-specific
(B) Visible to one session only
(C) A permanent table
(D) A view

✓ **Answer: (A) Temp table visible to all sessions, data session-specific**

Q374. What is a local temporary table?

(A) Visible only to current session/procedure
(B) Visible to all sessions
(C) A permanent table
(D) A CTE

✓ **Answer: (A) Visible only to current session/procedure**

Q375. Which prefix denotes local temp table in SQL Server?

(A) #
(B) ##
(C) @
(D) @@

✓ **Answer: (A) #**

Q376. Which prefix denotes global temp table in SQL Server?

(A) ##
(B) #
(C) @
(D) @@

✓ **Answer: (A) ##**

Q377. What is a table variable in SQL Server?

(A) A variable holding a table structure, scoped to batch
(B) A permanent table
(C) A global temp table
(D) A cursor

✓ **Answer: (A) A variable holding a table structure, scoped to batch**

Q378. Which is generally faster: temp table or table variable?

(A) Depends on data size; temp table better for large data
(B) Table variable always faster
(C) Temp table always faster
(D) They are identical

✓ **Answer: (A) Depends on data size; temp table better for large data**

Q379. Can stored procedures be nested?

(A) Yes
(B) No
(C) Only two levels
(D) Only in Oracle

✓ **Answer: (A) Yes**

Q380. What is the maximum nesting level for stored procedures in SQL Server?

(A) 32
(B) 64
(C) 128
(D) 16

✓ **Answer: (A) 32**

Section: Transactions, ACID, Concurrency & Recovery (Q 381–440)

Q381. What are the ACID properties?

(A) Atomicity, Consistency, Isolation, Durability
(B) Accuracy, Completeness, Integrity, Durability
(C) Atomicity, Concurrency, Integrity, Durability
(D) Availability, Consistency, Integrity, Durability

✓ **Answer: (A) Atomicity, Consistency, Isolation, Durability**

Q382. What does Atomicity mean?

(A) All operations in transaction succeed or all fail
(B) Operations can be partial
(C) Each operation is independent
(D) Transactions are fast

✓ **Answer: (A) All operations in transaction succeed or all fail**

Q383. What does Consistency mean?

(A) Database moves from one valid state to another
(B) Data is always correct
(C) Transactions are isolated
(D) Data is durable

✓ **Answer: (A) Database moves from one valid state to another**

Q384. What does Isolation mean?

(A) Concurrent transactions don't interfere
(B) All transactions run sequentially
(C) Transactions are atomic
(D) Transactions are durable

✓ **Answer: (A) Concurrent transactions don't interfere**

Q385. What does Durability mean?

(A) Committed data persists even after failure
(B) Data can be lost on failure
(C) Transactions are fast
(D) Data is always consistent

✓ **Answer: (A) Committed data persists even after failure**

Q386. What is a transaction?

(A) A unit of work that must complete entirely
(B) A single SQL statement
(C) A stored procedure
(D) A database connection

✓ **Answer: (A) A unit of work that must complete entirely**

Q387. Which isolation level allows dirty reads?

(A) READ UNCOMMITTED
(B) READ COMMITTED
(C) REPEATABLE READ
(D) SERIALIZABLE

✓ **Answer: (A) READ UNCOMMITTED**

Q388. Which isolation level is the most restrictive?

(A) SERIALIZABLE
(B) READ UNCOMMITTED
(C) READ COMMITTED
(D) REPEATABLE READ

✓ **Answer: (A) SERIALIZABLE**

Q389. What is a non-repeatable read?

(A) A row read twice in same transaction gives different values
(B) A phantom row
(C) A dirty read
(D) A lost update

✓ **Answer: (A) A row read twice in same transaction gives different values**

Q390. Which isolation level prevents non-repeatable reads?

(A) REPEATABLE READ
(B) READ COMMITTED
(C) READ UNCOMMITTED
(D) SNAPSHOT

✓ **Answer: (A) REPEATABLE READ**

Q391. What is a lost update?

(A) Two transactions overwrite each other's changes
(B) A deleted row
(C) A phantom row
(D) A dirty read

✓ **Answer: (A) Two transactions overwrite each other's changes**

Q392. What is snapshot isolation?

(A) Each transaction sees data as of its start time
(B) Same as SERIALIZABLE
(C) Same as READ COMMITTED
(D) Same as READ UNCOMMITTED

✓ **Answer: (A) Each transaction sees data as of its start time**

Q393. What is optimistic concurrency control?

(A) Assumes conflicts are rare; checks at commit
(B) Assumes conflicts are common; locks always
(C) Uses shared locks
(D) Uses exclusive locks

✓ **Answer: (A) Assumes conflicts are rare; checks at commit**

Q394. What is pessimistic concurrency control?

(A) Acquires locks before accessing data
(B) Assumes no conflicts
(C) Checks at commit time
(D) Uses MVCC

✓ **Answer: (A) Acquires locks before accessing data**

Q395. What is MVCC?

(A) Multi-Version Concurrency Control
(B) Multi-Value Consistency Check
(C) Multiple Version Copy Control
(D) Multi-Value Concurrency Control

✓ **Answer: (A) Multi-Version Concurrency Control**

Q396. Which databases use MVCC?

(A) PostgreSQL, Oracle, MySQL InnoDB
(B) SQL Server only
(C) SQLite only
(D) MySQL MyISAM

✓ **Answer: (A) PostgreSQL, Oracle, MySQL InnoDB**

Q397. What is a shared lock?

(A) Allows reads but prevents writes
(B) Prevents reads and writes
(C) Allows writes
(D) Exclusive lock

✓ **Answer: (A) Allows reads but prevents writes**

Q398. What is an exclusive lock?

(A) Prevents other reads and writes
(B) Allows reads only
(C) Allows writes only
(D) A shared lock

✓ **Answer: (A) Prevents other reads and writes**

Q399. What is a deadlock detection?

(A) Database detects circular lock dependency and kills one transaction
(B) Prevents locks
(C) Allows all transactions
(D) Logs errors only

✓ **Answer: (A) Database detects circular lock dependency and kills one transaction**

Q400. What is lock escalation?

(A) Converting many fine-grain locks to fewer coarse-grain locks
(B) Adding more locks
(C) Removing locks
(D) Preventing deadlocks

✓ **Answer: (A) Converting many fine-grain locks to fewer coarse-grain locks**

Q401. Which command sets isolation level?

(A) SET TRANSACTION ISOLATION LEVEL
(B) SET ISOLATION
(C) CHANGE ISOLATION
(D) ALTER ISOLATION

✓ **Answer: (A) SET TRANSACTION ISOLATION LEVEL**

Q402. What is a two-phase commit (2PC)?

(A) Protocol to ensure distributed transaction atomicity
(B) Two COMMIT statements
(C) Two-phase locking
(D) Two databases committing

✓ **Answer: (A) Protocol to ensure distributed transaction atomicity**

Q403. What is two-phase locking (2PL)?

(A) Acquire all locks before releasing any
(B) Two types of locks
(C) Two lock phases
(D) Lock and unlock alternately

✓ **Answer: (A) Acquire all locks before releasing any**

Q404. What is a checkpoint in database recovery?

(A) Point where all dirty pages written to disk
(B) A SAVEPOINT
(C) A COMMIT
(D) A ROLLBACK

✓ **Answer: (A) Point where all dirty pages written to disk**

Q405. What is write-ahead logging (WAL)?

(A) Log changes before writing to data pages
(B) Write data before logging
(C) A recovery method
(D) A backup strategy

✓ **Answer: (A) Log changes before writing to data pages**

Q406. What is the redo log?

(A) Records changes for forward recovery
(B) Records changes for backward recovery
(C) A backup log
(D) An error log

✓ **Answer: (A) Records changes for forward recovery**

Q407. What is the undo log?

(A) Records original values for rollback
(B) Records new values
(C) A backup log
(D) A transaction log

✓ **Answer: (A) Records original values for rollback**

Q408. What is a transaction log?

(A) Sequential record of all transactions
(B) A backup
(C) An error log
(D) An audit log

✓ **Answer: (A) Sequential record of all transactions**

Q409. What is log shipping?

(A) Copying transaction logs to secondary server
(B) Shipping backups
(C) Copying data files
(D) A replication method

✓ **Answer: (A) Copying transaction logs to secondary server**

Q410. What is a distributed transaction?

(A) Transaction spanning multiple databases
(B) A transaction with many statements
(C) A long transaction
(D) A stored procedure

✓ **Answer: (A) Transaction spanning multiple databases**

Q411. What is the difference between COMMIT and ROLLBACK?

(A) COMMIT saves, ROLLBACK undoes
(B) ROLLBACK saves, COMMIT undoes
(C) They are identical
(D) COMMIT undoes partial changes

✓ **Answer: (A) COMMIT saves, ROLLBACK undoes**

Q412. What happens to locks after COMMIT?

(A) All locks are released
(B) Locks remain
(C) Locks escalate
(D) Locks are converted to shared

✓ **Answer: (A) All locks are released**

Q413. Can you rollback to a specific SAVEPOINT?

(A) Yes
(B) No
(C) Only in Oracle
(D) Only in PostgreSQL

✓ **Answer: (A) Yes**

Q414. What is implicit commit?

(A) Automatic commit after DDL in some databases
(B) Manual commit
(C) A partial commit
(D) A rollback

✓ **Answer: (A) Automatic commit after DDL in some databases**

Q415. Which databases auto-commit DDL statements?

(A) Oracle, MySQL
(B) PostgreSQL
(C) SQL Server
(D) SQLite

✓ **Answer: (A) Oracle, MySQL**

Q416. What is autocommit mode?

(A) Each statement is automatically committed
(B) Manual commit required
(C) Transactions are disabled
(D) Rollback is disabled

✓ **Answer: (A) Each statement is automatically committed**

Q417. What is a long-running transaction risk?

(A) Lock contention, blocking other transactions
(B) Faster execution
(C) More data written
(D) Better consistency

✓ **Answer: (A) Lock contention, blocking other transactions**

Q418. What is a compensating transaction?

(A) A transaction that undoes effects of another
(B) A long transaction
(C) A distributed transaction
(D) A batch transaction

✓ **Answer: (A) A transaction that undoes effects of another**

Q419. What is transaction timeout?

(A) Maximum allowed transaction duration
(B) Maximum rows

(C) Maximum locks
(D) Maximum log size

✓ **Answer: (A) Maximum allowed transaction duration**

Q420. What is a saga pattern?

(A) Sequence of local transactions with compensations
(B) A single long transaction
(C) A distributed lock
(D) A two-phase commit

✓ **Answer: (A) Sequence of local transactions with compensations**

Q421. What is row versioning?

(A) Maintaining old row versions for MVCC
(B) A form of indexing
(C) A backup strategy
(D) A replication method

✓ **Answer: (A) Maintaining old row versions for MVCC**

Q422. Which SQL Server feature provides optimistic concurrency?

(A) READ COMMITTED SNAPSHOT ISOLATION (RCSI)
(B) NOLOCK
(C) HOLDLOCK
(D) ROWLOCK

✓ **Answer: (A) READ COMMITTED SNAPSHOT ISOLATION (RCSI)**

Q423. What does HOLDLOCK hint do?

(A) Holds shared lock until end of transaction
(B) Releases lock immediately
(C) Escalates to exclusive
(D) Prevents lock

✓ **Answer: (A) Holds shared lock until end of transaction**

Q424. What does UPDLOCK hint do?

(A) Acquires update lock to prevent deadlocks on update
(B) Shared lock
(C) Exclusive lock
(D) No lock

✓ **Answer: (A) Acquires update lock to prevent deadlocks on update**

Q425. What is the lock hierarchy?

(A) Row -> Page -> Extent -> Table -> Database
(B) Table -> Row
(C) Row -> Table
(D) Page -> Row -> Table

✓ **Answer: (A) Row -> Page -> Extent -> Table -> Database**

Q426. What is a latch?

(A) Short-duration lock on in-memory structures
(B) A table lock
(C) A row lock
(D) A page lock

✓ **Answer: (A) Short-duration lock on in-memory structures**

Q427. What is spinlock?

(A) CPU-level synchronization for in-memory structures
(B) A database lock
(C) A row lock
(D) A page lock

✓ **Answer: (A) CPU-level synchronization for in-memory structures**

Q428. What is blocking?

(A) One transaction waiting for another to release a lock
(B) A deadlock
(C) An error
(D) A timeout

✓ **Answer: (A) One transaction waiting for another to release a lock**

Q429. What is the difference between blocking and deadlock?

(A) Blocking resolves when lock released; deadlock is circular
(B) They are identical
(C) Deadlock resolves automatically
(D) Blocking is circular

✓ **Answer: (A) Blocking resolves when lock released; deadlock is circular**

Q430. What is lock-free concurrency?

(A) Using atomic operations instead of locks
(B) A type of table lock
(C) A deadlock-free design
(D) An optimistic approach

✓ **Answer: (A) Using atomic operations instead of locks**

Q431. What is a dirty page?

(A) A modified page not yet written to disk
(B) A corrupted page
(C) A locked page
(D) A cached page

✓ **Answer: (A) A modified page not yet written to disk**

Q432. What is crash recovery?

(A) Restoring database to consistent state after failure
(B) A backup restore
(C) A ROLLBACK
(D) A COMMIT

✓ **Answer: (A) Restoring database to consistent state after failure**

Q433. What is ARIES recovery algorithm?

(A) Analysis, Redo, Undo phases for crash recovery
(B) A backup algorithm
(C) A replication method
(D) An index rebuild method

✓ **Answer: (A) Analysis, Redo, Undo phases for crash recovery**

Q434. What is the difference between hot and cold backup?

(A) Hot backup taken while database is running; cold with database offline
(B) They are identical
(C) Hot backup is faster
(D) Cold backup is online

✓ **Answer: (A) Hot backup taken while database is running; cold with database offline**

Q435. What is point-in-time recovery?

(A) Restoring database to a specific past moment
(B) A full restore
(C) A differential restore
(D) A log restore

✓ **Answer: (A) Restoring database to a specific past moment**

Q436. What is a full backup?

(A) Complete copy of all database data
(B) Only changed data
(C) Only transaction logs
(D) An incremental backup

✓ **Answer: (A) Complete copy of all database data**

Q437. What is a differential backup?

(A) Changes since last full backup
(B) Changes since last backup
(C) A full backup
(D) A transaction log backup

✓ **Answer: (A) Changes since last full backup**

Q438. What is an incremental backup?

(A) Changes since last backup of any type
(B) A full backup
(C) A differential backup
(D) A log backup

✓ **Answer: (A) Changes since last backup of any type**

Q439. What is database mirroring?

(A) Real-time copy to a mirror server
(B) Log shipping
(C) Replication
(D) A backup method

✓ **Answer: (A) Real-time copy to a mirror server**

Q440. What is Always On Availability Groups (SQL Server)?

(A) High-availability and disaster recovery solution
(B) A backup method
(C) Log shipping
(D) A replication method

✓ **Answer: (A) High-availability and disaster recovery solution**

Section: Window Functions (Q 441–500)

Q441. What are window functions?

(A) Functions that perform calculations across a set of related rows
(B) Aggregate functions
(C) Scalar functions
(D) Table functions

✓ **Answer: (A) Functions that perform calculations across a set of related rows**

Q442. Which clause defines the window for window functions?

(A) OVER()
(B) PARTITION
(C) WINDOW
(D) FRAME

✓ **Answer: (A) OVER()**

Q443. What does PARTITION BY do in window functions?

(A) Divides rows into groups for window calculation
(B) Groups rows like GROUP BY
(C) Filters rows
(D) Sorts rows

✓ **Answer: (A) Divides rows into groups for window calculation**

Q444. What does ORDER BY do inside OVER()?

(A) Defines row ordering within the window
(B) Sorts the final result
(C) Groups rows
(D) Filters rows

✓ **Answer: (A) Defines row ordering within the window**

Q445. What is ROW_NUMBER()?

(A) Assigns a unique sequential number to each row
(B) Assigns rank with gaps
(C) Assigns dense rank
(D) Returns row count

✓ **Answer: (A) Assigns a unique sequential number to each row**

Q446. What is RANK()?

(A) Assigns rank with gaps for ties
(B) Assigns sequential numbers
(C) Assigns dense rank
(D) Returns row count

✓ **Answer: (A) Assigns rank with gaps for ties**

Q447. What is DENSE_RANK()?

(A) Assigns rank without gaps for ties
(B) Assigns rank with gaps
(C) Assigns sequential numbers
(D) Returns row count

✓ **Answer: (A) Assigns rank without gaps for ties**

Q448. What is NTILE(n)?

(A) Divides rows into n equal buckets
(B) Assigns rank
(C) Returns nth row
(D) Counts rows

✓ **Answer: (A) Divides rows into n equal buckets**

Q449. What is LAG()?

(A) Access a previous row's value
(B) Access next row's value
(C) Access first row's value
(D) Access last row's value

✓ **Answer: (A) Access a previous row's value**

Q450. What is LEAD()?

(A) Access a next row's value
(B) Access previous row's value
(C) Access first row's value
(D) Access last row's value

✓ **Answer: (A) Access a next row's value**

Q451. What is FIRST_VALUE()?

(A) Returns value from first row in the window
(B) Returns value from last row
(C) Returns minimum value
(D) Returns first aggregate

✓ **Answer: (A) Returns value from first row in the window**

Q452. What is LAST_VALUE()?

(A) Returns value from last row in the window
(B) Returns value from first row
(C) Returns maximum value
(D) Returns last aggregate

✓ **Answer: (A) Returns value from last row in the window**

Q453. What is NTH_VALUE()?

(A) Returns value from nth row in the window
(B) Returns nth aggregate
(C) Returns nth rank
(D) Returns nth bucket

✓ **Answer: (A) Returns value from nth row in the window**

Q454. What is CUME_DIST()?

(A) Cumulative distribution of a value in a group
(B) Rank without gaps
(C) Rank with gaps
(D) Distribution rank

✓ **Answer: (A) Cumulative distribution of a value in a group**

Q455. What is PERCENT_RANK()?

(A) Relative rank as percentage
(B) Cumulative distribution
(C) Dense rank
(D) Row number

✓ **Answer: (A) Relative rank as percentage**

Q456. What is the difference between ROW_NUMBER and RANK?

(A) ROW_NUMBER has no ties; RANK has gaps for ties
(B) They are identical
(C) RANK has no gaps
(D) ROW_NUMBER has gaps

✓ **Answer: (A) ROW_NUMBER has no ties; RANK has gaps for ties**

Q457. Can window functions be used in WHERE clause?

(A) No (use subquery or CTE)
(B) Yes
(C) Only in HAVING
(D) Only with GROUP BY

✓ **Answer: (A) No (use subquery or CTE)**

Q458. Can you use GROUP BY and window functions together?

(A) Yes
(B) No
(C) Only in FROM
(D) Only in HAVING

✓ **Answer: (A) Yes**

Q459. What is a window frame?

(A) Subset of rows within partition for calculation
(B) The PARTITION BY definition
(C) The ORDER BY definition
(D) The OVER() clause

✓ **Answer: (A) Subset of rows within partition for calculation**

Q460. What does ROWS BETWEEN specify?

(A) Physical rows for the frame
(B) Logical values for the frame
(C) Partition definition
(D) Order definition

✓ **Answer: (A) Physical rows for the frame**

Q461. What does RANGE BETWEEN specify?

(A) Logical value range for the frame
(B) Physical rows
(C) Partition definition
(D) Order definition

✓ **Answer: (A) Logical value range for the frame**

Q462. What does UNBOUNDED PRECEDING mean?

(A) From the start of the partition
(B) From the current row
(C) From the previous row
(D) From a fixed offset

✓ **Answer: (A) From the start of the partition**

Q463. What does CURRENT ROW mean in frame?

(A) The current row being processed
(B) The first row
(C) The last row
(D) An arbitrary row

✓ **Answer: (A) The current row being processed**

Q464. What does UNBOUNDED FOLLOWING mean?

(A) To the end of the partition
(B) To the current row
(C) To the next row
(D) To a fixed offset

✓ **Answer: (A) To the end of the partition**

Q465. What is the default frame for ORDER BY in OVER()?

(A) RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
(B) ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING
(C) All rows in partition
(D) No frame

✓ **Answer: (A) RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW**

Q466. What is a running total?

(A) Cumulative sum using SUM() with window function
(B) A simple SUM()
(C) A GROUP BY sum
(D) A subquery sum

✓ **Answer: (A) Cumulative sum using SUM() with window function**

Q467. How to calculate a moving average?

(A) AVG() with ROWS BETWEEN frame
(B) Simple AVG()
(C) GROUP BY AVG()
(D) Subquery AVG()

✓ **Answer: (A) AVG() with ROWS BETWEEN frame**

Q468. Which function returns the median?

(A) PERCENTILE_CONT(0.5) or application logic
(B) MEDIAN()
(C) AVG()
(D) MID()

✓ **Answer: (A) PERCENTILE_CONT(0.5) or application logic**

Q469. What is PERCENTILE_CONT?

(A) Continuous percentile interpolating between values
(B) Discrete percentile
(C) A rank function
(D) An aggregate function

✓ **Answer: (A) Continuous percentile interpolating between values**

Q470. What is PERCENTILE_DISC?

(A) Discrete percentile returning actual row value
(B) Continuous percentile
(C) A rank function
(D) An aggregate function

✓ **Answer: (A) Discrete percentile returning actual row value**

Q471. Can you use multiple window functions in one SELECT?

(A) Yes
(B) No
(C) Only two
(D) Only with same OVER()

✓ **Answer: (A) Yes**

Q472. What is the named window clause?

(A) WINDOW clause defining reusable window definition
(B) PARTITION BY clause
(C) OVER() clause
(D) FRAME clause

✓ **Answer: (A) WINDOW clause defining reusable window definition**

Q473. Which keyword defines a named window?

(A) WINDOW ... AS (...)
(B) DEFINE WINDOW
(C) NAMED WINDOW
(D) REUSE WINDOW

✓ **Answer: (A) WINDOW ... AS (...)**

Q474. What is the order of processing: WHERE, window, HAVING?

(A) WHERE -> GROUP BY/HAVING -> window functions
(B) Window -> WHERE -> HAVING
(C) HAVING -> WHERE -> Window
(D) WHERE -> Window -> HAVING

✓ **Answer: (A) WHERE -> GROUP BY/HAVING -> window functions**

Q475. What is the difference between SUM() with GROUP BY and SUM() with OVER()?

(A) GROUP BY collapses rows; OVER() keeps all rows
(B) They produce identical results
(C) OVER() is faster
(D) GROUP BY keeps all rows

✓ **Answer: (A) GROUP BY collapses rows; OVER() keeps all rows**

Q476. Can you use DISTINCT inside window functions?

(A) No (most databases)
(B) Yes
(C) Only COUNT(DISTINCT)
(D) Only SUM(DISTINCT)

✓ **Answer: (A) No (most databases)**

Q477. Which window function can detect gaps in sequential data?

(A) ROW_NUMBER() comparison technique
(B) GAP()
(C) SEQUENCE()
(D) RANK()

✓ **Answer: (A) ROW_NUMBER() comparison technique**

Q478. What is the islands and gaps problem?

(A) Finding consecutive sequences and gaps in data
(B) A join problem
(C) A grouping problem
(D) A sorting problem

✓ **Answer: (A) Finding consecutive sequences and gaps in data**

Q479. Can window functions be used with CTEs?

(A) Yes
(B) No

(C) Only in FROM

(D) Only in SELECT

✓ **Answer: (A) Yes**

Q480. Which is more efficient: subquery or window function for ranking?

(A) Window function (single pass)

(B) Subquery

(C) They are identical

(D) CTE

✓ **Answer: (A) Window function (single pass)**

Q481. What does LAG(col, 2) do?

(A) Returns value from 2 rows before current

(B) Returns value from 2 rows after

(C) Returns 2nd value in partition

(D) Returns the second column

✓ **Answer: (A) Returns value from 2 rows before current**

Q482. Can you specify a default in LAG/LEAD?

(A) Yes, LAG(col, n, default)

(B) No

(C) Only in PostgreSQL

(D) Only in Oracle

✓ **Answer: (A) Yes, LAG(col, n, default)**

Q483. What is RATIO_TO_REPORT (Oracle)?

(A) Ratio of value to sum of values in partition

(B) A window rank function

(C) A cumulative distribution

(D) A percentile function

✓ **Answer: (A) Ratio of value to sum of values in partition**

Q484. Which window function is used to eliminate duplicates keeping only one per group?

(A) ROW_NUMBER() = 1 in CTE/subquery

(B) RANK() = 1

(C) DISTINCT

(D) DENSE_RANK() = 1

✓ **Answer: (A) ROW_NUMBER() = 1 in CTE/subquery**

Q485. What is the difference between RANK and DENSE_RANK for values 10,10,20?

(A) RANK: 1,1,3; DENSE_RANK: 1,1,2

(B) They are identical

(C) RANK: 1,1,2; DENSE_RANK: 1,1,3

(D) RANK: 1,2,3; DENSE_RANK: 1,2,3

✓ **Answer: (A) RANK: 1,1,3; DENSE_RANK: 1,1,2**

Q486. Can you partition by multiple columns?

(A) Yes, PARTITION BY col1, col2

(B) No

(C) Only one column

(D) Only in Oracle

✓ **Answer: (A) Yes, PARTITION BY col1, col2**

Q487. What is the purpose of ORDER BY in OVER()?

(A) Determines ordering for ranking/offset functions
(B) Sorts the final output
(C) Filters the window
(D) Defines the partition

✓ **Answer: (A) Determines ordering for ranking/offset functions**

Q488. Can SUM() be used as a window function?

(A) Yes
(B) No
(C) Only as aggregate
(D) Only in GROUP BY

✓ **Answer: (A) Yes**

Q489. Which function is used to number rows within groups resetting per group?

(A) ROW_NUMBER() OVER (PARTITION BY group ORDER BY col)
(B) ROWNUM
(C) RANK()
(D) SEQUENCE()

✓ **Answer: (A) ROW_NUMBER() OVER (PARTITION BY group ORDER BY col)**

Q490. What does EXCLUDE CURRENT ROW do in a frame?

(A) Excludes the current row from aggregate calculation
(B) Includes only current row
(C) Skips the frame
(D) Resets the frame

✓ **Answer: (A) Excludes the current row from aggregate calculation**

Q491. What is the GROUPS frame unit?

(A) Groups of peer rows (same ORDER BY value)
(B) Physical rows
(C) Logical value ranges
(D) Partitions

✓ **Answer: (A) Groups of peer rows (same ORDER BY value)**

Q492. Can you use window functions inside aggregate functions?

(A) No
(B) Yes
(C) Only in HAVING
(D) Only in GROUP BY

✓ **Answer: (A) No**

Q493. Can aggregate functions be used inside window functions?

(A) No (window functions are at same level)
(B) Yes
(C) Only COUNT
(D) Only SUM

✓ **Answer: (A) No (window functions are at same level)**

Q494. What is the performance consideration for window functions?

(A) Additional sort and computation overhead
(B) Always faster than subqueries
(C) No overhead
(D) Slower than GROUP BY always

✓ **Answer: (A) Additional sort and computation overhead**

Q495. Which window function gives percentage of total?

(A) SUM(val) OVER() to compute ratio
(B) PERCENT()
(C) RATIO()
(D) TOTAL()

✓ **Answer: (A) SUM(val) OVER() to compute ratio**

Q496. What is a top-N per group query?

(A) Getting top N rows within each partition
(B) A simple TOP query
(C) A LIMIT query
(D) A GROUP BY query

✓ **Answer: (A) Getting top N rows within each partition**

Q497. Which pattern implements top-N per group?

(A) ROW_NUMBER() OVER(PARTITION BY ... ORDER BY ...) <= N
(B) LIMIT N
(C) TOP N
(D) ROWNUM <= N

✓ **Answer: (A) ROW_NUMBER() OVER(PARTITION BY ... ORDER BY ...) <= N**

Q498. Can window functions work without PARTITION BY?

(A) Yes, treats entire result as one partition
(B) No
(C) Only with ORDER BY
(D) Only in SELECT

✓ **Answer: (A) Yes, treats entire result as one partition**

Q499. What is the difference between ROWS and RANGE frame?

(A) ROWS uses physical row count; RANGE uses value range
(B) They are identical
(C) RANGE is physical
(D) ROWS is logical

✓ **Answer: (A) ROWS uses physical row count; RANGE uses value range**

Q500. Which function returns the running count including ties?

(A) DENSE_RANK()
(B) ROW_NUMBER()
(C) RANK()
(D) NTILE()

✓ **Answer: (A) DENSE_RANK()**

Section: Normalization & Database Design (Q 501–570)

Q501. What is normalization?

(A) Organizing data to reduce redundancy and improve integrity
(B) Denormalizing data
(C) Adding indexes
(D) Creating views

✓ **Answer: (A) Organizing data to reduce redundancy and improve integrity**

Q502. What is the First Normal Form (1NF)?

(A) Atomic values, no repeating groups, unique rows
(B) No partial dependencies
(C) No transitive dependencies
(D) No multi-valued dependencies

✓ **Answer: (A) Atomic values, no repeating groups, unique rows**

Q503. What is the Second Normal Form (2NF)?

(A) 1NF + no partial dependencies on composite key
(B) 1NF + atomic values
(C) 1NF + no transitive dependencies
(D) 1NF + no repeating groups

✓ **Answer: (A) 1NF + no partial dependencies on composite key**

Q504. What is the Third Normal Form (3NF)?

(A) 2NF + no transitive dependencies
(B) 2NF + no partial dependencies
(C) 1NF + atomic values
(D) BCNF

✓ **Answer: (A) 2NF + no transitive dependencies**

Q505. What is BCNF (Boyce-Codd Normal Form)?

(A) Stronger version of 3NF: every determinant is a candidate key
(B) Same as 3NF
(C) 4NF
(D) 5NF

✓ **Answer: (A) Stronger version of 3NF: every determinant is a candidate key**

Q506. What is the Fourth Normal Form (4NF)?

(A) BCNF + no multi-valued dependencies
(B) BCNF + no join dependencies
(C) 3NF + no transitive
(D) 2NF + no partial

✓ **Answer: (A) BCNF + no multi-valued dependencies**

Q507. What is the Fifth Normal Form (5NF)?

(A) 4NF + no join dependencies
(B) 4NF + no multi-valued
(C) 3NF + no transitive
(D) BCNF

✓ **Answer: (A) 4NF + no join dependencies**

Q508. What is denormalization?

(A) Intentionally adding redundancy for performance
(B) Removing redundancy
(C) A form of normalization
(D) Adding indexes

✓ **Answer: (A) Intentionally adding redundancy for performance**

Q509. What is a functional dependency?

(A) X -> Y: X determines Y
(B) A foreign key relationship
(C) A constraint
(D) An index

✓ **Answer: (A) X -> Y: X determines Y**

Q510. What is a partial dependency?

(A) Non-key attribute depends on part of composite key
(B) Full dependency
(C) Transitive dependency
(D) Multi-valued dependency

✓ **Answer: (A) Non-key attribute depends on part of composite key**

Q511. What is a transitive dependency?

(A) Non-key attribute depends on another non-key attribute
(B) Partial dependency
(C) Full dependency
(D) Multi-valued dependency

✓ **Answer: (A) Non-key attribute depends on another non-key attribute**

Q512. What is an entity in ER modeling?

(A) A real-world object represented as a table
(B) An attribute
(C) A relationship
(D) A foreign key

✓ **Answer: (A) A real-world object represented as a table**

Q513. What is an attribute in ER modeling?

(A) A property of an entity
(B) An entity
(C) A relationship
(D) A table

✓ **Answer: (A) A property of an entity**

Q514. What is a relationship in ER modeling?

(A) Association between entities
(B) A foreign key
(C) An attribute
(D) A table

✓ **Answer: (A) Association between entities**

Q515. What is cardinality in ER modeling?

(A) Number of entity instances in a relationship
(B) Number of attributes
(C) Number of rows
(D) Number of tables

✓ **Answer: (A) Number of entity instances in a relationship**

Q516. What is a one-to-many relationship?

(A) One entity relates to many of another
(B) One-to-one
(C) Many-to-many
(D) Self-referencing

✓ **Answer: (A) One entity relates to many of another**

Q517. What is a many-to-many relationship?

(A) Many instances on both sides
(B) One-to-one
(C) One-to-many
(D) Self-referencing

✓ **Answer: (A) Many instances on both sides**

Q518. How to implement many-to-many in SQL?

(A) Junction/bridge table with two foreign keys
(B) One foreign key
(C) A view
(D) An index

✓ **Answer: (A) Junction/bridge table with two foreign keys**

Q519. What is a surrogate key?

(A) System-generated unique identifier
(B) A natural key
(C) A composite key
(D) A foreign key

✓ **Answer: (A) System-generated unique identifier**

Q520. What is a natural key?

(A) Key derived from real-world attribute
(B) A surrogate key
(C) A composite key
(D) A foreign key

✓ **Answer: (A) Key derived from real-world attribute**

Q521. What is a composite key?

(A) Key made of multiple columns
(B) A single-column key
(C) A surrogate key
(D) A foreign key

✓ **Answer: (A) Key made of multiple columns**

Q522. What is a candidate key?

(A) Any column(s) that could be primary key
(B) The chosen primary key
(C) A foreign key
(D) A composite key

✓ **Answer: (A) Any column(s) that could be primary key**

Q523. What is a super key?

(A) Any set of columns that uniquely identifies rows
(B) A candidate key
(C) A primary key
(D) A composite key

✓ **Answer: (A) Any set of columns that uniquely identifies rows**

Q524. What is referential integrity?

(A) Foreign key values must exist in referenced table
(B) Unique values in all columns
(C) No NULL values
(D) Consistent data types

✓ **Answer: (A) Foreign key values must exist in referenced table**

Q525. What is a self-referencing (recursive) table?

(A) A table with foreign key to itself
(B) A table with no keys
(C) A joined table
(D) A temporary table

✓ **Answer: (A) A table with foreign key to itself**

Q526. What is an ER diagram?

(A) Visual representation of entities and relationships
(B) A query plan
(C) An index diagram
(D) A normalization chart

✓ **Answer: (A) Visual representation of entities and relationships**

Q527. What is a weak entity?

(A) Entity that depends on another entity for identification
(B) An entity without a primary key
(C) An entity with no attributes
(D) A temporary entity

✓ **Answer: (A) Entity that depends on another entity for identification**

Q528. What is a strong entity?

(A) Entity with its own primary key
(B) An entity without relationships
(C) A denormalized entity
(D) An entity with foreign keys

✓ **Answer: (A) Entity with its own primary key**

Q529. What is a junction table?

(A) Table resolving many-to-many relationship
(B) A view
(C) A temporary table
(D) A derived table

✓ **Answer: (A) Table resolving many-to-many relationship**

Q530. What is a star schema?

(A) Central fact table connected to dimension tables
(B) A normalized schema
(C) A snowflake schema
(D) An ER schema

✓ **Answer: (A) Central fact table connected to dimension tables**

Q531. What is a snowflake schema?

(A) Star schema with normalized dimension tables
(B) A denormalized schema
(C) A flat schema
(D) A star schema

✓ **Answer: (A) Star schema with normalized dimension tables**

Q532. What is a fact table?

(A) Table containing measurable business facts
(B) A dimension table
(C) A lookup table
(D) A bridge table

✓ **Answer: (A) Table containing measurable business facts**

Q533. What is a dimension table?

(A) Table with descriptive attributes
(B) A fact table
(C) A bridge table
(D) A derived table

✓ **Answer: (A) Table with descriptive attributes**

Q534. What is data warehousing?

(A) Central repository of integrated data for analytics
(B) A database backup
(C) A replication method
(D) A normalization technique

✓ **Answer: (A) Central repository of integrated data for analytics**

Q535. What is OLTP?

(A) Online Transaction Processing: many small fast transactions
(B) Online Analytical Processing
(C) A database schema
(D) A normalization level

✓ **Answer: (A) Online Transaction Processing: many small fast transactions**

Q536. What is OLAP?

(A) Online Analytical Processing: complex queries on large data
(B) Online Transaction Processing
(C) A database type
(D) A storage format

✓ **Answer: (A) Online Analytical Processing: complex queries on large data**

Q537. What is an SCD (Slowly Changing Dimension)?

(A) Dimension data that changes over time
(B) A fast-changing dimension
(C) A static dimension
(D) A fact table

✓ **Answer: (A) Dimension data that changes over time**

Q538. What is SCD Type 1?

(A) Overwrite old data with new
(B) Keep history with new row
(C) Add effective dates
(D) Keep both old and new

✓ **Answer: (A) Overwrite old data with new**

Q539. What is SCD Type 2?

(A) Add new row with history; effective date columns
(B) Overwrite old data

(C) Add column for changes

(D) Delete old row

✓ **Answer: (A) Add new row with history; effective date columns**

Q540. What is a data mart?

(A) Subject-specific subset of data warehouse

(B) A full data warehouse

(C) A transactional database

(D) A temporary table

✓ **Answer: (A) Subject-specific subset of data warehouse**

Q541. What is ETL?

(A) Extract, Transform, Load

(B) Extract, Transfer, Load

(C) Evaluate, Transform, Load

(D) Export, Transform, Log

✓ **Answer: (A) Extract, Transform, Load**

Q542. What is ELT?

(A) Extract, Load, Transform (in data lake)

(B) Extract, Load, Test

(C) Evaluate, Load, Transform

(D) Export, Load, Transform

✓ **Answer: (A) Extract, Load, Transform (in data lake)**

Q543. What is a data lake?

(A) Raw data storage in native format

(B) A data warehouse

(C) A relational database

(D) A dimensional model

✓ **Answer: (A) Raw data storage in native format**

Q544. What is database partitioning?

(A) Dividing a table into parts for performance

(B) Creating a new table

(C) Indexing a table

(D) Normalizing a table

✓ **Answer: (A) Dividing a table into parts for performance**

Q545. What is horizontal partitioning (sharding)?

(A) Splitting rows across partitions

(B) Splitting columns

(C) Creating views

(D) Indexing

✓ **Answer: (A) Splitting rows across partitions**

Q546. What is vertical partitioning?

(A) Splitting columns across tables

(B) Splitting rows

(C) Creating indexes

(D) Normalizing

✓ **Answer: (A) Splitting columns across tables**

Q547. What is range partitioning?

(A) Partitioning by range of values
(B) Partitioning by hash
(C) Partitioning by list
(D) Random partitioning

✓ Answer: (A) Partitioning by range of values

Q548. What is hash partitioning?

(A) Partitioning by hash of a key
(B) Partitioning by range
(C) Partitioning by list
(D) Random partitioning

✓ Answer: (A) Partitioning by hash of a key

Q549. What is list partitioning?

(A) Partitioning by explicit list of values
(B) Partitioning by range
(C) Partitioning by hash
(D) Random partitioning

✓ Answer: (A) Partitioning by explicit list of values

Q550. What is composite partitioning?

(A) Combination of partitioning methods
(B) Single method partitioning
(C) A view
(D) An index partition

✓ Answer: (A) Combination of partitioning methods

Q551. What is a partitioned index?

(A) Index built on a partitioned table
(B) An index with partitions
(C) A composite index
(D) A bitmap index

✓ Answer: (A) Index built on a partitioned table

Q552. What is table compression?

(A) Reducing storage by encoding data
(B) Adding indexes
(C) Normalizing tables
(D) Partitioning tables

✓ Answer: (A) Reducing storage by encoding data

Q553. What is page compression (SQL Server)?

(A) Row compression + prefix/dictionary compression
(B) Only row compression
(C) Only prefix compression
(D) A backup compression

✓ Answer: (A) Row compression + prefix/dictionary compression

Q554. What is columnar storage?

(A) Storing data column-by-column for analytics
(B) Row-based storage
(C) A partitioning method
(D) An indexing method

✓ Answer: (A) Storing data column-by-column for analytics

Q555. What is a columnstore index?

(A) Index that stores data in columnar format
(B) A B-Tree index
(C) A bitmap index
(D) A full-text index

✓ **Answer: (A) Index that stores data in columnar format**

Q556. What is a materialized view?

(A) A view whose result is stored physically
(B) A virtual view
(C) A CTE
(D) A derived table

✓ **Answer: (A) A view whose result is stored physically**

Q557. What is view refresh?

(A) Updating materialized view with new data
(B) Refreshing the query plan
(C) Rebuilding an index
(D) Updating statistics

✓ **Answer: (A) Updating materialized view with new data**

Q558. What is a virtual view?

(A) A stored query without physical data
(B) A materialized view
(C) A CTE
(D) A derived table

✓ **Answer: (A) A stored query without physical data**

Q559. Can a view be updated?

(A) Sometimes (simple views without aggregates)
(B) Always
(C) Never
(D) Only with INSTEAD OF trigger

✓ **Answer: (A) Sometimes (simple views without aggregates)**

Q560. What is an updatable view?

(A) A view that can be used in DML if it maps directly to a table
(B) A view with computed columns
(C) A view with GROUP BY
(D) A view with multiple tables

✓ **Answer: (A) A view that can be used in DML if it maps directly to a table**

Q561. What is a recursive relationship?

(A) An entity related to itself
(B) An entity with two keys
(C) A foreign key relationship
(D) A self-join

✓ **Answer: (A) An entity related to itself**

Q562. What is database metadata?

(A) Data about the structure of the database
(B) Actual data values
(C) Index data
(D) Query plans

✓ **Answer: (A) Data about the structure of the database**

Q563. What is the information schema?

(A) Standard set of views with database metadata
(B) A system table
(C) A schema type
(D) A database object

✓ **Answer: (A) Standard set of views with database metadata**

Q564. What is a system catalog?

(A) Database's internal metadata tables
(B) A user table
(C) A view
(D) An index

✓ **Answer: (A) Database's internal metadata tables**

Q565. What is data integrity?

(A) Accuracy and consistency of data in a database
(B) Data speed
(C) Data storage
(D) Data compression

✓ **Answer: (A) Accuracy and consistency of data in a database**

Q566. What are domain constraints?

(A) Limits on valid values for a column
(B) Foreign key constraints
(C) Table-level constraints
(D) Index constraints

✓ **Answer: (A) Limits on valid values for a column**

Q567. What is entity integrity?

(A) Primary key must be unique and not NULL
(B) Referential integrity
(C) Domain constraint
(D) Check constraint

✓ **Answer: (A) Primary key must be unique and not NULL**

Q568. What is a lookup table?

(A) Table with reference data (code/description pairs)
(B) A junction table
(C) A fact table
(D) A dimension table

✓ **Answer: (A) Table with reference data (code/description pairs)**

Q569. What is database replication?

(A) Copying data to one or more servers
(B) A backup method
(C) Log shipping
(D) A partitioning method

✓ **Answer: (A) Copying data to one or more servers**

Q570. What is master-slave (primary-replica) replication?

(A) One primary accepts writes, replicas read-only
(B) All servers equal
(C) A backup method
(D) A partitioning method

✓ **Answer: (A) One primary accepts writes, replicas read-only**

Section: Advanced SQL Topics (Q 571–700)

Q571. What is a MERGE statement?

(A) Performs INSERT, UPDATE, DELETE based on condition
(B) Merges two tables
(C) Combines SELECT results
(D) A union operation

✓ **Answer: (A) Performs INSERT, UPDATE, DELETE based on condition**

Q572. What is the UPSERT operation?

(A) Insert if not exists, update if exists
(B) Always insert
(C) Always update
(D) Delete and insert

✓ **Answer: (A) Insert if not exists, update if exists**

Q573. How to do UPSERT in PostgreSQL?

(A) INSERT ... ON CONFLICT DO UPDATE
(B) MERGE
(C) REPLACE INTO
(D) INSERT OR REPLACE

✓ **Answer: (A) INSERT ... ON CONFLICT DO UPDATE**

Q574. How to do UPSERT in MySQL?

(A) INSERT ... ON DUPLICATE KEY UPDATE or REPLACE INTO
(B) MERGE
(C) ON CONFLICT
(D) INSERT OR UPDATE

✓ **Answer: (A) INSERT ... ON DUPLICATE KEY UPDATE or REPLACE INTO**

Q575. What does COALESCE() do?

(A) Returns first non-NULL value from list
(B) Returns last NULL
(C) Returns NULL if any NULL
(D) Returns the sum

✓ **Answer: (A) Returns first non-NULL value from list**

Q576. What does NULLIF(a,b) do?

(A) Returns NULL if a=b, else returns a
(B) Returns a always
(C) Returns b always
(D) Returns NULL always

✓ **Answer: (A) Returns NULL if a=b, else returns a**

Q577. What does ISNULL(a,b) do (SQL Server)?

(A) Returns b if a is NULL, else a
(B) Returns NULL
(C) Returns a always
(D) Returns b always

✓ **Answer: (A) Returns b if a is NULL, else a**

Q578. What is the NVL function (Oracle)?

(A) Returns second argument if first is NULL
(B) Same as COALESCE
(C) Returns NULL
(D) Returns zero

✓ **Answer: (A) Returns second argument if first is NULL**

Q579. What does CASE expression do?

(A) Conditional expression returning different values
(B) A loop
(C) A stored procedure
(D) A trigger

✓ **Answer: (A) Conditional expression returning different values**

Q580. What are the two forms of CASE?

(A) Simple CASE and Searched CASE
(B) IF and CASE
(C) WHEN and THEN
(D) SWITCH and CASE

✓ **Answer: (A) Simple CASE and Searched CASE**

Q581. What is a simple CASE expression?

(A) Compares an expression to values
(B) Tests conditions
(C) A loop
(D) An IF statement

✓ **Answer: (A) Compares an expression to values**

Q582. What is a searched CASE expression?

(A) Tests boolean conditions
(B) Compares to values
(C) A loop
(D) An IF statement

✓ **Answer: (A) Tests boolean conditions**

Q583. What is DECODE() (Oracle)?

(A) Conditional expression similar to CASE
(B) A decoding function
(C) An encryption function
(D) A CAST function

✓ **Answer: (A) Conditional expression similar to CASE**

Q584. What does IIF() do (SQL Server)?

(A) Inline IF: returns one of two values based on condition
(B) A CASE expression
(C) A loop
(D) A stored procedure

✓ **Answer: (A) Inline IF: returns one of two values based on condition**

Q585. What is the PIVOT operator?

(A) Rotates rows to columns
(B) Rotates columns to rows
(C) A join type
(D) A filter

✓ **Answer: (A) Rotates rows to columns**

Q586. What is the UNPIVOT operator?

(A) Rotates columns to rows
(B) Rotates rows to columns
(C) A join type
(D) A filter

✓ **Answer: (A) Rotates columns to rows**

Q587. What is CROSS APPLY (SQL Server)?

(A) Like INNER JOIN with table-valued function
(B) A CROSS JOIN
(C) A LATERAL JOIN
(D) An OUTER JOIN

✓ **Answer: (A) Like INNER JOIN with table-valued function**

Q588. What is OUTER APPLY (SQL Server)?

(A) Like LEFT JOIN with table-valued function
(B) A CROSS JOIN
(C) A LATERAL JOIN
(D) An INNER JOIN

✓ **Answer: (A) Like LEFT JOIN with table-valued function**

Q589. What does FOR XML (SQL Server) do?

(A) Returns query results as XML
(B) Returns JSON
(C) A pivot operation
(D) A string function

✓ **Answer: (A) Returns query results as XML**

Q590. What does FOR JSON (SQL Server) do?

(A) Returns query results as JSON
(B) Returns XML
(C) A pivot operation
(D) A string function

✓ **Answer: (A) Returns query results as JSON**

Q591. What is JSON_VALUE() (SQL Server)?

(A) Extracts scalar value from JSON string
(B) Creates JSON
(C) Validates JSON
(D) Parses XML

✓ **Answer: (A) Extracts scalar value from JSON string**

Q592. What is JSON_QUERY() (SQL Server)?

(A) Extracts object or array from JSON
(B) Extracts scalar value
(C) Creates JSON
(D) Validates JSON

✓ **Answer: (A) Extracts object or array from JSON**

Q593. What does STRING_AGG() do?

(A) Concatenates values into a string with delimiter
(B) Splits a string
(C) Counts strings
(D) Aggregates integers

✓ **Answer: (A) Concatenates values into a string with delimiter**

Q594. What does GROUP_CONCAT() do (MySQL)?

(A) Concatenates grouped values with delimiter
(B) Groups values
(C) Counts groups
(D) Splits string

✓ **Answer: (A) Concatenates grouped values with delimiter**

Q595. What does LISTAGG() do (Oracle)?

(A) Concatenates grouped values with delimiter
(B) Counts groups
(C) Groups values
(D) Splits string

✓ **Answer: (A) Concatenates grouped values with delimiter**

Q596. What does ARRAY_AGG() do (PostgreSQL)?

(A) Aggregates values into an array
(B) Creates a string
(C) Counts values
(D) Groups values

✓ **Answer: (A) Aggregates values into an array**

Q597. What is the GENERATE_SERIES function (PostgreSQL)?

(A) Generates a series of numbers or dates
(B) Generates random numbers
(C) A recursive CTE
(D) A loop construct

✓ **Answer: (A) Generates a series of numbers or dates**

Q598. What is ROWNUM in Oracle?

(A) Pseudo-column for sequential row number before sorting
(B) A window function
(C) An index
(D) A partition key

✓ **Answer: (A) Pseudo-column for sequential row number before sorting**

Q599. What is ROW_NUMBER() advantage over ROWNUM?

(A) Works correctly with ORDER BY in same query
(B) They are identical
(C) ROWNUM is better
(D) Both are equivalent

✓ **Answer: (A) Works correctly with ORDER BY in same query**

Q600. What is FETCH FIRST n ROWS ONLY (SQL standard)?

(A) Limits result to n rows
(B) Orders rows
(C) Groups rows
(D) Filters rows

✓ **Answer: (A) Limits result to n rows**

Q601. What is QUALIFY clause (BigQuery/Snowflake)?

(A) Filters window function results like HAVING filters aggregates
(B) A WHERE clause
(C) A HAVING clause
(D) A GROUP BY clause

✓ **Answer: (A) Filters window function results like HAVING filters aggregates**

Q602. What is a lateral column alias?

(A) Using a SELECT alias in a later SELECT expression
(B) A table alias
(C) A column alias in WHERE
(D) An alias in ORDER BY

✓ **Answer: (A) Using a SELECT alias in a later SELECT expression**

Q603. What does TABLESAMPLE do?

(A) Returns a random sample of rows from a table
(B) Filters rows
(C) Orders rows
(D) Groups rows

✓ **Answer: (A) Returns a random sample of rows from a table**

Q604. What is a sparse column (SQL Server)?

(A) Optimized storage for columns with many NULLs
(B) A partial index
(C) A compressed column
(D) A virtual column

✓ **Answer: (A) Optimized storage for columns with many NULLs**

Q605. What is a computed/generated column?

(A) A column whose value is derived from other columns
(B) An indexed column
(C) A compressed column
(D) A sparse column

✓ **Answer: (A) A column whose value is derived from other columns**

Q606. What is a virtual generated column?

(A) Computed on-the-fly, not stored
(B) Stored generated column
(C) A sparse column
(D) A compressed column

✓ **Answer: (A) Computed on-the-fly, not stored**

Q607. What is a stored generated column?

(A) Pre-computed and physically stored
(B) Computed on-the-fly
(C) A sparse column
(D) A virtual column

✓ **Answer: (A) Pre-computed and physically stored**

Q608. What is sequence in SQL?

(A) An object generating sequential numbers
(B) An index
(C) An auto-increment column
(D) A random number generator

✓ **Answer: (A) An object generating sequential numbers**

Q609. What is IDENTITY column?

(A) Auto-incrementing column for primary keys
(B) A sequence

(C) An indexed column

(D) A computed column

✓ **Answer: (A) Auto-incrementing column for primary keys**

Q610. What is UUID?

(A) Universally Unique Identifier

(B) A sequence

(C) An auto-increment

(D) An identity column

✓ **Answer: (A) Universally Unique Identifier**

Q611. What is the RETURNING clause (PostgreSQL)?

(A) Returns modified rows from DML statements

(B) A SELECT clause

(C) A HAVING clause

(D) A filter clause

✓ **Answer: (A) Returns modified rows from DML statements**

Q612. What is OUTPUT clause (SQL Server)?

(A) Returns rows affected by DML statements

(B) A SELECT clause

(C) A HAVING clause

(D) A filter clause

✓ **Answer: (A) Returns rows affected by DML statements**

Q613. What is a CTE used for delete?

(A) WITH cte AS (...) DELETE FROM cte

(B) CTEs can't be used with DELETE

(C) Only in Oracle

(D) Only in PostgreSQL

✓ **Answer: (A) WITH cte AS (...) DELETE FROM cte**

Q614. What is a multi-table UPDATE?

(A) UPDATE with JOIN to update based on another table

(B) UPDATE with multiple WHERE conditions

(C) A stored procedure

(D) A trigger

✓ **Answer: (A) UPDATE with JOIN to update based on another table**

Q615. What is a writeable CTE?

(A) CTE used in DML statement target

(B) A CTE with INSERT

(C) A CTE with SELECT

(D) A CTE with GROUP BY

✓ **Answer: (A) CTE used in DML statement target**

Q616. What does TRUNCATE vs DELETE differ in?

(A) TRUNCATE is DDL, faster, no WHERE; DELETE is DML, slower, supports WHERE

(B) They are identical

(C) DELETE is faster

(D) TRUNCATE supports WHERE

✓ **Answer: (A) TRUNCATE is DDL, faster, no WHERE; DELETE is DML, slower, supports WHERE**

Q617. Can TRUNCATE be rolled back?

(A) Depends on database (yes in PostgreSQL, no in Oracle)
(B) Always yes
(C) Always no
(D) Only in SQL Server

✓ **Answer: (A) Depends on database (yes in PostgreSQL, no in Oracle)**

Q618. What is a foreign key with CASCADE DELETE?

(A) Deletes related child rows when parent is deleted
(B) Prevents deletion
(C) Sets child to NULL
(D) Does nothing

✓ **Answer: (A) Deletes related child rows when parent is deleted**

Q619. What is SET NULL on delete?

(A) Sets FK column to NULL when parent is deleted
(B) Deletes child rows
(C) Prevents deletion
(D) Sets to default

✓ **Answer: (A) Sets FK column to NULL when parent is deleted**

Q620. What is SET DEFAULT on delete?

(A) Sets FK column to default value when parent deleted
(B) Sets to NULL
(C) Deletes child
(D) Prevents deletion

✓ **Answer: (A) Sets FK column to default value when parent deleted**

Q621. What is ON DELETE RESTRICT?

(A) Prevents deletion if child rows exist
(B) Deletes child rows
(C) Sets to NULL
(D) Sets to default

✓ **Answer: (A) Prevents deletion if child rows exist**

Q622. What is ON DELETE NO ACTION?

(A) Similar to RESTRICT, error if child rows exist
(B) Deletes child rows
(C) Sets to NULL
(D) Sets to default

✓ **Answer: (A) Similar to RESTRICT, error if child rows exist**

Q623. What is deferred constraint checking?

(A) Constraint checked at COMMIT instead of per statement
(B) Immediate checking
(C) No constraint checking
(D) Per-row checking

✓ **Answer: (A) Constraint checked at COMMIT instead of per statement**

Q624. What is DEFERRABLE INITIALLY DEFERRED?

(A) Constraint deferred by default until COMMIT
(B) Checked immediately
(C) Not deferrable
(D) Checked per statement

✓ **Answer: (A) Constraint deferred by default until COMMIT**

Q625. What is DEFERRABLE INITIALLY IMMEDIATE?

(A) Constraint checked immediately by default but can be deferred
(B) Always deferred
(C) Not deferrable
(D) Always immediate

✓ **Answer: (A) Constraint checked immediately by default but can be deferred**

Q626. What is a domain in SQL?

(A) User-defined data type with constraints
(B) A schema
(C) A table space
(D) A constraint type

✓ **Answer: (A) User-defined data type with constraints**

Q627. What is a tablespace?

(A) Physical storage location for database objects
(B) A schema
(C) A database
(D) A partition

✓ **Answer: (A) Physical storage location for database objects**

Q628. What is a database link (Oracle)?

(A) Connection to another database
(B) A foreign key
(C) A replication link
(D) A view

✓ **Answer: (A) Connection to another database**

Q629. What is a synonym (Oracle/SQL Server)?

(A) Alias for a database object
(B) A view
(C) A stored procedure
(D) A function

✓ **Answer: (A) Alias for a database object**

Q630. What is a public synonym?

(A) Synonym available to all users
(B) Private synonym
(C) A view
(D) An alias

✓ **Answer: (A) Synonym available to all users**

Q631. What does EXPLAIN ANALYZE do (PostgreSQL)?

(A) Executes query and shows actual execution statistics
(B) Only shows plan
(C) Creates an index
(D) Optimizes the query

✓ **Answer: (A) Executes query and shows actual execution statistics**

Q632. What is a query plan cache?

(A) Cached execution plans for repeated queries
(B) A result cache
(C) An index cache
(D) A data cache

✓ **Answer: (A) Cached execution plans for repeated queries**

Q633. What is parameter sniffing?

(A) Optimizer uses initial parameter values to create plan
(B) A security feature
(C) A query optimization
(D) An index strategy

✓ **Answer: (A) Optimizer uses initial parameter values to create plan**

Q634. What is plan regression?

(A) Previously good plan becomes bad due to data changes
(B) A query error
(C) A missing index
(D) A deadlock

✓ **Answer: (A) Previously good plan becomes bad due to data changes**

Q635. What is adaptive query processing?

(A) Optimizer adjusts plan mid-execution based on actual data
(B) A static query plan
(C) A cached plan
(D) An indexed plan

✓ **Answer: (A) Optimizer adjusts plan mid-execution based on actual data**

Q636. What is batch mode processing?

(A) Processing multiple rows at once (columnar)
(B) Processing one row at a time
(C) A GROUP BY method
(D) A JOIN method

✓ **Answer: (A) Processing multiple rows at once (columnar)**

Q637. What is row mode processing?

(A) Processing one row at a time (traditional)
(B) Processing multiple rows
(C) A columnar method
(D) A batch method

✓ **Answer: (A) Processing one row at a time (traditional)**

Q638. What is parallel query execution?

(A) Query split across multiple CPU cores
(B) Sequential execution
(C) A join method
(D) An index method

✓ **Answer: (A) Query split across multiple CPU cores**

Q639. What is degree of parallelism (DOP)?

(A) Number of parallel threads for a query
(B) Number of indexes
(C) Number of rows
(D) Number of partitions

✓ **Answer: (A) Number of parallel threads for a query**

Q640. What is query store (SQL Server)?

(A) Captures query plans and performance data over time
(B) A result cache
(C) An index cache
(D) A data cache

✓ **Answer: (A) Captures query plans and performance data over time**

Q641. What is pg_stat_statements (PostgreSQL)?

(A) Extension tracking query execution statistics
(B) A system table
(C) A view
(D) A function

✓ **Answer: (A) Extension tracking query execution statistics**

Q642. What is a hot backup?

(A) Backup taken while database is running
(B) Backup while offline
(C) A fast backup
(D) An incremental backup

✓ **Answer: (A) Backup taken while database is running**

Q643. What is RMAN (Oracle)?

(A) Recovery Manager: Oracle backup and recovery tool
(B) A replication tool
(C) An ETL tool
(D) A query optimizer

✓ **Answer: (A) Recovery Manager: Oracle backup and recovery tool**

Q644. What is pg_dump (PostgreSQL)?

(A) Utility to back up a PostgreSQL database
(B) A query tool
(C) A monitoring tool
(D) A replication tool

✓ **Answer: (A) Utility to back up a PostgreSQL database**

Q645. What is mysqldump?

(A) Utility to back up a MySQL database
(B) A query tool
(C) A monitoring tool
(D) A replication tool

✓ **Answer: (A) Utility to back up a MySQL database**

Q646. What is SQL injection?

(A) Inserting malicious SQL into input fields
(B) A type of JOIN
(C) A query optimization
(D) A backup method

✓ **Answer: (A) Inserting malicious SQL into input fields**

Q647. How to prevent SQL injection?

(A) Parameterized queries / prepared statements
(B) Input validation only
(C) Escaping quotes only
(D) Using views

✓ **Answer: (A) Parameterized queries / prepared statements**

Q648. What is role-based access control (RBAC)?

(A) Permissions assigned to roles, roles to users
(B) Per-user permissions

(C) Column-level security
(D) Row-level security

✓ **Answer: (A) Permissions assigned to roles, roles to users**

Q649. What is row-level security (RLS)?

(A) Restricts rows visible to users based on policy
(B) Table-level security
(C) Column-level security
(D) Schema-level security

✓ **Answer: (A) Restricts rows visible to users based on policy**

Q650. What is column-level security?

(A) Restricts access to specific columns
(B) Row-level security
(C) Table-level security
(D) Schema security

✓ **Answer: (A) Restricts access to specific columns**

Q651. What is dynamic data masking?

(A) Hides sensitive data from unauthorized users without changing stored data
(B) Encrypts data
(C) Deletes data
(D) Compresses data

✓ **Answer: (A) Hides sensitive data from unauthorized users without changing stored data**

Q652. What is transparent data encryption (TDE)?

(A) Encrypts data files at rest
(B) Encrypts queries
(C) Masks data
(D) Compresses data

✓ **Answer: (A) Encrypts data files at rest**

Q653. What is SSL/TLS in database connections?

(A) Encrypts data in transit
(B) Encrypts data at rest
(C) Masks data
(D) Compresses data

✓ **Answer: (A) Encrypts data in transit**

Q654. What is an audit log?

(A) Record of database activities for compliance
(B) A query log
(C) An error log
(D) A backup log

✓ **Answer: (A) Record of database activities for compliance**

Q655. What is a database proxy?

(A) Intermediary between application and database
(B) A firewall
(C) A load balancer
(D) A cache layer

✓ **Answer: (A) Intermediary between application and database**

Q656. What is connection pooling?

(A) Reusing database connections for efficiency
(B) Creating new connections
(C) Closing connections
(D) Caching queries

✓ **Answer: (A) Reusing database connections for efficiency**

Q657. What is PgBouncer?

(A) PostgreSQL connection pooler
(B) A backup tool
(C) A replication tool
(D) A monitoring tool

✓ **Answer: (A) PostgreSQL connection pooler**

Q658. What is ProxySQL?

(A) MySQL proxy with connection pooling and routing
(B) A backup tool
(C) A replication tool
(D) A monitoring tool

✓ **Answer: (A) MySQL proxy with connection pooling and routing**

Q659. What is database sharding?

(A) Horizontal partitioning across multiple servers
(B) Vertical partitioning
(C) A replication method
(D) A backup method

✓ **Answer: (A) Horizontal partitioning across multiple servers**

Q660. What is consistent hashing?

(A) Distributing data across nodes minimizing rebalancing
(B) A hash index
(C) A join algorithm
(D) A sort algorithm

✓ **Answer: (A) Distributing data across nodes minimizing rebalancing**

Q661. What is a NoSQL database?

(A) Non-relational database for unstructured/semi-structured data
(B) A relational database
(C) A SQL database
(D) A normalized database

✓ **Answer: (A) Non-relational database for unstructured/semi-structured data**

Q662. What is NewSQL?

(A) Scalable database with ACID guarantees
(B) A NoSQL database
(C) A relational database only
(D) An in-memory database

✓ **Answer: (A) Scalable database with ACID guarantees**

Q663. What is an in-memory database?

(A) Database storing data primarily in RAM
(B) A disk-based database
(C) A columnar database
(D) A distributed database

✓ **Answer: (A) Database storing data primarily in RAM**

Q664. What is Redis?

(A) In-memory data structure store
(B) A relational database
(C) A document database
(D) A graph database

✓ **Answer: (A) In-memory data structure store**

Q665. What is a graph database?

(A) Stores data as nodes and edges
(B) A relational database
(C) A document database
(D) A key-value store

✓ **Answer: (A) Stores data as nodes and edges**

Q666. What is a document database?

(A) Stores semi-structured documents (JSON/BSON)
(B) A relational database
(C) A graph database
(D) A key-value store

✓ **Answer: (A) Stores semi-structured documents (JSON/BSON)**

Q667. What is MongoDB?

(A) A document-oriented NoSQL database
(B) A relational database
(C) A graph database
(D) An in-memory database

✓ **Answer: (A) A document-oriented NoSQL database**

Q668. What is CAP theorem?

(A) Consistency, Availability, Partition tolerance: can have at most 2
(B) ACID properties
(C) Normalization levels
(D) Isolation levels

✓ **Answer: (A) Consistency, Availability, Partition tolerance: can have at most 2**

Q669. What is eventual consistency?

(A) System will become consistent given enough time
(B) Immediate consistency
(C) Strong consistency
(D) ACID consistency

✓ **Answer: (A) System will become consistent given enough time**

Q670. What is a time-series database?

(A) Optimized for time-stamped data
(B) A relational database
(C) A document database
(D) A graph database

✓ **Answer: (A) Optimized for time-stamped data**

Q671. What is HTAP?

(A) Hybrid Transactional and Analytical Processing
(B) A backup type
(C) A replication method
(D) A join algorithm

✓ **Answer: (A) Hybrid Transactional and Analytical Processing**

Q672. What is a federated database?

(A) Multiple autonomous databases with unified access
(B) A single large database
(C) A replicated database
(D) A partitioned database

✓ **Answer: (A) Multiple autonomous databases with unified access**

Q673. What is SQL Server Always Encrypted?

(A) Encrypts data at column level, never decrypted on server
(B) TDE
(C) Dynamic data masking
(D) Row-level security

✓ **Answer: (A) Encrypts data at column level, never decrypted on server**

Q674. What is homomorphic encryption?

(A) Computing on encrypted data without decrypting
(B) Standard encryption
(C) TDE
(D) Column encryption

✓ **Answer: (A) Computing on encrypted data without decrypting**

Q675. What is data lineage?

(A) Tracking origin and transformation of data over time
(B) Data backup
(C) Data migration
(D) Data compression

✓ **Answer: (A) Tracking origin and transformation of data over time**

Q676. What is a data dictionary?

(A) Repository of metadata about database structure
(B) A lookup table
(C) A system catalog
(D) A view

✓ **Answer: (A) Repository of metadata about database structure**

Q677. What is schema evolution?

(A) Changing database schema without losing data
(B) Normalization
(C) Denormalization
(D) Schema creation

✓ **Answer: (A) Changing database schema without losing data**

Q678. What is a migration script?

(A) SQL script to change database schema version
(B) A backup script
(C) A replication script
(D) A monitoring script

✓ **Answer: (A) SQL script to change database schema version**

Q679. What is Flyway?

(A) Database migration tool
(B) A query optimizer
(C) A monitoring tool
(D) A backup tool

✓ **Answer: (A) Database migration tool**

Q680. What is Liquibase?

(A) Database change management tool
(B) A monitoring tool
(C) A backup tool
(D) A query optimizer

✓ **Answer: (A) Database change management tool**

Q681. What is ORM?

(A) Object-Relational Mapping: maps objects to tables
(B) A query language
(C) A database type
(D) A normalization technique

✓ **Answer: (A) Object-Relational Mapping: maps objects to tables**

Q682. What is Hibernate?

(A) A Java ORM framework
(B) A database
(C) A query language
(D) A backup tool

✓ **Answer: (A) A Java ORM framework**

Q683. What is SQLAlchemy?

(A) A Python SQL toolkit and ORM
(B) A database
(C) A query optimizer
(D) A backup tool

✓ **Answer: (A) A Python SQL toolkit and ORM**

Q684. What is JDBC?

(A) Java Database Connectivity: API for Java-database interaction
(B) A Python library
(C) A .NET library
(D) A query language

✓ **Answer: (A) Java Database Connectivity: API for Java-database interaction**

Q685. What is ODBC?

(A) Open Database Connectivity: standard API for databases
(B) A Java API
(C) A query language
(D) A database type

✓ **Answer: (A) Open Database Connectivity: standard API for databases**

Q686. What is ADO.NET?

(A) Microsoft .NET data access framework
(B) A Java framework
(C) A Python framework
(D) A query language

✓ **Answer: (A) Microsoft .NET data access framework**

Q687. What is a prepared statement?

(A) Pre-compiled SQL with parameters for security and performance
(B) A stored procedure

(C) A dynamic SQL
(D) A cursor

✓ **Answer: (A) Pre-compiled SQL with parameters for security and performance**

Q688. What is a bind variable?

(A) Parameter placeholder in SQL statement
(B) A view
(C) A synonym
(D) An index

✓ **Answer: (A) Parameter placeholder in SQL statement**

Q689. What is soft delete?

(A) Marking rows as deleted without removing them
(B) Physical deletion
(C) TRUNCATE
(D) Archiving

✓ **Answer: (A) Marking rows as deleted without removing them**

Q690. What is hard delete?

(A) Physically removing rows from a table
(B) Soft delete
(C) Archiving
(D) Marking as inactive

✓ **Answer: (A) Physically removing rows from a table**

Q691. What is data archiving?

(A) Moving old data to cheaper storage
(B) Deleting old data
(C) Compressing data
(D) Partitioning data

✓ **Answer: (A) Moving old data to cheaper storage**

Q692. What is temporal table (SQL Server 2016+)?

(A) Table with built-in history tracking for all changes
(B) A time-series table
(C) A partitioned table
(D) A view

✓ **Answer: (A) Table with built-in history tracking for all changes**

Q693. What is system-versioned temporal table?

(A) Database manages period columns for history
(B) Application-managed periods
(C) A partitioned table
(D) A materialized view

✓ **Answer: (A) Database manages period columns for history**

Q694. What is FOR SYSTEM_TIME AS OF?

(A) Query table as it appeared at a specific past time
(B) A current query
(C) A future query
(D) A snapshot query

✓ **Answer: (A) Query table as it appeared at a specific past time**

Q695. What is bi-temporal data?

(A) Data with both transaction time and valid time

(B) Data with one time dimension

(C) A temporal table

(D) A partitioned table

✓ **Answer: (A) Data with both transaction time and valid time**

Q696. What is valid time?

(A) When data was true in the real world

(B) When data was stored

(C) When data was deleted

(D) When data was archived

✓ **Answer: (A) When data was true in the real world**

Q697. What is transaction time?

(A) When data was stored in the database

(B) When data was true

(C) When data was deleted

(D) When data was archived

✓ **Answer: (A) When data was stored in the database**

Q698. What is a graph query language?

(A) A language for querying graph databases (e.g., Cypher, SPARQL)

(B) SQL for graphs

(C) A stored procedure language

(D) A scripting language

✓ **Answer: (A) A language for querying graph databases (e.g., Cypher, SPARQL)**

Q699. What is SQL:2023 standard?

(A) Latest SQL standard including graph queries

(B) SQL standard from 2019

(C) SQL standard from 2016

(D) SQL standard from 2011

✓ **Answer: (A) Latest SQL standard including graph queries**

Q700. What is ISO/IEC 9075?

(A) The ISO standard for SQL

(B) A security standard

(C) A networking standard

(D) A storage standard

✓ **Answer: (A) The ISO standard for SQL**

Section: MySQL Specific (Q 701–760)

Q701. Which storage engine is the default in MySQL 5.5+?

(A) InnoDB
(B) MyISAM
(C) MEMORY
(D) CSV

✓ **Answer: (A) InnoDB**

Q702. Which MySQL storage engine supports transactions?

(A) InnoDB
(B) MyISAM
(C) MEMORY
(D) ARCHIVE

✓ **Answer: (A) InnoDB**

Q703. Which MySQL storage engine is fastest for reads with no transactions?

(A) MyISAM
(B) InnoDB
(C) MEMORY
(D) CSV

✓ **Answer: (A) MyISAM**

Q704. What is the MySQL data directory?

(A) Directory where MySQL stores database files
(B) /var/www
(C) /etc/mysql
(D) /home/mysql

✓ **Answer: (A) Directory where MySQL stores database files**

Q705. What is AUTO_INCREMENT in MySQL?

(A) Automatically increments integer primary key
(B) A sequence
(C) A trigger
(D) An index

✓ **Answer: (A) Automatically increments integer primary key**

Q706. What is the SHOW TABLES command?

(A) Lists all tables in current database
(B) Shows table structure
(C) Shows indexes
(D) Shows create statement

✓ **Answer: (A) Lists all tables in current database**

Q707. What is DESCRIBE (DESC) in MySQL?

(A) Shows column structure of a table
(B) Shows all tables
(C) Shows indexes
(D) Shows create statement

✓ **Answer: (A) Shows column structure of a table**

Q708. What is SHOW CREATE TABLE?

(A) Shows the CREATE TABLE statement for a table
(B) Shows table data
(C) Shows indexes
(D) Shows column info

✓ Answer: (A) Shows the CREATE TABLE statement for a table

Q709. What is SHOW INDEX FROM table?

(A) Shows all indexes on a table
(B) Shows table structure
(C) Shows foreign keys
(D) Shows constraints

✓ Answer: (A) Shows all indexes on a table

Q710. What is the difference between CHAR and VARCHAR in MySQL?

(A) CHAR is fixed length, VARCHAR is variable
(B) Both are variable length
(C) Both are fixed length
(D) CHAR is variable, VARCHAR is fixed

✓ Answer: (A) CHAR is fixed length, VARCHAR is variable

Q711. What is the TEXT data type in MySQL?

(A) Variable-length string up to 65,535 bytes
(B) Fixed-length string
(C) A BLOB type
(D) A numeric type

✓ Answer: (A) Variable-length string up to 65,535 bytes

Q712. What is the BLOB data type?

(A) Binary Large Object for binary data
(B) A text type
(C) A numeric type
(D) A date type

✓ Answer: (A) Binary Large Object for binary data

Q713. What is JSON data type in MySQL?

(A) Native JSON data type (MySQL 5.7+)
(B) A VARCHAR type
(C) A TEXT type
(D) A BLOB type

✓ Answer: (A) Native JSON data type (MySQL 5.7+)

Q714. What does ENUM do in MySQL?

(A) Restricts column to predefined string values
(B) A numeric type
(C) A boolean type
(D) A set type

✓ Answer: (A) Restricts column to predefined string values

Q715. What does SET do in MySQL?

(A) Allows multiple values from predefined list
(B) A single-value ENUM
(C) A numeric type
(D) A boolean type

✓ Answer: (A) Allows multiple values from predefined list

Q716. What is GROUP_CONCAT in MySQL?

(A) Concatenates values within a group
(B) Groups values
(C) Counts groups
(D) A window function

✓ **Answer: (A) Concatenates values within a group**

Q717. What is FIND_IN_SET in MySQL?

(A) Searches for a value in a comma-separated string
(B) Finds in JSON
(C) Finds in array
(D) A search function

✓ **Answer: (A) Searches for a value in a comma-separated string**

Q718. What does ON DUPLICATE KEY UPDATE do?

(A) Updates existing row on duplicate key conflict
(B) Inserts always
(C) Deletes duplicate
(D) Errors on duplicate

✓ **Answer: (A) Updates existing row on duplicate key conflict**

Q719. What is REPLACE INTO in MySQL?

(A) Deletes existing row and inserts new one on key conflict
(B) An UPDATE
(C) A safe INSERT
(D) A MERGE

✓ **Answer: (A) Deletes existing row and inserts new one on key conflict**

Q720. What is the difference between REPLACE INTO and ON DUPLICATE KEY UPDATE?

(A) REPLACE deletes+inserts; ON DUPLICATE updates only
(B) They are identical
(C) REPLACE updates only
(D) ON DUPLICATE deletes+inserts

✓ **Answer: (A) REPLACE deletes+inserts; ON DUPLICATE updates only**

Q721. What is MySQL's LIMIT syntax?

(A) LIMIT count or LIMIT offset, count
(B) LIMIT OFFSET
(C) TOP n
(D) FETCH FIRST

✓ **Answer: (A) LIMIT count or LIMIT offset, count**

Q722. What is MySQL's equivalent of SQL Server's TOP?

(A) LIMIT
(B) TOP
(C) FETCH
(D) FIRST

✓ **Answer: (A) LIMIT**

Q723. What is IFNULL() in MySQL?

(A) Returns second arg if first is NULL
(B) Same as COALESCE
(C) Returns NULL
(D) Returns zero

✓ **Answer: (A) Returns second arg if first is NULL**

Q724. What is IF() in MySQL?

(A) IF(condition, true_val, false_val)
(B) A stored procedure
(C) A CASE expression
(D) A trigger

✓ **Answer: (A) IF(condition, true_val, false_val)**

Q725. What is the DUAL table in MySQL?

(A) A dummy table for SELECT without FROM
(B) A system table
(C) A temporary table
(D) A view

✓ **Answer: (A) A dummy table for SELECT without FROM**

Q726. What is MySQL's syntax for string concatenation?

(A) CONCAT() function
(B) || operator
(C) + operator
(D) & operator

✓ **Answer: (A) CONCAT() function**

Q727. What is MySQL's default character set?

(A) utf8mb4 (MySQL 8.0+)
(B) latin1
(C) utf8
(D) ascii

✓ **Answer: (A) utf8mb4 (MySQL 8.0+)**

Q728. What is the difference between utf8 and utf8mb4 in MySQL?

(A) utf8mb4 supports full Unicode including emoji; utf8 is 3-byte only
(B) They are identical
(C) utf8 supports more characters
(D) utf8mb4 is faster

✓ **Answer: (A) utf8mb4 supports full Unicode including emoji; utf8 is 3-byte only**

Q729. What does STRICT_TRANS_TABLES mode do?

(A) Enforces strict data validation, rejects invalid data
(B) Allows invalid data
(C) A transaction mode
(D) An isolation level

✓ **Answer: (A) Enforces strict data validation, rejects invalid data**

Q730. What is MySQL's information_schema?

(A) Virtual database with metadata about all databases
(B) A user database
(C) A system database
(D) A temp database

✓ **Answer: (A) Virtual database with metadata about all databases**

Q731. What is MySQL's performance_schema?

(A) Database for performance monitoring
(B) A metadata database
(C) A user database
(D) A temp database

✓ **Answer: (A) Database for performance monitoring**

Q732. What is MySQL's sys schema?

(A) Views simplifying performance_schema data
(B) A user schema
(C) A metadata schema
(D) A temp schema

✓ **Answer: (A) Views simplifying performance_schema data**

Q733. What is binlog in MySQL?

(A) Binary log recording all changes for replication/recovery
(B) An error log
(C) A query log
(D) An audit log

✓ **Answer: (A) Binary log recording all changes for replication/recovery**

Q734. What is MySQL slow query log?

(A) Log of queries exceeding long_query_time threshold
(B) An error log
(C) A binary log
(D) An audit log

✓ **Answer: (A) Log of queries exceeding long_query_time threshold**

Q735. What is MySQL's EXPLAIN FORMAT=JSON?

(A) Shows execution plan in JSON format
(B) Shows plan as text
(C) A backup command
(D) A query hint

✓ **Answer: (A) Shows execution plan in JSON format**

Q736. What is MySQL's query_cache (deprecated in 8.0)?

(A) Cache for SELECT results
(B) A plan cache
(C) An index cache
(D) A data cache

✓ **Answer: (A) Cache for SELECT results**

Q737. What is MySQL's InnoDB buffer pool?

(A) In-memory cache for InnoDB data and indexes
(B) A query cache
(C) A plan cache
(D) A log buffer

✓ **Answer: (A) In-memory cache for InnoDB data and indexes**

Q738. What does innodb_buffer_pool_size control?

(A) Size of InnoDB buffer pool
(B) Log file size
(C) Sort buffer size
(D) Join buffer size

✓ **Answer: (A) Size of InnoDB buffer pool**

Q739. What is MySQL replication lag?

(A) Delay between primary and replica
(B) A deadlock

(C) A slow query

(D) A lock timeout

✓ **Answer: (A) Delay between primary and replica**

Q740. What is GTID in MySQL replication?

(A) Global Transaction Identifier for tracking transactions

(B) A table ID

(C) A row ID

(D) A session ID

✓ **Answer: (A) Global Transaction Identifier for tracking transactions**

Q741. What is MySQL Group Replication?

(A) Multi-primary replication with automatic failover

(B) Single-primary replication

(C) Log shipping

(D) A backup method

✓ **Answer: (A) Multi-primary replication with automatic failover**

Q742. What is Galera Cluster?

(A) Synchronous multi-master MySQL clustering

(B) An async replication

(C) A backup solution

(D) A monitoring tool

✓ **Answer: (A) Synchronous multi-master MySQL clustering**

Q743. What is ProxySQL used for?

(A) MySQL proxy with load balancing and query routing

(B) A backup tool

(C) A monitoring tool

(D) A replication tool

✓ **Answer: (A) MySQL proxy with load balancing and query routing**

Q744. What is MySQL Workbench?

(A) Graphical tool for MySQL design, admin, query

(B) A command-line tool

(C) A backup tool

(D) A replication tool

✓ **Answer: (A) Graphical tool for MySQL design, admin, query**

Q745. What is mysqldump used for?

(A) Logical backup of MySQL databases

(B) Physical backup

(C) Replication

(D) Monitoring

✓ **Answer: (A) Logical backup of MySQL databases**

Q746. What is mysqlpump?

(A) Parallel mysqldump replacement

(B) A monitoring tool

(C) A replication tool

(D) A query optimizer

✓ **Answer: (A) Parallel mysqldump replacement**

Q747. What is MySQL Shell?

(A) Advanced client with scripting support
(B) A monitoring tool
(C) A backup tool
(D) A GUI tool

✓ **Answer: (A) Advanced client with scripting support**

Q748. What is MySQL Router?

(A) Routing middleware for MySQL InnoDB Cluster
(B) A proxy
(C) A backup tool
(D) A monitoring tool

✓ **Answer: (A) Routing middleware for MySQL InnoDB Cluster**

Q749. What is InnoDB Cluster?

(A) High-availability solution with Group Replication + Router + Shell
(B) A single-server setup
(C) A backup solution
(D) A monitoring tool

✓ **Answer: (A) High-availability solution with Group Replication + Router + Shell**

Q750. What does ANALYZE TABLE do in MySQL?

(A) Updates table statistics for optimizer
(B) Rebuilds indexes
(C) Defragments table
(D) Backs up table

✓ **Answer: (A) Updates table statistics for optimizer**

Q751. What does OPTIMIZE TABLE do in MySQL?

(A) Defragments table and reclaims space
(B) Updates statistics
(C) Rebuilds indexes
(D) Analyzes table

✓ **Answer: (A) Defragments table and reclaims space**

Q752. What does CHECK TABLE do in MySQL?

(A) Checks table for corruption
(B) Defragments table
(C) Updates statistics
(D) Analyzes table

✓ **Answer: (A) Checks table for corruption**

Q753. What does REPAIR TABLE do in MySQL?

(A) Repairs a corrupted MyISAM table
(B) Defragments table
(C) Updates statistics
(D) Analyzes table

✓ **Answer: (A) Repairs a corrupted MyISAM table**

Q754. What is the default port for MySQL?

(A) 3306
(B) 5432
(C) 1433
(D) 27017

✓ **Answer: (A) 3306**

Q755. What is MySQL's WITH ROLLUP modifier?

(A) Adds super-aggregate rows to GROUP BY result
(B) A CTE
(C) A recursive query
(D) A pivot operation

✓ **Answer: (A) Adds super-aggregate rows to GROUP BY result**

Q756. What is MySQL 8.0 window function support?

(A) Full window function support added in 8.0
(B) Not supported
(C) Partial support
(D) Only ROW_NUMBER

✓ **Answer: (A) Full window function support added in 8.0**

Q757. What is MySQL's LATERAL keyword (8.0)?

(A) Allows subquery to reference preceding FROM tables
(B) A JOIN type
(C) A filter
(D) A sort

✓ **Answer: (A) Allows subquery to reference preceding FROM tables**

Q758. What is invisible index in MySQL 8.0?

(A) Index ignored by optimizer unless hinted; useful for testing
(B) A dropped index
(C) A disabled index
(D) A partial index

✓ **Answer: (A) Index ignored by optimizer unless hinted; useful for testing**

Q759. What is descending index in MySQL 8.0?

(A) Index sorted in descending order natively
(B) A reverse key index
(C) A partial index
(D) A hash index

✓ **Answer: (A) Index sorted in descending order natively**

Q760. What is MySQL's generated column?

(A) Column computed from other columns
(B) An identity column
(C) A sequence column
(D) A sparse column

✓ **Answer: (A) Column computed from other columns**

Section: PostgreSQL Specific (Q 761–820)

Q761. What is PostgreSQL's equivalent of AUTO_INCREMENT?

(A) SERIAL / GENERATED AS IDENTITY
(B) AUTO_INCREMENT
(C) SEQUENCE only
(D) IDENTITY

✓ **Answer: (A) SERIAL / GENERATED AS IDENTITY**

Q762. What is the default port for PostgreSQL?

(A) 5432
(B) 3306
(C) 1433
(D) 27017

✓ **Answer: (A) 5432**

Q763. What is psql?

(A) PostgreSQL command-line interface
(B) A GUI tool
(C) A backup tool
(D) A monitoring tool

✓ **Answer: (A) PostgreSQL command-line interface**

Q764. What is pg_dump?

(A) Utility to back up a PostgreSQL database
(B) A query tool
(C) A monitoring tool
(D) A replication tool

✓ **Answer: (A) Utility to back up a PostgreSQL database**

Q765. What is pg_restore?

(A) Restores a PostgreSQL backup created by pg_dump
(B) A backup tool
(C) A monitoring tool
(D) A replication tool

✓ **Answer: (A) Restores a PostgreSQL backup created by pg_dump**

Q766. What is PostgreSQL's TOAST?

(A) The Oversized-Attribute Storage Technique for large values
(B) A caching mechanism
(C) An index type
(D) A replication feature

✓ **Answer: (A) The Oversized-Attribute Storage Technique for large values**

Q767. What is PostgreSQL's VACUUM?

(A) Reclaims storage from dead tuples
(B) Defragments table
(C) Updates statistics
(D) Rebuilds indexes

✓ **Answer: (A) Reclaims storage from dead tuples**

Q768. What is AUTOVACUUM in PostgreSQL?

(A) Automatic background VACUUM process
(B) A manual vacuum
(C) A backup process
(D) A replication process

✓ **Answer: (A) Automatic background VACUUM process**

Q769. What is MVCC in PostgreSQL?

(A) Multi-Version Concurrency Control using row versions
(B) A locking mechanism
(C) A backup method
(D) A replication method

✓ **Answer: (A) Multi-Version Concurrency Control using row versions**

Q770. What is a dead tuple in PostgreSQL?

(A) An old row version no longer visible to any transaction
(B) A corrupted row
(C) A NULL row
(D) A deleted index entry

✓ **Answer: (A) An old row version no longer visible to any transaction**

Q771. What is bloat in PostgreSQL?

(A) Accumulated dead tuples increasing table/index size
(B) Fragmentation
(C) Corruption
(D) Compression

✓ **Answer: (A) Accumulated dead tuples increasing table/index size**

Q772. What is pg_stat_activity?

(A) View showing current database connections and queries
(B) A backup view
(C) A replication view
(D) An index view

✓ **Answer: (A) View showing current database connections and queries**

Q773. What is pg_stat_user_tables?

(A) View showing table statistics
(B) An index view
(C) A query view
(D) A connection view

✓ **Answer: (A) View showing table statistics**

Q774. What is EXPLAIN ANALYZE in PostgreSQL?

(A) Executes query and shows actual vs estimated statistics
(B) Shows plan only
(C) Creates index
(D) Optimizes query

✓ **Answer: (A) Executes query and shows actual vs estimated statistics**

Q775. What is PostgreSQL's LISTEN/NOTIFY?

(A) Asynchronous notification between sessions
(B) A replication feature
(C) A trigger type
(D) A backup feature

✓ **Answer: (A) Asynchronous notification between sessions**

Q776. What are PostgreSQL extensions?

(A) Add-on modules extending PostgreSQL functionality
(B) Built-in functions
(C) System views
(D) Schemas

✓ **Answer: (A) Add-on modules extending PostgreSQL functionality**

Q777. What is PostGIS?

(A) PostgreSQL extension for geospatial data
(B) A monitoring tool
(C) A backup tool
(D) A replication tool

✓ **Answer: (A) PostgreSQL extension for geospatial data**

Q778. What is pg_trgm?

(A) PostgreSQL extension for trigram-based similarity search
(B) A full-text search extension
(C) A monitoring extension
(D) A backup extension

✓ **Answer: (A) PostgreSQL extension for trigram-based similarity search**

Q779. What is the hstore extension?

(A) Key-value store within a PostgreSQL column
(B) A JSON extension
(C) A full-text extension
(D) A monitoring extension

✓ **Answer: (A) Key-value store within a PostgreSQL column**

Q780. What is PostgreSQL's JSONB?

(A) Binary JSON format for efficient querying
(B) Plain JSON text
(C) A BLOB type
(D) A TEXT type

✓ **Answer: (A) Binary JSON format for efficient querying**

Q781. What is the difference between JSON and JSONB in PostgreSQL?

(A) JSON stores text as-is; JSONB parses and stores binary, faster for queries
(B) They are identical
(C) JSON is faster
(D) JSONB is larger

✓ **Answer: (A) JSON stores text as-is; JSONB parses and stores binary, faster for queries**

Q782. What is GIN index?

(A) Generalized Inverted Index: for arrays, JSON, full-text
(B) A B-Tree index
(C) A hash index
(D) A BRIN index

✓ **Answer: (A) Generalized Inverted Index: for arrays, JSON, full-text**

Q783. What is GiST index?

(A) Generalized Search Tree: for geometric and custom types
(B) A B-Tree index
(C) A hash index
(D) A GIN index

✓ **Answer: (A) Generalized Search Tree: for geometric and custom types**

Q784. What is BRIN index?

(A) Block Range Index: for naturally ordered large tables
(B) A B-Tree index
(C) A GIN index
(D) A GiST index

✓ **Answer: (A) Block Range Index: for naturally ordered large tables**

Q785. What is PostgreSQL's COPY command?

(A) Bulk load/export data to/from files
(B) A backup command
(C) A replication command
(D) A query command

✓ **Answer: (A) Bulk load/export data to/from files**

Q786. What is PostgreSQL's foreign data wrapper (FDW)?

(A) Allows querying external data sources
(B) A foreign key
(C) A view
(D) An extension

✓ **Answer: (A) Allows querying external data sources**

Q787. What is postgres_fdw?

(A) FDW to query other PostgreSQL servers
(B) A local query
(C) A backup tool
(D) A monitoring tool

✓ **Answer: (A) FDW to query other PostgreSQL servers**

Q788. What is table inheritance in PostgreSQL?

(A) A table can inherit columns from a parent table
(B) A normalization technique
(C) A partitioning method
(D) A replication method

✓ **Answer: (A) A table can inherit columns from a parent table**

Q789. What is declarative partitioning in PostgreSQL (10+)?

(A) Built-in table partitioning using PARTITION BY
(B) External partitioning
(C) Manual partitioning
(D) A trigger-based partitioning

✓ **Answer: (A) Built-in table partitioning using PARTITION BY**

Q790. What is PostgreSQL's logical replication?

(A) Replicates data changes based on logical level
(B) Physical replication
(C) Streaming replication
(D) A backup method

✓ **Answer: (A) Replicates data changes based on logical level**

Q791. What is streaming replication in PostgreSQL?

(A) Real-time WAL streaming to standby servers
(B) Logical replication
(C) A backup method
(D) Log shipping

✓ **Answer: (A) Real-time WAL streaming to standby servers**

Q792. What is Patroni?

(A) High-availability PostgreSQL template using DCS
(B) A backup tool
(C) A monitoring tool
(D) A query optimizer

✓ **Answer: (A) High-availability PostgreSQL template using DCS**

Q793. What is PgBouncer?

(A) PostgreSQL connection pooler
(B) A backup tool
(C) A monitoring tool
(D) A replication tool

✓ **Answer: (A) PostgreSQL connection pooler**

Q794. What is pgAdmin?

(A) GUI administration tool for PostgreSQL
(B) A command-line tool
(C) A backup tool
(D) A replication tool

✓ **Answer: (A) GUI administration tool for PostgreSQL**

Q795. What is PostgreSQL's COALESCE alternative?

(A) COALESCE is standard SQL, also NVL is not in PG; use COALESCE
(B) NVL()
(C) ISNULL()
(D) IIF()

✓ **Answer: (A) COALESCE is standard SQL, also NVL is not in PG; use COALESCE**

Q796. What is PostgreSQL's string concatenation operator?

(A) || operator
(B) + operator
(C) CONCAT() only
(D) & operator

✓ **Answer: (A) || operator**

Q797. What is ILIKE in PostgreSQL?

(A) Case-insensitive LIKE
(B) Case-sensitive LIKE
(C) A regular expression match
(D) A full-text search

✓ **Answer: (A) Case-insensitive LIKE**

Q798. What is ~ operator in PostgreSQL?

(A) Regular expression match (case-sensitive)
(B) A LIKE operator
(C) A comparison operator
(D) A bitwise operator

✓ **Answer: (A) Regular expression match (case-sensitive)**

Q799. What is PostgreSQL's ARRAY type?

(A) Native array data type
(B) A JSON type

(C) A BLOB type

(D) A TEXT type

✓ **Answer: (A) Native array data type**

Q800. What is unnest() in PostgreSQL?

(A) Expands an array to a set of rows

(B) Collapses rows to array

(C) A PIVOT function

(D) A string function

✓ **Answer: (A) Expands an array to a set of rows**

Q801. What is PostgreSQL's WITH RECURSIVE syntax?

(A) Standard recursive CTE syntax

(B) A MySQL syntax

(C) A SQL Server syntax

(D) An Oracle syntax

✓ **Answer: (A) Standard recursive CTE syntax**

Q802. What is row constructor in PostgreSQL?

(A) (val1, val2, ...) syntax for comparing multiple columns

(B) A tuple type

(C) A composite type

(D) An array type

✓ **Answer: (A) (val1, val2, ...) syntax for comparing multiple columns**

Q803. What is a composite type in PostgreSQL?

(A) User-defined type with multiple fields

(B) An array type

(C) A JSON type

(D) A BLOB type

✓ **Answer: (A) User-defined type with multiple fields**

Q804. What is PostgreSQL's RETURNING clause?

(A) Returns rows affected by DML

(B) A SELECT clause

(C) A HAVING clause

(D) A filter clause

✓ **Answer: (A) Returns rows affected by DML**

Q805. What is PostgreSQL's DO block?

(A) Anonymous PL/pgSQL block for one-time execution

(B) A stored procedure

(C) A function

(D) A trigger

✓ **Answer: (A) Anonymous PL/pgSQL block for one-time execution**

Q806. What is PL/pgSQL?

(A) PostgreSQL's procedural language for stored procedures and functions

(B) A query language

(C) A scripting language

(D) A markup language

✓ **Answer: (A) PostgreSQL's procedural language for stored procedures and functions**

Q807. What is PostgreSQL's range type?

(A) Data type representing a range of values
(B) An array type
(C) A JSON type
(D) A date type

✓ Answer: (A) Data type representing a range of values

Q808. What is PostgreSQL's tsrange?

(A) Range of timestamp values
(B) A text range
(C) A numeric range
(D) A date range

✓ Answer: (A) Range of timestamp values

Q809. What is GIN index used for in PostgreSQL?

(A) Full-text search, arrays, JSONB
(B) B-Tree replacement
(C) Geographic data
(D) Low-cardinality data

✓ Answer: (A) Full-text search, arrays, JSONB

Q810. What is a sequence in PostgreSQL?

(A) Object generating sequential numbers
(B) An AUTO_INCREMENT
(C) An IDENTITY column
(D) An index

✓ Answer: (A) Object generating sequential numbers

Q811. What is NEXTVAL() in PostgreSQL?

(A) Gets next value from a sequence
(B) Gets current value
(C) Resets sequence
(D) Creates sequence

✓ Answer: (A) Gets next value from a sequence

Q812. What is CURRVAL() in PostgreSQL?

(A) Gets last value returned by sequence in current session
(B) Gets next value
(C) Resets sequence
(D) Gets min value

✓ Answer: (A) Gets last value returned by sequence in current session

Q813. What is SETVAL() in PostgreSQL?

(A) Sets sequence to a specific value
(B) Gets current value
(C) Gets next value
(D) Resets sequence

✓ Answer: (A) Sets sequence to a specific value

Q814. What is pg_class in PostgreSQL?

(A) System catalog table with information about tables and indexes
(B) A user table
(C) A view
(D) A function

✓ Answer: (A) System catalog table with information about tables and indexes

Q815. What is pg_attribute?

(A) System catalog with column information
(B) A user table
(C) A view
(D) A function

✓ **Answer: (A) System catalog with column information**

Q816. What is pg_constraint?

(A) System catalog with constraint information
(B) A user table
(C) A view
(D) A function

✓ **Answer: (A) System catalog with constraint information**

Q817. What is PostgreSQL's TABLEFUNC extension?

(A) Provides CROSSTAB for pivot operations
(B) A monitoring extension
(C) A backup extension
(D) A replication extension

✓ **Answer: (A) Provides CROSSTAB for pivot operations**

Q818. What is CROSSTAB in PostgreSQL?

(A) Function to pivot rows to columns
(B) A join type
(C) A window function
(D) A set operation

✓ **Answer: (A) Function to pivot rows to columns**

Q819. What is PostgreSQL's CUBE extension in GROUP BY?

(A) Generates all combinations of GROUP BY columns
(B) A pivot operation
(C) A recursive CTE
(D) A window function

✓ **Answer: (A) Generates all combinations of GROUP BY columns**

Q820. What is GROUPING SETS?

(A) Allows multiple GROUP BY specifications in one query
(B) A window function
(C) A pivot operation
(D) A set operation

✓ **Answer: (A) Allows multiple GROUP BY specifications in one query**

Section: SQL Server Specific (Q 821–880)

Q821. What is the default port for SQL Server?

(A) 1433
(B) 3306
(C) 5432
(D) 27017

✓ **Answer: (A) 1433**

Q822. What is SQL Server Management Studio (SSMS)?

(A) GUI tool for managing SQL Server
(B) A command-line tool
(C) A backup tool
(D) A replication tool

✓ **Answer: (A) GUI tool for managing SQL Server**

Q823. What is T-SQL?

(A) Transact-SQL: Microsoft's SQL Server dialect
(B) Standard SQL
(C) Oracle's PL/SQL
(D) MySQL's SQL

✓ **Answer: (A) Transact-SQL: Microsoft's SQL Server dialect**

Q824. What is the IDENTITY property in SQL Server?

(A) Auto-incrementing column
(B) A sequence
(C) A trigger
(D) An index

✓ **Answer: (A) Auto-incrementing column**

Q825. What is @@IDENTITY?

(A) Last identity value inserted in current session
(B) Last identity globally
(C) A sequence
(D) A system variable

✓ **Answer: (A) Last identity value inserted in current session**

Q826. What is SCOPE_IDENTITY()?

(A) Last identity inserted in current scope
(B) Last identity globally
(C) @@IDENTITY
(D) IDENT_CURRENT

✓ **Answer: (A) Last identity inserted in current scope**

Q827. What is IDENT_CURRENT()?

(A) Last identity for a specific table regardless of session
(B) Current session identity
(C) Current scope identity
(D) A sequence

✓ **Answer: (A) Last identity for a specific table regardless of session**

Q828. What is SQL Server's TOP clause?

(A) Limits rows returned
(B) Groups rows
(C) Filters rows
(D) Sorts rows

✓ **Answer: (A) Limits rows returned**

Q829. What is SQL Server's TOP WITH TIES?

(A) Includes rows tied with the last row
(B) Limits to exact count
(C) Adds extra rows
(D) A window function

✓ **Answer: (A) Includes rows tied with the last row**

Q830. What is SQL Server's NOCOUNT option?

(A) Suppresses 'rows affected' messages
(B) Counts rows
(C) A query hint
(D) A lock hint

✓ **Answer: (A) Suppresses 'rows affected' messages**

Q831. What is SQL Server's SET NOCOUNT ON?

(A) Disables row count messages for performance
(B) Enables row counting
(C) A transaction setting
(D) A lock setting

✓ **Answer: (A) Disables row count messages for performance**

Q832. What is SQL Server's TRY...CATCH?

(A) Error handling block
(B) A loop construct
(C) A cursor
(D) A transaction block

✓ **Answer: (A) Error handling block**

Q833. What is THROW in SQL Server?

(A) Raises an error (SQL Server 2012+)
(B) Logs an error
(C) Ends a procedure
(D) Starts a transaction

✓ **Answer: (A) Raises an error (SQL Server 2012+)**

Q834. What is RAISERROR in SQL Server?

(A) Generates an error message (older syntax)
(B) Logs an error
(C) Ends a procedure
(D) Raises event

✓ **Answer: (A) Generates an error message (older syntax)**

Q835. What is SQL Server's PRINT statement?

(A) Outputs a message to client
(B) Logs to file
(C) A debugging breakpoint
(D) A trace flag

✓ **Answer: (A) Outputs a message to client**

Q836. What is SQL Server's dynamic management views (DMVs)?

(A) System views exposing server internal state
(B) User views
(C) System tables
(D) Metadata tables

✓ **Answer: (A) System views exposing server internal state**

Q837. What is sys.dm_exec_requests?

(A) DMV showing currently executing requests
(B) A user table
(C) An index view
(D) A statistics view

✓ **Answer: (A) DMV showing currently executing requests**

Q838. What is sys.dm_exec_query_plan?

(A) DMV returning query execution plan
(B) A statistics view
(C) A lock view
(D) A connection view

✓ **Answer: (A) DMV returning query execution plan**

Q839. What is SQL Server Profiler?

(A) Tool for tracing SQL Server events
(B) A query optimizer
(C) A backup tool
(D) A monitoring tool

✓ **Answer: (A) Tool for tracing SQL Server events**

Q840. What replaced SQL Server Profiler?

(A) Extended Events (XE)
(B) SQL Trace
(C) DMVs
(D) Query Store

✓ **Answer: (A) Extended Events (XE)**

Q841. What is SQL Server Extended Events?

(A) Lightweight event tracing system
(B) The old Profiler
(C) A DMV
(D) A query hint

✓ **Answer: (A) Lightweight event tracing system**

Q842. What is SQL Server Agent?

(A) Job scheduling service
(B) A replication tool
(C) A backup tool
(D) A monitoring tool

✓ **Answer: (A) Job scheduling service**

Q843. What is SQL Server Integration Services (SSIS)?

(A) ETL platform for data integration
(B) A reporting tool
(C) A backup tool
(D) A replication tool

✓ **Answer: (A) ETL platform for data integration**

Q844. What is SQL Server Reporting Services (SSRS)?

(A) Platform for creating and delivering reports
(B) An ETL tool
(C) A backup tool
(D) A replication tool

✓ **Answer: (A) Platform for creating and delivering reports**

Q845. What is SQL Server Analysis Services (SSAS)?

(A) OLAP and data mining platform
(B) An ETL tool
(C) A reporting tool
(D) A backup tool

✓ **Answer: (A) OLAP and data mining platform**

Q846. What is SQL Server's BULK INSERT?

(A) Imports data from file into a table
(B) A backup command
(C) A copy command
(D) An export command

✓ **Answer: (A) Imports data from file into a table**

Q847. What is BCP utility?

(A) Bulk Copy Program for importing/exporting data
(B) A backup utility
(C) A monitoring utility
(D) A replication utility

✓ **Answer: (A) Bulk Copy Program for importing/exporting data**

Q848. What is SQL Server's FILESTREAM?

(A) Stores BLOBs in NTFS file system
(B) An in-memory storage
(C) A column type
(D) An index type

✓ **Answer: (A) Stores BLOBs in NTFS file system**

Q849. What is SQL Server In-Memory OLTP (Hekaton)?

(A) Memory-optimized tables for high-performance OLTP
(B) An in-memory cache
(C) A columnar storage
(D) A replication feature

✓ **Answer: (A) Memory-optimized tables for high-performance OLTP**

Q850. What is a memory-optimized table?

(A) Table stored in RAM for high performance
(B) A cached table
(C) A partitioned table
(D) A columnstore table

✓ **Answer: (A) Table stored in RAM for high performance**

Q851. What is SQL Server's columnstore index?

(A) Columnar index for analytical queries
(B) A B-Tree index
(C) A bitmap index
(D) A hash index

✓ **Answer: (A) Columnar index for analytical queries**

Q852. What is batch mode on rowstore (SQL Server 2019)?

(A) Batch mode processing for row-based tables
(B) A new index type
(C) A new lock type
(D) A new join type

✓ **Answer: (A) Batch mode processing for row-based tables**

Q853. What is SQL Server's PIVOT operator?

(A) Rotates rows to columns using aggregation
(B) A join type
(C) A filter
(D) A set operation

✓ **Answer: (A) Rotates rows to columns using aggregation**

Q854. What is SQL Server's STRING_SPLIT function?

(A) Splits a string by delimiter into rows
(B) Joins strings
(C) Aggregates strings
(D) A window function

✓ **Answer: (A) Splits a string by delimiter into rows**

Q855. What is SQL Server's JSON_OBJECT function?

(A) Creates a JSON object from key-value pairs
(B) Parses JSON
(C) Validates JSON
(D) Queries JSON

✓ **Answer: (A) Creates a JSON object from key-value pairs**

Q856. What is SQL Server's OPENJSON?

(A) Parses JSON text into rows and columns
(B) Creates JSON
(C) Validates JSON
(D) A query function

✓ **Answer: (A) Parses JSON text into rows and columns**

Q857. What is SQL Server's OPENXML?

(A) Parses XML document into rows
(B) Creates XML
(C) Validates XML
(D) A query function

✓ **Answer: (A) Parses XML document into rows**

Q858. What is SQL Server's OPENQUERY?

(A) Executes a pass-through query on a linked server
(B) A local query
(C) A subquery
(D) A CTE

✓ **Answer: (A) Executes a pass-through query on a linked server**

Q859. What is a linked server in SQL Server?

(A) Connection to another data source for distributed queries
(B) A replication link

(C) A backup link
(D) A monitoring link

✓ **Answer: (A) Connection to another data source for distributed queries**

Q860. What is SQL Server's FOR XML PATH?

(A) Generates XML from query results
(B) A pivot
(C) A JSON generator
(D) A window function

✓ **Answer: (A) Generates XML from query results**

Q861. What is SQL Server's CROSS APPLY?

(A) Applies a table-valued function to each row (like INNER JOIN)
(B) A CROSS JOIN
(C) A LATERAL JOIN
(D) An OUTER JOIN

✓ **Answer: (A) Applies a table-valued function to each row (like INNER JOIN)**

Q862. What is SQL Server's OUTER APPLY?

(A) Applies table-valued function keeping non-matching rows (like LEFT JOIN)
(B) A CROSS JOIN
(C) A LATERAL JOIN
(D) An INNER JOIN

✓ **Answer: (A) Applies table-valued function keeping non-matching rows (like LEFT JOIN)**

Q863. What is SQL Server's MERGE statement?

(A) Performs UPSERT and DELETE in one statement
(B) An UPDATE statement
(C) An INSERT statement
(D) A join type

✓ **Answer: (A) Performs UPSERT and DELETE in one statement**

Q864. What is CHECKSUM_AGG in SQL Server?

(A) Aggregate function returning checksum of group values
(B) A hash function
(C) A window function
(D) A string function

✓ **Answer: (A) Aggregate function returning checksum of group values**

Q865. What is SQL Server's CHARINDEX?

(A) Returns starting position of substring
(B) A string split
(C) A pattern match
(D) A substring function

✓ **Answer: (A) Returns starting position of substring**

Q866. What is SQL Server's PATINDEX?

(A) Returns starting position of pattern in string
(B) A CHARINDEX variant
(C) A LIKE operator
(D) A regex function

✓ **Answer: (A) Returns starting position of pattern in string**

Q867. What is SQL Server's FORMAT function?

(A) Formats a value using .NET format strings
(B) A CAST function
(C) A CONVERT function
(D) A string function

✓ Answer: (A) Formats a value using .NET format strings

Q868. What is SQL Server's DATEFROMPARTS?

(A) Creates a date from year, month, day parts
(B) A date format function
(C) A date add function
(D) A date diff function

✓ Answer: (A) Creates a date from year, month, day parts

Q869. What is SQL Server's DATEDIFF?

(A) Returns difference between two dates
(B) A date add function
(C) A date format function
(D) A date extract function

✓ Answer: (A) Returns difference between two dates

Q870. What is SQL Server's DATEADD?

(A) Adds an interval to a date
(B) A date difference function
(C) A date format function
(D) A date extract function

✓ Answer: (A) Adds an interval to a date

Q871. What is SQL Server's GETDATE()?

(A) Returns current date and time
(B) Returns current date only
(C) Returns current time only
(D) Returns UTC time

✓ Answer: (A) Returns current date and time

Q872. What is SQL Server's GETUTCDATE()?

(A) Returns current UTC date and time
(B) Returns local time
(C) Returns date only
(D) Returns time only

✓ Answer: (A) Returns current UTC date and time

Q873. What is SQL Server's SYSDATETIMEOFFSET()?

(A) Returns current datetime with timezone offset
(B) Returns local datetime
(C) Returns UTC datetime
(D) Returns date only

✓ Answer: (A) Returns current datetime with timezone offset

Q874. What is SQL Server's TRY_CONVERT?

(A) Returns NULL if conversion fails instead of error
(B) Same as CONVERT
(C) Always succeeds
(D) Returns error

✓ Answer: (A) Returns NULL if conversion fails instead of error

Q875. What is SQL Server's TRY_CAST?

(A) Returns NULL if cast fails instead of error
(B) Same as CAST
(C) Always succeeds
(D) Returns error

✓ **Answer: (A) Returns NULL if cast fails instead of error**

Q876. What is SQL Server's ISJSON function?

(A) Returns 1 if string is valid JSON
(B) Parses JSON
(C) Creates JSON
(D) Validates XML

✓ **Answer: (A) Returns 1 if string is valid JSON**

Q877. What is SQL Server's Database Snapshots?

(A) Read-only static view of database at a point in time
(B) A backup
(C) A replica
(D) A copy

✓ **Answer: (A) Read-only static view of database at a point in time**

Q878. What is SQL Server's page verification (CHECKSUM)?

(A) Detects corrupted pages on disk
(B) A hashing function
(C) An encryption method
(D) A compression method

✓ **Answer: (A) Detects corrupted pages on disk**

Q879. What is SQL Server's Resource Governor?

(A) Manages workload by limiting CPU and memory per group
(B) A query optimizer
(C) A monitoring tool
(D) A backup tool

✓ **Answer: (A) Manages workload by limiting CPU and memory per group**

Q880. What is SQL Server's Policy-Based Management?

(A) Enforces configuration standards across servers
(B) A security policy
(C) A backup policy
(D) A replication policy

✓ **Answer: (A) Enforces configuration standards across servers**

Section: Oracle Specific & Expert Optimization (Q 881–1000)

Q881. What is Oracle's dual table?

(A) A dummy one-row table for SELECT expressions
(B) A system table
(C) A temp table
(D) A view

✓ **Answer: (A) A dummy one-row table for SELECT expressions**

Q882. What is Oracle's ROWNUM?

(A) Pseudo-column assigning row number before ORDER BY
(B) A window function
(C) An index
(D) A partition key

✓ **Answer: (A) Pseudo-column assigning row number before ORDER BY**

Q883. What is Oracle's ROWID?

(A) Physical address of a row in database
(B) A primary key
(C) A sequence
(D) A UUID

✓ **Answer: (A) Physical address of a row in database**

Q884. What is Oracle's SEQUENCE?

(A) Object generating unique numeric values
(B) An AUTO_INCREMENT
(C) An IDENTITY column
(D) A trigger

✓ **Answer: (A) Object generating unique numeric values**

Q885. What is Oracle's NVL function?

(A) Returns second arg if first is NULL
(B) Same as COALESCE
(C) Returns NULL
(D) Returns zero

✓ **Answer: (A) Returns second arg if first is NULL**

Q886. What is Oracle's NVL2 function?

(A) Returns second arg if first not NULL, else third
(B) Same as NVL
(C) Same as COALESCE
(D) Returns NULL

✓ **Answer: (A) Returns second arg if first not NULL, else third**

Q887. What is Oracle's DECODE function?

(A) Conditional expression: decode(val, search, result, default)
(B) A decode function
(C) An encryption function
(D) A CAST function

✓ **Answer: (A) Conditional expression: decode(val, search, result, default)**

Q888. What is Oracle's CONNECT BY?

(A) Hierarchical query clause
(B) A join type
(C) A filter
(D) A group by

✓ Answer: (A) Hierarchical query clause

Q889. What is START WITH in Oracle hierarchical queries?

(A) Specifies root rows of hierarchy
(B) An ORDER BY
(C) A GROUP BY
(D) A WHERE clause

✓ Answer: (A) Specifies root rows of hierarchy

Q890. What is LEVEL in Oracle hierarchical queries?

(A) Pseudo-column showing depth in hierarchy
(B) A row number
(C) A rank
(D) A partition key

✓ Answer: (A) Pseudo-column showing depth in hierarchy

Q891. What is SYS_CONNECT_BY_PATH?

(A) Returns path from root to current row as string
(B) A hierarchy function
(C) A string function
(D) A window function

✓ Answer: (A) Returns path from root to current row as string

Q892. What is Oracle's MERGE statement called?

(A) MERGE (also known as UPSERT)
(B) UPDATE OR INSERT
(C) UPSERT
(D) INSERT OR UPDATE

✓ Answer: (A) MERGE (also known as UPSERT)

Q893. What is Oracle's PL/SQL?

(A) Procedural Language/SQL: Oracle's procedural extension
(B) A query language
(C) A scripting language
(D) A markup language

✓ Answer: (A) Procedural Language/SQL: Oracle's procedural extension

Q894. What is Oracle's %TYPE attribute?

(A) Declares variable with same type as a column
(B) A table type
(C) A cursor type
(D) A collection type

✓ Answer: (A) Declares variable with same type as a column

Q895. What is Oracle's %ROWTYPE attribute?

(A) Declares variable with same structure as a table row
(B) A column type
(C) A cursor type
(D) A collection type

✓ Answer: (A) Declares variable with same structure as a table row

Q896. What is Oracle's VARRAY?

(A) Variable-size array collection type
(B) A nested table
(C) An associative array
(D) A cursor type

✓ **Answer: (A) Variable-size array collection type**

Q897. What is Oracle's nested table?

(A) Unbounded collection stored in separate storage
(B) A VARRAY
(C) An associative array
(D) A cursor type

✓ **Answer: (A) Unbounded collection stored in separate storage**

Q898. What is Oracle's associative array (INDEX BY table)?

(A) Key-value collection in PL/SQL
(B) A VARRAY
(C) A nested table
(D) A cursor type

✓ **Answer: (A) Key-value collection in PL/SQL**

Q899. What is Oracle's BULK COLLECT?

(A) Fetches multiple rows at once into collection
(B) A single row fetch
(C) A cursor open
(D) A DML operation

✓ **Answer: (A) Fetches multiple rows at once into collection**

Q900. What is Oracle's FORALL?

(A) Sends bulk DML operations for collection elements
(B) A loop
(C) A cursor
(D) A collection fetch

✓ **Answer: (A) Sends bulk DML operations for collection elements**

Q901. What is Oracle's DBMS_OUTPUT.PUT_LINE?

(A) Prints output to console for debugging
(B) A logging function
(C) An error handler
(D) A return statement

✓ **Answer: (A) Prints output to console for debugging**

Q902. What is Oracle's DBMS_SCHEDULER?

(A) Package for job scheduling
(B) A monitoring package
(C) A backup package
(D) A replication package

✓ **Answer: (A) Package for job scheduling**

Q903. What is Oracle's AWR?

(A) Automatic Workload Repository: performance statistics
(B) A backup tool
(C) A replication tool
(D) A monitoring view

✓ **Answer: (A) Automatic Workload Repository: performance statistics**

Q904. What is Oracle's ASH?

(A) Active Session History: recent session activity
(B) A backup tool
(C) A replication tool
(D) A monitoring view

✓ **Answer: (A) Active Session History: recent session activity**

Q905. What is Oracle's ADDM?

(A) Automatic Database Diagnostic Monitor: performance advisor
(B) A backup tool
(C) A monitoring tool
(D) A replication tool

✓ **Answer: (A) Automatic Database Diagnostic Monitor: performance advisor**

Q906. What is Oracle's RAC?

(A) Real Application Clusters: multiple instances on shared storage
(B) A replication method
(C) A backup method
(D) A standby method

✓ **Answer: (A) Real Application Clusters: multiple instances on shared storage**

Q907. What is Oracle Data Guard?

(A) Disaster recovery with standby databases
(B) A backup tool
(C) A replication tool
(D) A monitoring tool

✓ **Answer: (A) Disaster recovery with standby databases**

Q908. What is Oracle's Flashback feature?

(A) Recover data from a past point without restore
(B) A backup
(C) A replication
(D) A standby

✓ **Answer: (A) Recover data from a past point without restore**

Q909. What is Oracle's UNDO tablespace?

(A) Stores before-images for MVCC and rollback
(B) A data tablespace
(C) A temp tablespace
(D) A system tablespace

✓ **Answer: (A) Stores before-images for MVCC and rollback**

Q910. What is Oracle's TEMP tablespace?

(A) Stores temporary data for sort operations
(B) A data tablespace
(C) An UNDO tablespace
(D) A system tablespace

✓ **Answer: (A) Stores temporary data for sort operations**

Q911. What is Oracle's SYSTEM tablespace?

(A) Core tablespace containing data dictionary
(B) A data tablespace
(C) A temp tablespace
(D) An UNDO tablespace

✓ **Answer: (A) Core tablespace containing data dictionary**

Q912. What is Oracle's SGA?

(A) System Global Area: shared memory structures
(B) A session-specific memory
(C) A backup area
(D) A replication area

✓ **Answer: (A) System Global Area: shared memory structures**

Q913. What is Oracle's PGA?

(A) Program Global Area: session-specific memory
(B) Shared memory
(C) A backup area
(D) A replication area

✓ **Answer: (A) Program Global Area: session-specific memory**

Q914. What is Oracle's shared pool?

(A) SGA component caching SQL plans and dictionary
(B) A data buffer
(C) A log buffer
(D) A temp buffer

✓ **Answer: (A) SGA component caching SQL plans and dictionary**

Q915. What is Oracle's buffer cache?

(A) SGA component caching data blocks
(B) A plan cache
(C) A log buffer
(D) A temp buffer

✓ **Answer: (A) SGA component caching data blocks**

Q916. What is Oracle's redo log buffer?

(A) SGA component buffering redo log entries
(B) A data cache
(C) A plan cache
(D) A temp buffer

✓ **Answer: (A) SGA component buffering redo log entries**

Q917. What is a hint in Oracle SQL?

(A) Comment directive to influence optimizer
(B) A stored procedure
(C) A trigger
(D) An index

✓ **Answer: (A) Comment directive to influence optimizer**

Q918. What is Oracle's /*+ INDEX(table index) */ hint?

(A) Forces use of specified index
(B) Forces a full scan
(C) A join hint
(D) A parallel hint

✓ **Answer: (A) Forces use of specified index**

Q919. What is Oracle's EXPLAIN PLAN FOR?

(A) Shows execution plan for a query
(B) Executes the query

(C) Creates an index
(D) Analyzes the query

✓ **Answer: (A) Shows execution plan for a query**

Q920. What is Oracle's DBMS_STATS?

(A) Package for gathering optimizer statistics
(B) A backup package
(C) A replication package
(D) A monitoring package

✓ **Answer: (A) Package for gathering optimizer statistics**

Q921. What is Oracle's partitioning option?

(A) Divides large tables/indexes for performance and manageability
(B) A normalization technique
(C) A replication method
(D) A backup method

✓ **Answer: (A) Divides large tables/indexes for performance and manageability**

Q922. What is Oracle's interval partitioning?

(A) Automatically creates partitions for new ranges
(B) Manual range partitioning
(C) List partitioning
(D) Hash partitioning

✓ **Answer: (A) Automatically creates partitions for new ranges**

Q923. What is Oracle's reference partitioning?

(A) Partitions child table by parent's partition key via FK
(B) A hash partition
(C) A range partition
(D) A list partition

✓ **Answer: (A) Partitions child table by parent's partition key via FK**

Q924. What is Oracle's virtual column partitioning?

(A) Partitioning based on expression column
(B) A range partition
(C) A list partition
(D) A hash partition

✓ **Answer: (A) Partitioning based on expression column**

Q925. What is Oracle's full outer join syntax alternative?

(A) (+) operator on both sides (rarely used)
(B) FULL JOIN
(C) CROSS JOIN
(D) NATURAL JOIN

✓ **Answer: (A) (+) operator on both sides (rarely used)**

Q926. What is Oracle's (+) operator?

(A) Outer join syntax (deprecated in favor of standard)
(B) An addition operator
(C) A hint
(D) A pseudo-column

✓ **Answer: (A) Outer join syntax (deprecated in favor of standard)**

Q927. What is Oracle's LAG/LEAD support?

(A) Fully supported as standard SQL window functions
(B) Not supported
(C) Partial support
(D) Only in 12c+

✓ **Answer: (A) Fully supported as standard SQL window functions**

Q928. What is Oracle's LISTAGG function?

(A) Aggregates strings within a group with delimiter
(B) A list function
(C) A count function
(D) A window function

✓ **Answer: (A) Aggregates strings within a group with delimiter**

Q929. What is Oracle's WITHIN GROUP clause?

(A) Used with ordered-set aggregate functions like LISTAGG
(B) A GROUP BY alternative
(C) A HAVING clause
(D) A filter

✓ **Answer: (A) Used with ordered-set aggregate functions like LISTAGG**

Q930. What is Oracle's PIVOT XML?

(A) Returns pivot results as XML for unknown values
(B) A standard pivot
(C) A JSON pivot
(D) A window function

✓ **Answer: (A) Returns pivot results as XML for unknown values**

Q931. What does SQL standard FETCH FIRST n ROWS ONLY do?

(A) Limits result to n rows (SQL:2008 standard)
(B) Orders rows
(C) Groups rows
(D) Filters rows

✓ **Answer: (A) Limits result to n rows (SQL:2008 standard)**

Q932. What is the SQL standard OVER() clause used for?

(A) Window functions
(B) GROUP BY replacement
(C) ORDER BY alternative
(D) A join type

✓ **Answer: (A) Window functions**

Q933. What is the FILTER clause in aggregate functions?

(A) Filters rows for aggregate with condition (SQL standard)
(B) A HAVING clause
(C) A WHERE clause
(D) A window filter

✓ **Answer: (A) Filters rows for aggregate with condition (SQL standard)**

Q934. What is the DISTINCT ON clause (PostgreSQL)?

(A) Returns first row per distinct value group
(B) A DISTINCT filter
(C) A window function
(D) A GROUP BY

✓ **Answer: (A) Returns first row per distinct value group**

Q935. What is a covering index in practice?

(A) Index contains all columns the query needs, no table lookup needed
(B) An index on all columns
(C) A clustered index
(D) A unique index

✓ **Answer: (A) Index contains all columns the query needs, no table lookup needed**

Q936. What is index intersection?

(A) Using multiple indexes together to satisfy a query
(B) A composite index
(C) A covering index
(D) An index merge

✓ **Answer: (A) Using multiple indexes together to satisfy a query**

Q937. What is index merge (MySQL)?

(A) MySQL can use multiple indexes for a single table query
(B) A composite index
(C) A covering index
(D) A partial index

✓ **Answer: (A) MySQL can use multiple indexes for a single table query**

Q938. What is the SQL execution order?

(A) FROM->WHERE->GROUP BY->HAVING->SELECT->ORDER BY->LIMIT
(B) SELECT first
(C) ORDER BY first
(D) HAVING before WHERE

✓ **Answer: (A) FROM->WHERE->GROUP BY->HAVING->SELECT->ORDER BY->LIMIT**

Q939. What is a sargable predicate?

(A) A predicate that can use an index (Search ARGument ABLE)
(B) Any predicate
(C) A computed predicate
(D) A function predicate

✓ **Answer: (A) A predicate that can use an index (Search ARGument ABLE)**

Q940. What makes a predicate non-sargable?

(A) Applying a function to the indexed column
(B) Using = operator
(C) Using BETWEEN
(D) Using IN

✓ **Answer: (A) Applying a function to the indexed column**

Q941. What is predicate pushdown?

(A) Optimizer moves filter conditions closer to data source
(B) Pushing indexes
(C) A join optimization
(D) A sort optimization

✓ **Answer: (A) Optimizer moves filter conditions closer to data source**

Q942. What is cost-based optimization?

(A) Optimizer uses statistics to estimate cheapest plan
(B) Rule-based optimization
(C) A fixed plan selection
(D) A hint-based selection

✓ **Answer: (A) Optimizer uses statistics to estimate cheapest plan**

Q943. What is rule-based optimization?

(A) Uses fixed rules to select plan regardless of data distribution
(B) Cost-based optimization
(C) Statistics-based
(D) Adaptive

✓ **Answer: (A) Uses fixed rules to select plan regardless of data distribution**

Q944. What is a hash aggregate?

(A) Aggregation using hash table for group identification
(B) A sort aggregate
(C) A window function
(D) A scalar aggregate

✓ **Answer: (A) Aggregation using hash table for group identification**

Q945. What is a sort aggregate?

(A) Aggregation after sorting data by group keys
(B) A hash aggregate
(C) A window function
(D) A scalar aggregate

✓ **Answer: (A) Aggregation after sorting data by group keys**

Q946. What is loop join?

(A) For each row in outer table, probe inner table
(B) A hash join
(C) A merge join
(D) A sort merge join

✓ **Answer: (A) For each row in outer table, probe inner table**

Q947. What is a lookup join?

(A) Join using index to retrieve single rows from inner table
(B) A hash join
(C) A merge join
(D) A loop join

✓ **Answer: (A) Join using index to retrieve single rows from inner table**

Q948. What is join reordering?

(A) Optimizer changes join order for efficiency
(B) A query hint
(C) A fixed join order
(D) A sort operation

✓ **Answer: (A) Optimizer changes join order for efficiency**

Q949. What is the optimizer's join cardinality estimation?

(A) Estimates rows output from a join for planning
(B) Counting rows
(C) Sorting rows
(D) Grouping rows

✓ **Answer: (A) Estimates rows output from a join for planning**

Q950. What is statistics histogram?

(A) Distribution of data values in a column for optimizer
(B) A chart
(C) An index
(D) A partition

✓ **Answer: (A) Distribution of data values in a column for optimizer**

Q951. What is multi-column statistics?

(A) Statistics on combinations of columns for better estimates
(B) Single-column statistics
(C) An index
(D) A constraint

✓ **Answer: (A) Statistics on combinations of columns for better estimates**

Q952. What is automatic query rewrite?

(A) Optimizer rewrites query to use materialized views
(B) Manual rewriting
(C) A hint
(D) A stored procedure

✓ **Answer: (A) Optimizer rewrites query to use materialized views**

Q953. What is result set caching?

(A) Caching query results for identical repeat queries
(B) A plan cache
(C) An index cache
(D) A data cache

✓ **Answer: (A) Caching query results for identical repeat queries**

Q954. What is database partitioning pruning?

(A) Optimizer skips irrelevant partitions based on WHERE clause
(B) Partition deletion
(C) Partition compression
(D) Partition indexing

✓ **Answer: (A) Optimizer skips irrelevant partitions based on WHERE clause**

Q955. What is row estimation error?

(A) When optimizer's cardinality estimate differs significantly from actual
(B) A join error
(C) An index error
(D) A filter error

✓ **Answer: (A) When optimizer's cardinality estimate differs significantly from actual**

Q956. What is a selectivity hint?

(A) Manually providing selectivity estimate to optimizer
(B) A standard hint
(C) An index hint
(D) A join hint

✓ **Answer: (A) Manually providing selectivity estimate to optimizer**

Q957. What is adaptive statistics (Oracle 12c+)?

(A) Optimizer gathers additional stats during execution
(B) Static statistics
(C) Manual statistics
(D) Cached statistics

✓ **Answer: (A) Optimizer gathers additional stats during execution**

Q958. What is SQL plan baseline?

(A) Accepted plans that optimizer must use
(B) A query hint

(C) A stored procedure
(D) An index

✓ **Answer: (A) Accepted plans that optimizer must use**

Q959. What is SQL plan management?

(A) Controlling which plans the optimizer uses
(B) Query management
(C) Index management
(D) Statistics management

✓ **Answer: (A) Controlling which plans the optimizer uses**

Q960. What is cardinality feedback (Oracle)?

(A) Adjusts estimates based on actual cardinality after first execution
(B) Static cardinality
(C) Manual cardinality
(D) Cached cardinality

✓ **Answer: (A) Adjusts estimates based on actual cardinality after first execution**

Q961. What is the purpose of the ANALYZE command (PostgreSQL)?

(A) Updates statistics for the query planner
(B) Checks for corruption
(C) Defragments tables
(D) Rebuilds indexes

✓ **Answer: (A) Updates statistics for the query planner**

Q962. What is the purpose of UPDATE STATISTICS (SQL Server)?

(A) Updates column statistics for query optimizer
(B) Rebuilds indexes
(C) Defragments tables
(D) Checks corruption

✓ **Answer: (A) Updates column statistics for query optimizer**

Q963. What is the default statistics sampling rate in SQL Server?

(A) Automatic (varies by table size)
(B) 100%
(C) 10%
(D) 1%

✓ **Answer: (A) Automatic (varies by table size)**

Q964. What is full statistics scan?

(A) Sampling 100% of rows for accurate statistics
(B) Default sampling
(C) Automatic sampling
(D) A partial scan

✓ **Answer: (A) Sampling 100% of rows for accurate statistics**

Q965. What is incremental statistics (SQL Server 2014+)?

(A) Per-partition statistics for partitioned tables
(B) Full statistics
(C) Sampled statistics
(D) Automatic statistics

✓ **Answer: (A) Per-partition statistics for partitioned tables**

Q966. What is the difference between a view and a table?

(A) View is a stored query; table stores physical data
(B) They are identical
(C) Table is faster always
(D) View stores data

✓ **Answer: (A) View is a stored query; table stores physical data**

Q967. What is a recursive view?

(A) A view that references itself
(B) A materialized view
(C) A simple view
(D) A view with GROUP BY

✓ **Answer: (A) A view that references itself**

Q968. What is schema binding in views (SQL Server)?

(A) Prevents changes to referenced objects
(B) Binds data to view
(C) Creates a materialized view
(D) An index on the view

✓ **Answer: (A) Prevents changes to referenced objects**

Q969. What is an indexed view (SQL Server)?

(A) A view with a clustered index (materialized)
(B) A simple view with indexes
(C) A view on indexed table
(D) A schema-bound view

✓ **Answer: (A) A view with a clustered index (materialized)**

Q970. What is view encapsulation?

(A) Hiding complex query logic behind a simple view
(B) A security feature
(C) A normalization technique
(D) A compression method

✓ **Answer: (A) Hiding complex query logic behind a simple view**

Q971. What is column aliasing in a view?

(A) Renaming columns in view definition
(B) Creating new columns
(C) Hiding columns
(D) Indexing columns

✓ **Answer: (A) Renaming columns in view definition**

Q972. What is a recursive query?

(A) A query that references itself (usually via CTE)
(B) A join query
(C) A subquery
(D) A window function

✓ **Answer: (A) A query that references itself (usually via CTE)**

Q973. What is the base case in a recursive query?

(A) The non-recursive anchor member
(B) The recursive member
(C) The final SELECT
(D) The UNION ALL

✓ **Answer: (A) The non-recursive anchor member**

Q974. What is the recursive member in a CTE?

(A) The query that references the CTE itself
(B) The anchor member
(C) The final SELECT
(D) The base case

✓ **Answer: (A) The query that references the CTE itself**

Q975. What is UNION ALL used for in recursive CTE?

(A) Combines anchor and recursive members
(B) Removes duplicates
(C) Joins results
(D) Filters results

✓ **Answer: (A) Combines anchor and recursive members**

Q976. What is depth-first traversal in hierarchical SQL?

(A) Processing nodes by going deep before wide
(B) Processing level by level
(C) Random traversal
(D) Sorted traversal

✓ **Answer: (A) Processing nodes by going deep before wide**

Q977. What is breadth-first traversal in hierarchical SQL?

(A) Processing level by level (same depth before going deeper)
(B) Going deep first
(C) Random traversal
(D) Sorted traversal

✓ **Answer: (A) Processing level by level (same depth before going deeper)**

Q978. What is cycle detection in recursive CTE?

(A) Preventing infinite loops in circular hierarchies
(B) Detecting duplicate rows
(C) Finding NULL values
(D) Checking constraints

✓ **Answer: (A) Preventing infinite loops in circular hierarchies**

Q979. What is path enumeration?

(A) Storing the hierarchy path as a string column
(B) A traversal method
(C) A CTE technique
(D) A join method

✓ **Answer: (A) Storing the hierarchy path as a string column**

Q980. What is nested set model?

(A) Representing hierarchy using left/right values
(B) A join model
(C) A CTE model
(D) A partitioning model

✓ **Answer: (A) Representing hierarchy using left/right values**

Q981. What is closure table?

(A) A table storing all ancestor-descendant pairs
(B) A junction table
(C) A bridge table
(D) A fact table

✓ **Answer: (A) A table storing all ancestor-descendant pairs**

Q982. What is adjacency list?

(A) Storing parent_id in each row
(B) A nested set
(C) A closure table
(D) A path enumeration

✓ **Answer: (A) Storing parent_id in each row**

Q983. What is the simplest hierarchy representation?

(A) Adjacency list (parent_id column)
(B) Nested set
(C) Closure table
(D) Path enumeration

✓ **Answer: (A) Adjacency list (parent_id column)**

Q984. What is a balanced tree in B-Tree index?

(A) All leaf nodes at same depth
(B) Unbalanced tree
(C) A heap
(D) A sequential file

✓ **Answer: (A) All leaf nodes at same depth**

Q985. What is index depth in B-Tree?

(A) Number of levels from root to leaf
(B) Number of indexes
(C) Number of pages
(D) Number of rows per page

✓ **Answer: (A) Number of levels from root to leaf**

Q986. What is a page in SQL databases?

(A) Fixed-size block of storage (typically 8KB)
(B) A row
(C) A column
(D) An index entry

✓ **Answer: (A) Fixed-size block of storage (typically 8KB)**

Q987. What is page splitting?

(A) Dividing a full page into two when inserting
(B) Merging pages
(C) Compressing pages
(D) Deleting from a page

✓ **Answer: (A) Dividing a full page into two when inserting**

Q988. What is SQL's SIMILAR TO operator (PostgreSQL)?

(A) Pattern matching using SQL regular expression syntax
(B) LIKE operator
(C) ~ operator
(D) ILIKE operator

✓ **Answer: (A) Pattern matching using SQL regular expression syntax**

Q989. What is pg_trgm's similarity function?

(A) Returns similarity score between two strings (0-1)
(B) A LIKE function
(C) A regex function
(D) A distance function

✓ **Answer: (A) Returns similarity score between two strings (0-1)**

Q990. What is Levenshtein distance in SQL?

(A) Edit distance between two strings (via extension)
(B) A similarity score
(C) A regex match
(D) A pattern match

✓ **Answer: (A) Edit distance between two strings (via extension)**

Q991. What is full-text search?

(A) Searching text content using linguistic analysis
(B) A LIKE search
(C) A regex search
(D) An exact match

✓ **Answer: (A) Searching text content using linguistic analysis**

Q992. What is a tsvector in PostgreSQL?

(A) Preprocessed document for full-text search
(B) A vector type
(C) An array type
(D) A JSON type

✓ **Answer: (A) Preprocessed document for full-text search**

Q993. What is a tsquery in PostgreSQL?

(A) A full-text search query
(B) A time query
(C) A text query
(D) A vector query

✓ **Answer: (A) A full-text search query**

Q994. What is to_tsvector() in PostgreSQL?

(A) Converts text to tsvector for full-text search
(B) A conversion function
(C) A hashing function
(D) A sorting function

✓ **Answer: (A) Converts text to tsvector for full-text search**

Q995. What is the @@ operator in PostgreSQL?

(A) Full-text search match between tsvector and tsquery
(B) A comparison operator
(C) A regex operator
(D) A LIKE operator

✓ **Answer: (A) Full-text search match between tsvector and tsquery**

Q996. What is SQL's MATCH AGAINST (MySQL)?

(A) Full-text search syntax in MySQL
(B) A pattern match
(C) A LIKE operator
(D) A regex operator

✓ **Answer: (A) Full-text search syntax in MySQL**

Q997. What is CONTAINS in SQL Server full-text search?

(A) Searches for words and phrases in full-text indexed columns
(B) A LIKE operator

(C) A regex operator

(D) A pattern match

✓ **Answer: (A) Searches for words and phrases in full-text indexed columns**

Q998. What is FREETEXT in SQL Server?

(A) Full-text search using meaning of words

(B) An exact match

(C) A LIKE search

(D) A CONTAINS variant

✓ **Answer: (A) Full-text search using meaning of words**

Q999. What is a stop word in full-text search?

(A) Common words ignored in search (the, a, is)

(B) A search keyword

(C) A search operator

(D) A search index

✓ **Answer: (A) Common words ignored in search (the, a, is)**

Q1000. What is the purpose of stemming in full-text search?

(A) Reduces words to their root form for broader matching

(B) Removes stop words

(C) Indexes all words

(D) Creates synonyms

✓ **Answer: (A) Reduces words to their root form for broader matching**
